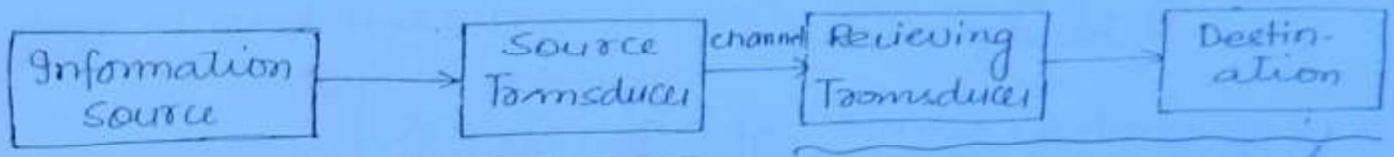
Defn:
Communication is the process of transformation of information from source to destination or from transmitter to Reciever.

(3)

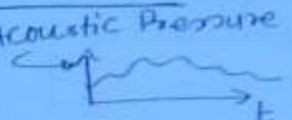
* Basic Block diagram of Commn system: (wired Commn System)



TRANSMITTER

→ Preferable for short dist. Commn.

RECIEVER



Note:
Voice → 300Hz – 3.5KHz } subset
Audio → 20Hz – 20KHz }
Video → 0 to 4.5MHz {Variation of light intensity with time}.

{Variation of Acoustic Pressure with time}.

Information Source:
*Information source is the source of information

Source Transducer: Source xducer converts a physical signal to electrical signal equivalent. Eg Mic

channel:
channel is the medium through which signal is transmitted from one place to another.

Note:
1. wired Commn system, is preferred for short distance commn. where the channel will be:
   a) Co-axial Cable
   b) Parralled wire.
   c) Twisted pair etc.

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: For long distance comm^{n} wireless comm^{n} is preferred where the channel will be free space.

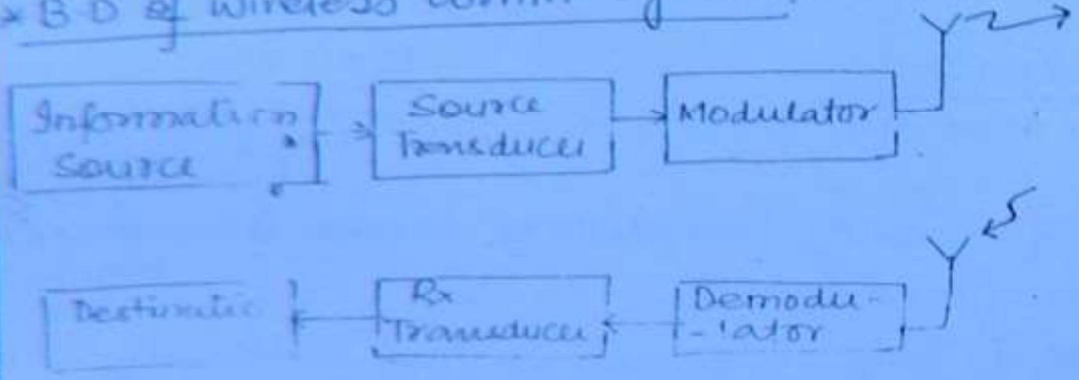
(4)

Receiver circuit

. It converts electrical signal as physical signal

Eg: loudspeaker.

* B.D of wireless comm^{n} system :-



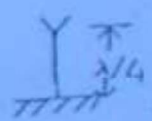
- For transmission of a signal to very much long distance through free space, modulation has to be used.

* Need of Modulation :-

¿ transmission for long distance

1. Reducing of Antenna height ; for faithful transmission, the height of Antenna from the ground should be $\lambda/4$

$$h_t = \lambda/4$$



$$\lambda = \frac{v}{f}$$

Note :-

As, only EM wave travel in free space and travels with speed of light. So

$$\lambda = c/f$$

$$\text{So, } h_t = \frac{c}{4f}, \quad h_t \propto \frac{1}{f}$$

let $f = 15KHz$, so, $h_t = 5Km$, which is not possible

Hence frequency has to be increased and hence modulation is introduced.

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15 KHz → [Modulator] → 1 MHz

$h_t = \frac{c}{4f} = \frac{3\times10^8}{4\times10^6}$

$h_t = 75\,m$, practically possible

(5)

Note:

1. Transmitting Antenna converts electrical signal as EM signal and will travel with the velocity of light

So,

$\lambda = c/f$

$h_t = c/4f$

2. Modulation is the Process of increasing the frequency of the signal to reduce Antenna height.

a) $f = 15\,KHz$ —(After modulation)→ b) $f = 1\,MHz$

$h_t = c/4f$

$h_t = 5000\,m$

$h_t = c/4f$

$h_t = 75\,M$

2. Multiplexing:

It is the process of transmission of multiple no of signal through a single channel at the same time

* Generally, without modulation, Multiplexing is not possible

* The lowest Possible frequency contained by a signal has to be taken into Reference to decide Antenna height.

Note:

1. Modulation is used in wired comm^n system for Multiple xing

FOURIER TRANSFORM:

Fourier Transform is a mathematical tool used to find the frequencies contained by given time domain signal

As,

$$x(t) \xrightarrow{F.T} x(f)$$

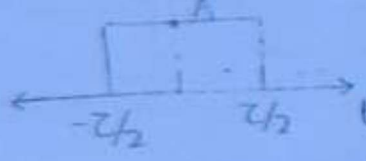
and mathematically,

$$\boxed{X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt}$$

→ fourier xform is performed by the spectrum Analyser and has 2 modes:

1) magnitude plot (b/w $|X(f)|$ & f)
2) phase plot (b/w $\angle X(f)$ & f).

let



$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt$$

$$= \int_{-\tau/2}^{\tau/2} A e^{-j2\pi ft}\, dt$$

$$= \frac{A}{-j2\pi f}\left[e^{-j2\pi ft}\right]_{-\tau/2}^{\tau/2}$$

$$= \frac{-A}{j2\pi f}\left[e^{-j2\pi f \times \tau/2} - e^{-j2\pi f \times -\tau/2}\right]$$

$$= \frac{-A}{j2\pi f}\left[e^{-j\pi f\tau} - e^{j\pi f\tau}\right]$$

$$= \frac{A}{j2\pi f}\left[e^{j\pi f\tau} - e^{-j\pi f\tau}\right]$$

$$= \frac{A}{\pi f}\left[\frac{e^{j\pi f\tau} - e^{-j\pi f\tau}}{2j}\right]$$

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Note:

1. The Rectangular Pulse covers the frequency from $+\infty$ to $-\infty$ but only +ve frequency is taken into consideration

2. Signal B.W is defined as:

$$\text{Signal B.W} = \text{Highest +ve freq}^{n} - \text{Lowest +ve freq}^{n}$$

NOW,

In above case,

$$\text{Signal B.W} = +\infty - 0$$
$$= +\infty$$

3. Channel Bandwidth is defined as the Range of frequency which the channel allows to pass without any distortion or attenuation.

Generally,

Co-axial cable = 0 to 600 MHz → B.W = 600 MHz

Parallel wire = 0 to 200 MHz → B.W = 200 MHz

Optical fibre = GHz → B.W = In GHz

Also,

$$\text{Channel B.W} \geq \text{Signal B.W}$$

4. In the above example, the B.W of signal was ∞. Hence no channel supports its transmission. Hence either has to be performed:

1) channel to be designed of B.W = ∞ X

2) Signal B.W reduced to some finite value. (Band limiting)

5. For Proper xmission of signal without any loss,

$$\text{Channel B.W} \geq \text{Signal B.W}$$

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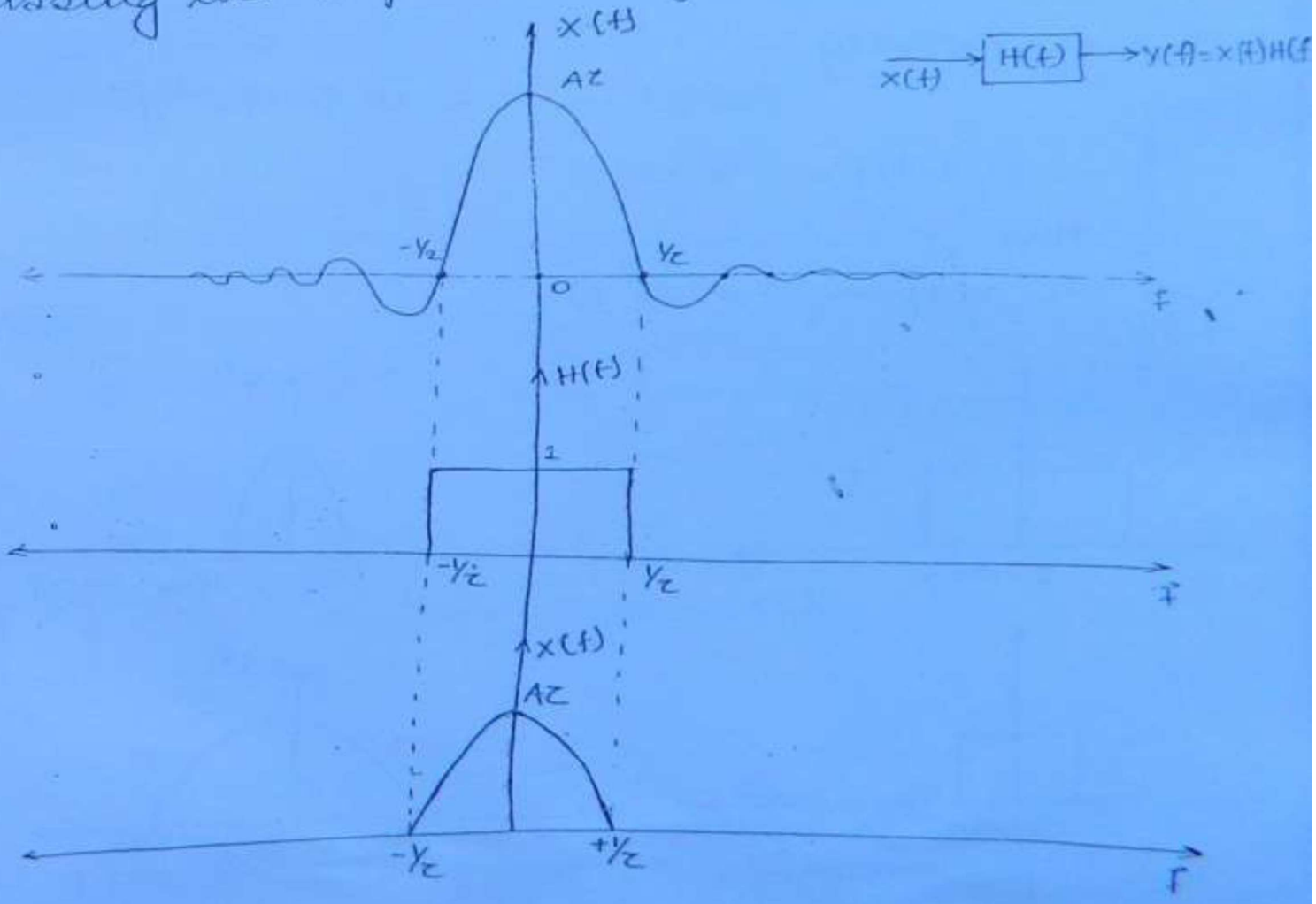
Since B.W of above signal is ∞, a channel having B.W ∞ is needed for transmission, but for practical channel, B.W will be finite.

Note:

1. The process of decreasing the B.W of a signal from ∞ to some finite value is called as "BAND LIMITING PROCESS".

2. The frequency ranging from $-1/2$ to $+1/2$ contains 95 to 99% of the strength of the total signal. Hence we have to eliminate the rest. This is done by passing the signal through a L.P.F.



*Generally for a signal, most of the strength will be retained by low frequencies, strength of high frequencies go on decreasing and finally becomes zero.

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2. For Band limiting a signal, all the significant frequencies should be Retained and Insignificant frequencies has to be eliminated.

3. For Band limiting, generally the signal will be passed through proper LPF

NOTE:

1. To use the channel B.W efficiently, we generally transmit Significant frequencies only.

* Properties of FOURIER TRANSFORM:

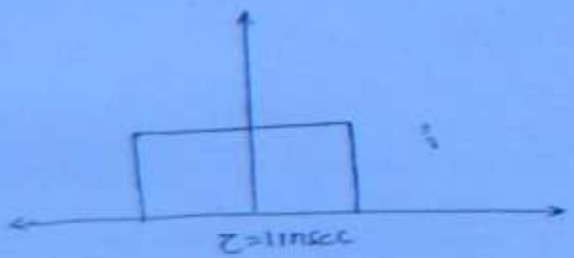
1. Duality Property:

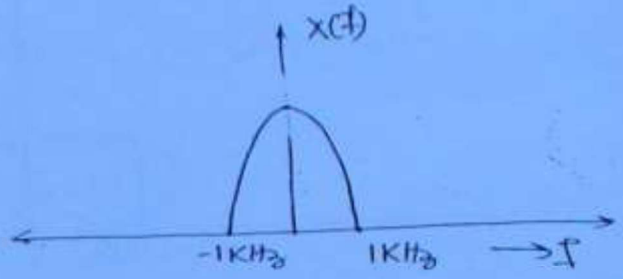
According to this property:

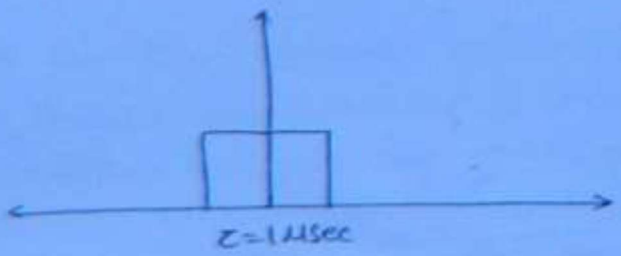
$$\mathcal{F}\{x(t)\} \longleftrightarrow x(f)$$

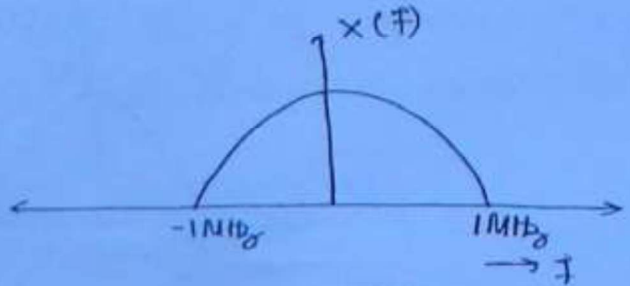
then,

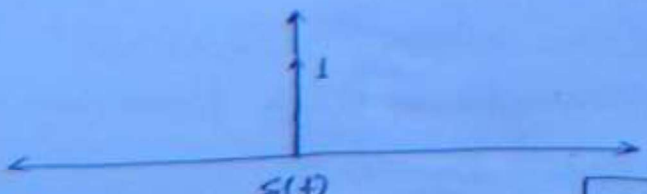
$$x(t) \longleftrightarrow x(-f)$$



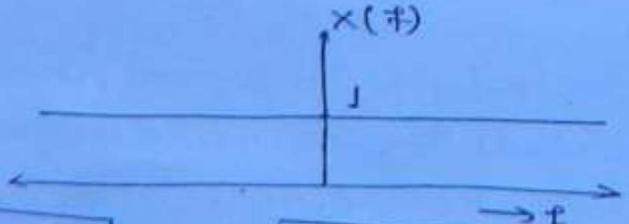








$\delta(t) \longleftrightarrow 1$



B.W = ∞

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Note:

1. $\delta(t) = \infty, \; t = 0$
$= 0 \; ; \; t \neq 0$
$\Rightarrow \int_{-\infty}^{\infty} \delta(t)\,dt = 1$ (II)

2. $2\delta(t) = \infty \; ; \; t = 0$
$= 0 \; ; \; t \neq 0$
$\Rightarrow \int_{-\infty}^{\infty} 2\delta(t)\,dt = 2$

Now,

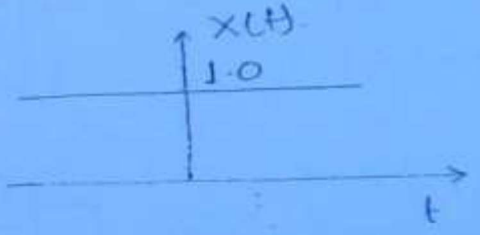
$x(t) \longrightarrow \delta(t) \longleftrightarrow 1 \times X(f)$

by duality property.

$1 \longleftrightarrow \delta(-f) = \delta(f)$
$x(t) \qquad x(-f)$

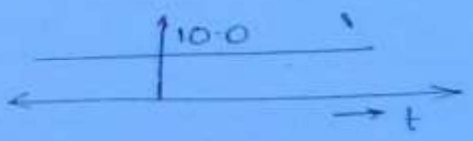
1. So,

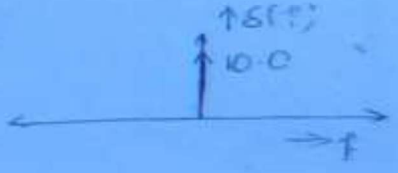
$1 \longleftrightarrow \delta(f)$



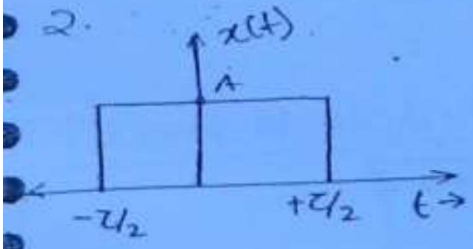
Also,

$10 \longleftrightarrow 10\delta(f)$





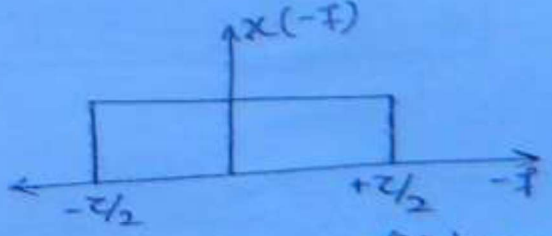
2.



$x(f) \xleftrightarrow{FT} A\tau \,\mathrm{sinc}(f\tau)$

So, $x(t) \xleftrightarrow{F.T} x(-f)$.

So, $A\tau \,\mathrm{sinc}(t\tau) \longleftrightarrow$



(∵ it is even func^n)

Frequency shifting property:

According to this property

if $x(t) \longleftrightarrow X(f)$

then

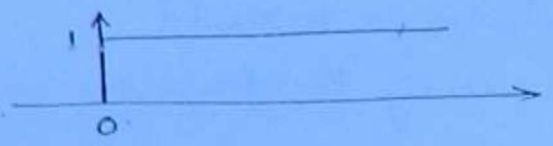
$$e^{j2\pi f_0 t} x(t) \longleftrightarrow X(f - f_0)$$

$$e^{-j2\pi f_0 t} x(t) \longleftrightarrow X(f + f_0)$$

let

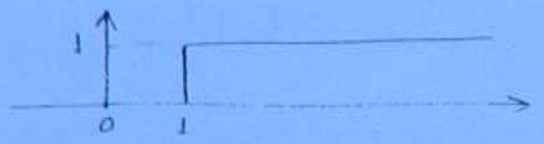
$u(t) = 1 ; t \geq 0$

$= 0 ; t < 0$



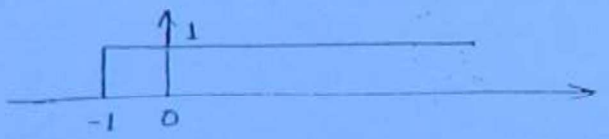
$u(t-1) = 1 ; t \geq 1$

$0 ; t < 1$



$u(t+1) = 1 ; t \geq -1$

$0 ; t < -1$

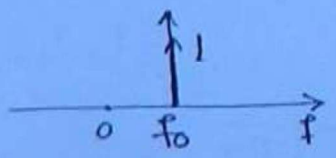


Note:

; $e^{j2\pi f_0 t} \longleftrightarrow ?$

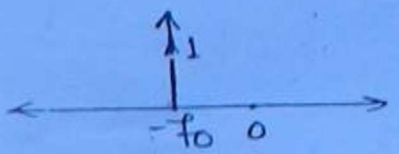
As, $1 \xrightarrow{x(t)} \longleftrightarrow \delta(f) \rightarrow X(f)$

$$e^{j2\pi f_0 t} \cdot 1 \longleftrightarrow \delta(f - f_0)$$



Also,

$$e^{-j2\pi f_0 t} \longleftrightarrow \delta(f + f_0)$$



Also,

$$\cos 2\pi f_0 t = \frac{e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}}{2} \longleftrightarrow \frac{1}{2}\{\delta(f - f_0) + \delta(f + f_0)\}.$$

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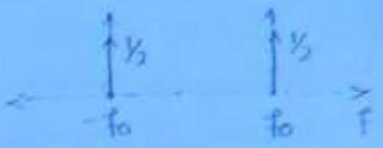
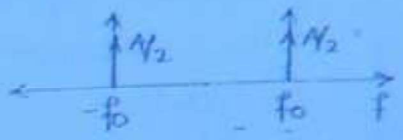
So,

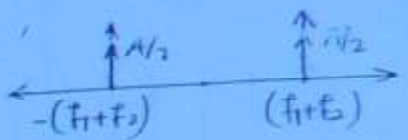
1. $\cos 2\pi f_0 t \longleftrightarrow \frac{\delta(f-f_0)+\delta(f+f_0)}{2}$



2. $A\cos 2\pi f_0 t \longleftrightarrow \frac{A}{2}\{\delta(f-f_0)+\delta(f+f_0)\}$



3. $A\cos 2\pi(f_1+f_2)t \longleftrightarrow \frac{A}{2}\{\delta(f-(f_1+f_2))+\delta(f+(f_1+f_2))\}$



3. Modulation Property:-

It states that:

If

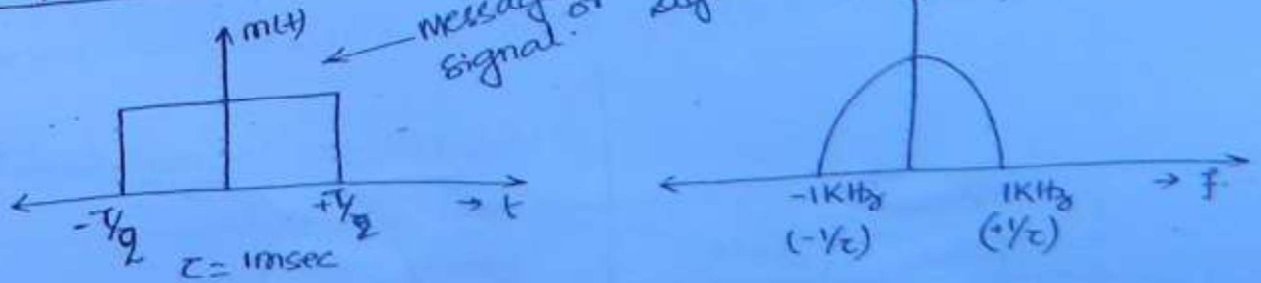
$$x(t) \longleftrightarrow X(f)$$

then

$$x(t)\cos 2\pi f_0 t \longleftrightarrow \frac{X(f-f_0)+X(f+f_0)}{2}$$

$$x(t)\left\{\frac{e^{j2\pi f_0 t}+e^{-j2\pi f_0 t}}{2}\right\}$$

Concept of Modulation:-



→ If a signal contain all the significant frequency low then such signals are called as "BASEBAND SIGNAL".

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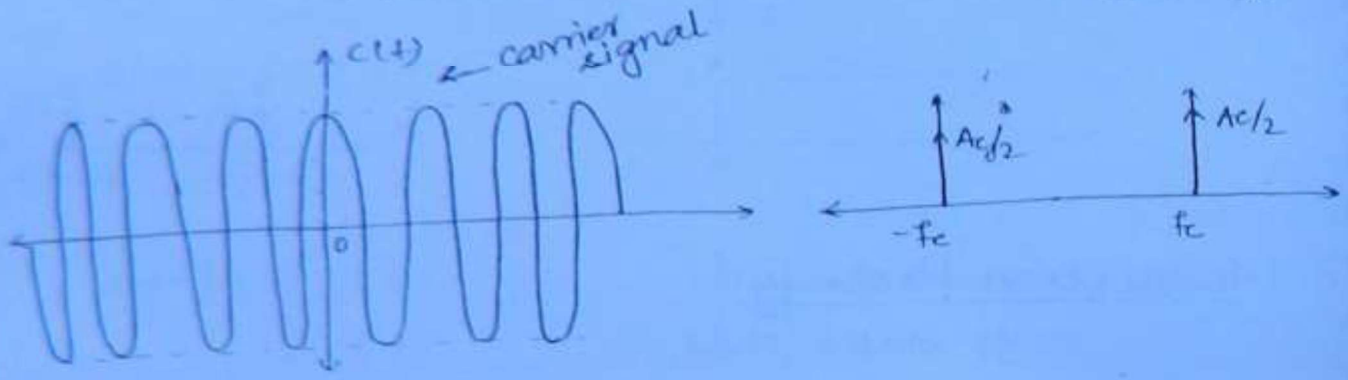
→ These BPF based signals have significant low frequencies. Hence they require huge Antenna heights, which is impossible to construct. Hence the process of Modulation is introduced such that the frequency is increased to Reduce Antenna heights.

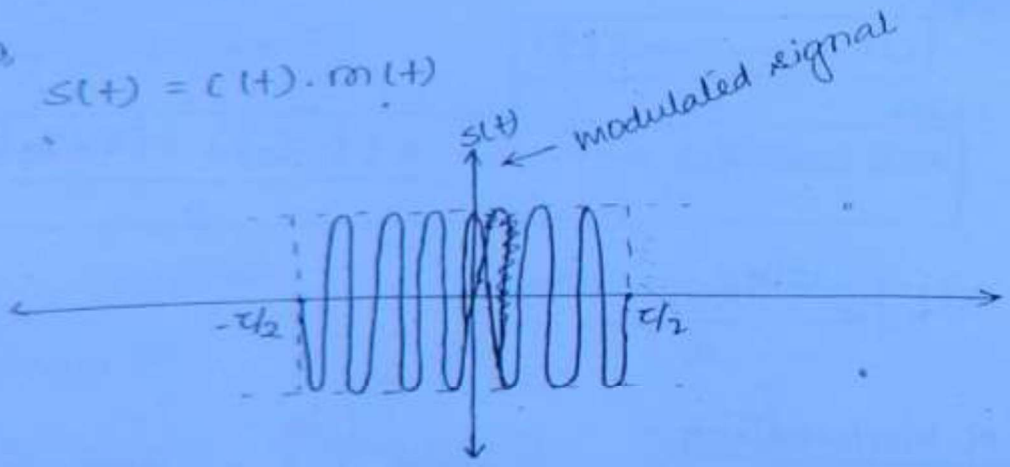
So, let

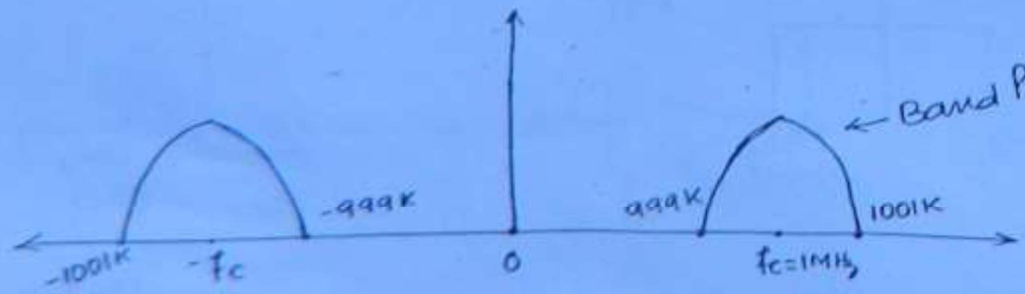
Carrier Signal; $c(t) = A_c \cos 2\pi f_c t$; $f_c = 1MHz = 1000KHz$.



So

$s(t) = c(t) \cdot m(t)$





→ If a signal contain only significant high frequencies then such signal are called BAND PASS SIGNAL.

Note:

A Base Band Signal can't be transmitted faithfully as it requires huge Antenna, but a Band Pass signal can be transmitted faithfully

(16)

* By MODULATION;-

i) the signal is translated from low frequency Region to high frequency Region.

ii) the Base Band Signal, is converted as Band Pass signal.

iii) A wide Band Signal, becomes narrow Band signal.

Note:

If; $\frac{\text{Highest Frequency (+ve)}}{\text{lowest Frequency (+ve)}} >>> 1$ ← wide Band signal

$\frac{\text{Highest +ve frequency}}{\text{lowest +ve frequency}} \approx 1$ ← Narrow Band signal

* Concept of Demodulation:-

→ Process of Recieving back the message signal, from the modulated signal.

→ Demodulation will be done at the Reciever.

* Classification of modulation:-

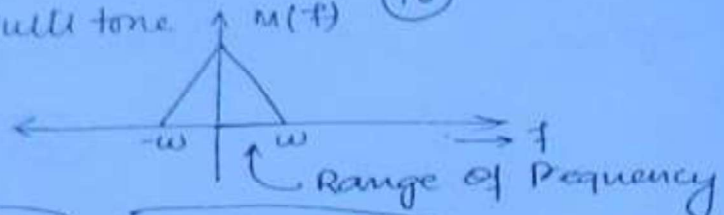
- Single tone modulation (Single freq^{n} msg signal modulation)
- Multitone modulation (Multi freq^{n} msg signal modulation)

eg.

1. $m(t) = A_m \cos 2\pi f_m t$ ← Single tone.

2. $m(t) \longleftrightarrow M(f)$ ← Multi tone

(Single message containing Range of freqⁿ)



or

$m(t) \longleftrightarrow A_{m1}\cos 2\pi f_{m1}t + A_{m2}\cos 2\pi f_{m2}t$

(group of messages having diff. freqⁿ)

Note :-

Modulation is basically of two types :

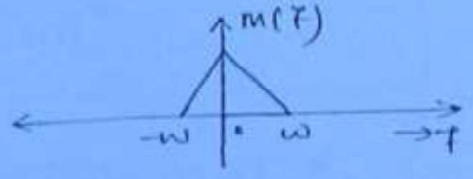
i) If the message signal having single freqⁿ then the corresponding frequency is called as **SINGLE TONE MODULATION**

eg $m(t) = A_m \cos 2\pi f_m t$

ii) If the message signal having multiple freqⁿ then the corresponding modulation is called as **MULTI TONE MODULATION**

eg

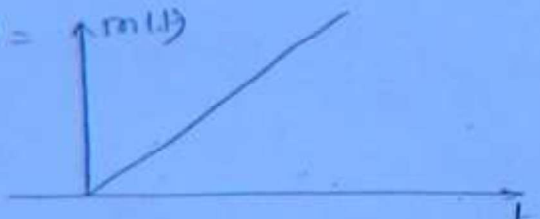
$m(t) \longleftrightarrow M(f)$

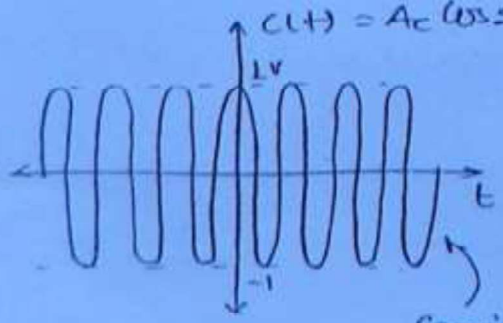


or

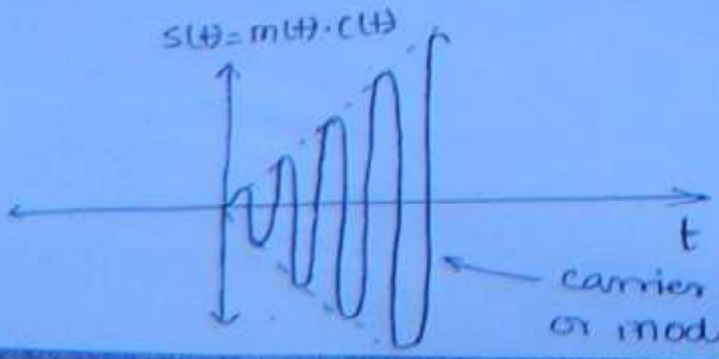
$m(t) = A_{m_1}\cos 2\pi f_{m_1}t + A_{m_2}\cos 2\pi f_{m_2}t$

eg



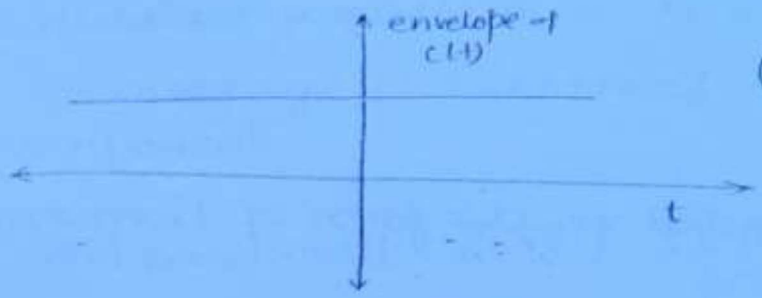


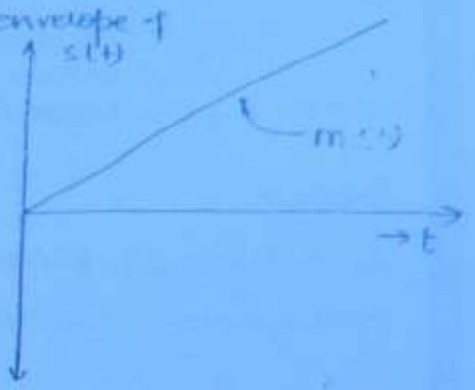
Carrier before modulation



Carrier after modulation or modulated signal.

Envelope: A line which touches all the +ve peak of the signal is called as ENVELOPE





Note :-

The above corresponds to Amplitude modulation where the message signal is stored in the form of Amplitude variations of carrier after modulation or in the form of envelope of the modulated signal

* MODULATION:

It is the process in which one of the Parameter (Amplitude, frequency or phase) of the carrier signal will be varied linearly in accordance with message signal amplitude variations

* AMPLITUDE MODULATION :-

Defn: It is the process in which Amplitude of the carrier signal will be changed (varied) linearly in accordance with message signal Amplitude variations

Assume,

$m(t)$ = msg signal

$A_c \cos 2\pi f_c t$ = carrier signal $c(t)$.

* General exp of AM signal:

$$\boxed{S_{AM}(t) = A_c\{1 + K_a m(t)\}\cos 2\pi f_c t}$$

K_a = Amplitude sensitivity of AM modulator.

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So,

$S_{AM}(t) = A_c \cos 2\pi f_c t + A_c K_a m(t) \cos 2\pi f_c t$ (18)

($A_c \cos 2\pi f_c t$ — carrier signal; $A_c K_a m(t) \cos 2\pi f_c t$ — modulated signal)

Disad:

Additional power is wasted in the form of transmitting of carrier signal.

Advantage:

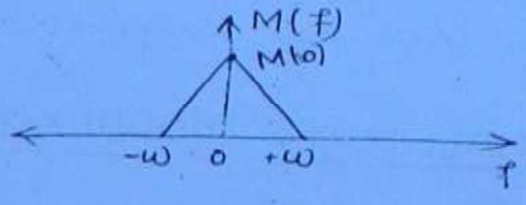
Due to the additional carrier signal, the demodulation of the AM signal becomes easier and cheaper.

Note:-

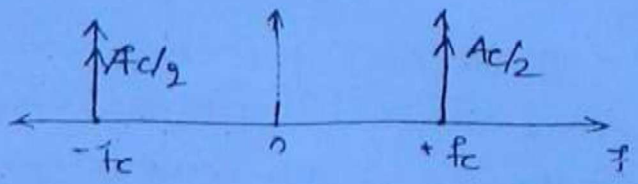
* AM signal consists of Additional carrier along with the Actual modulated signal

Because of this additional carrier transmitted power will be wasted but demodulation becomes simple.

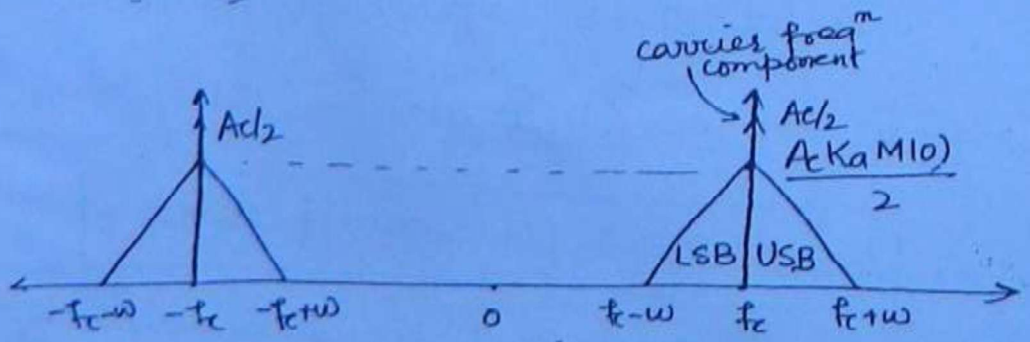
$m(t) \longleftrightarrow M(f)$



$c(t) = A_c \cos 2\pi f_c t \longleftrightarrow$



$S_{AM}(t) \longleftrightarrow$



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Note:

The original message signal is contained in the modulated signal ie in the USB & LSB

2. NO message is contained in the frequency (carrier) component

3. AM Bandwidth = $(f_c+w) - (f_c-w)$

$= 2w$

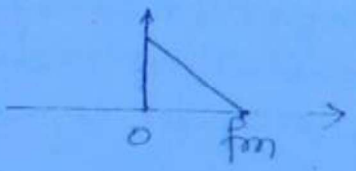
So, **AM Bandwidth = 2 x msg signal B.W**

* Importance of -ve frequency:

let only +ve frequency be considered

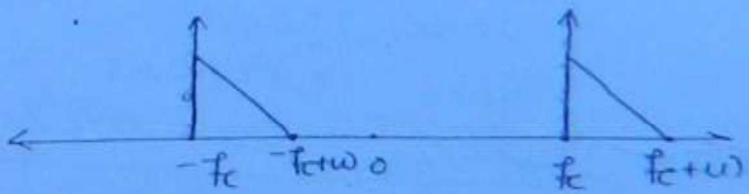
Hence,

$m(t) \longleftrightarrow M(f)$



So,

$S_{AM}(t) \longleftrightarrow$



B.W = w

Originally B.W of AM = 2w

Hence needed the -ve frequency in the discussion.

* AM spectrum consists of:

1) Carrier frequency component existing at fc.
2) USB existing above fc.
3) LSB existing below fc.

Note:

The Actual msg signal will be Retained by only Side Bands

* The B.W of each of the sideband is equal to w ie equal to the msg B.W

* SINGLE TONE AM:

let,

$$m(t) = A_m \cos 2\pi f_m t$$

&

$$S_{AM}(t) = A_c\{1 + K_a m(t)\}\cos 2\pi f_c t$$

Substituting we get:-

$$S_{AM}(t) = A_c\{1 + K_a \cdot A_m \cos 2\pi f_m t\}\cos 2\pi f_c t$$

where,

$$\mu = K_a \cdot A_m = \text{modulation index of AM.}$$

$$\mu \times 100\% = \%\text{ of modulation or depth of modulation}$$

→ The physical significance of depth of modulation is th the content of message signal that is stored in th carrier signal is called as 'depth of modulation'.

- $\mu < 1$ → under modulation ✓
- $\mu = 1$ → critical modulation ✓
- $\mu > 1$ → over modulation

} Generally used.

demodulation of AM signal becomes difficult.

* To what extent the carrier signal is modulated by the msg signal is specified by MODULATION INDEX.

* over modulation is not preferred because demodulati becomes complex.

*** SO,

$$S_{AM}(t) = A_c\{1 + \mu \cos 2\pi f_m t\}\cos 2\pi f_c t \quad \text{---} (1)$$

on expanding we get:-

$$S_{AM}(t) = A_c \cos 2\pi f_c t + A_c \mu \cos 2\pi f_m t \cos 2\pi f_c t \quad \text{---} (2)$$

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$$S_{AM}(t) = A_c \cos 2\pi f_c t + \frac{A_c \cdot \mu}{2} \cos 2\pi (f_c + f_m)t + \frac{A_c \mu}{2} \cos 2\pi (f_c - f_m)t$$

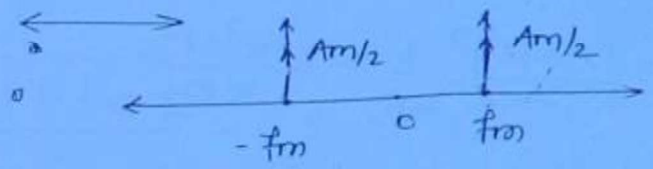
* Frequency Analysis of Single tone AM:

Ac,

$$S_{AM}(t) = A_c \cos 2\pi f_c t + \frac{A_c \cdot \mu}{2} \cos 2\pi (f_c + f_m)t + \frac{A_c \cdot \mu}{2} \cos 2\pi (f_c - f_m)t$$

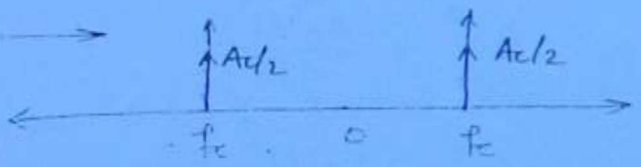
- $A_c \cos 2\pi f_c t$: carrier component
- $\frac{A_c \cdot \mu}{2} \cos 2\pi (f_c + f_m)t$: USB
- $\frac{A_c \cdot \mu}{2} \cos 2\pi (f_c - f_m)t$: LSB.

$m(t) = A_m \cos 2\pi f_m t$ $\longleftrightarrow$

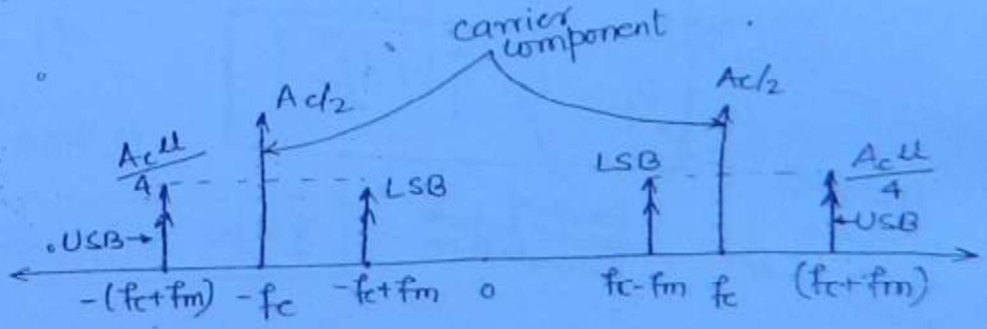


{only single +ve frequency (fm). Hence B.W = fm}

$c(t) = A_c \cos 2\pi f_c t$ $\longleftrightarrow$



$S_{AM}(t)$ $\longleftrightarrow$



So, $\boxed{\text{AM Bandwidth} = (f_c + f_m) - (f_c - f_m) = 2f_m}$

So, f_m = frequency of msg signal.

So, $\boxed{\text{AM Bandwidth} = 2 \times \text{frequency of msg signal}}$

* POWER OF AM SIGNAL :-

The total power of AM signal is given as:

$$P_t = P_c + P_{USB} + P_{LSB}$$

Now,

$$P_c = \frac{\left(\frac{A_c}{\sqrt{2}}\right)^2}{R} = \frac{A_c^2}{2R}$$

$$P_{USB} = \left(\frac{A_c \mu}{2}\right)^2 \cdot \frac{1}{2R} = \frac{A_c^2 \mu^2}{8R}$$

$$P_{LSB} = \left(\frac{A_c \mu}{2}\right)^2 \cdot \frac{1}{2R} = \frac{A_c^2 \mu^2}{8R}$$

$x(t) = V_m$; $x(t) = V_m \sin \omega t$

$P_{dc} = \frac{V_m^2}{R}$; $P_{ac} = \frac{V_{rms}^2}{R}$

$V_{rms} = V_m/\sqrt{2}$,

$P_{ac} = \frac{V_m^2}{2R}$

So,

$$P_t = \frac{A_c^2}{2R} + \frac{A_c^2 \mu^2}{4R}$$

$$P_t = \frac{A_c^2}{2R}\left\{1 + \frac{\mu^2}{2}\right\}$$

So, ***

$$\boxed{P_t = P_c\left\{1 + \frac{\mu^2}{2}\right\}}$$

P_t : Power of carrier after modulation

P_c : Power of carrier before modulation.

Now,

$$P_t = P_c + \frac{P_c \mu^2}{2} = P_c + P_{SB}$$

where, $\boxed{P_{SB} = \frac{P_c \mu^2}{2}}$; $\boxed{P_{USB} = P_{LSB} = \frac{P_c \mu^2}{4}}$

* The Power of carrier is independent of μ.

** The Side Band power depends on μ, and as the μ is increased, the P_{SB} also increases (Power of the modulated signal & total power of AM signal increases).

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Case 1:

$\mu = 0$; no modulation

Hence, $\boxed{P_t = P_c}$

as, $S_{AM}(t) = A_c \cos 2\pi f_c t$

Case 2:

$\mu = 1$; total modulation
(100% modulation)

Hence,

$$P_t = \frac{3}{2} P_c$$

$$\boxed{P_t = 1.5 P_c}$$

And, $S_{AM}(t) = A_c \cos 2\pi f_c t + A_c \cos 2\pi f_m t * \cos 2\pi f_c t$

Note:

As μ increases from 0 to 1, total AM power is increased by 50%.

Now,

$$P_c = \frac{2}{3} P_t$$

$$P_c = 0.666 P_t$$

So, $\boxed{P_c = 66.66\% \text{ of } P_t}$

Now, as,

$$P_t = P_c + P_{SB}$$

$$P_t = \frac{2}{3} P_t + P_{SB}$$

$$P_{SB} = \frac{1}{3} P_t$$

$$\boxed{P_{SB} = 33.33\% \text{ of } P_t}$$

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CONCLUSION:

1. If $\mu = 0 \Rightarrow P_c = 100\%$ of $P_t \Rightarrow P_{SB} = 0\%$ of P_t (24)
2. If $\mu = 1 \Rightarrow P_c = 66.66$ of $P_t \Rightarrow P_{SB} = 33.33\%$ of P_t.

Note:

In the efficient power distribution case, ie $\boxed{\mu = 1}$ still 66.66 % of P_t; is wasted in the form of transmission of additional carrier. This is the biggest drawback of AM.

* Modulation efficiency (η):

* It specifies share of sideband power in total power

Eg $\eta = 0.1 \longrightarrow 10\%$ of P_{SB} in P_t

$\eta = 0.3 \longrightarrow 30\%$ of P_{SB} in P_t

Mathematically

$$\boxed{\eta = \frac{P_{SB}}{P_t}}$$

So,

$$\eta = \frac{P_c \frac{\mu^2}{2}}{P_c\left\{1 + \frac{\mu^2}{2}\right\}}$$

***

$$\boxed{\eta = \frac{\mu^2}{2 + \mu^2}}$$

Case 1:

$\mu = 0$

for $\mu = 0$; $\eta = 0 \Rightarrow P_{SB} = 0\%$ of P_t

P_c = 100 % of P_t

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$\eta = 0.11 \Rightarrow P_{SB} = 11\%$ of P_t

$P_c = 89\%$ of P_t

Case 3: ($\mu = 0.707$)

$\eta = 0.2 \Rightarrow P_{SB} = 20\%$ of P_t

$P_c = 80\%$ of P_t

Case 4: ($\mu = 1$)

$\eta = 0.33 \Rightarrow P_{SB} = 33.3\%$ of P_t

$P_c = 66.7\%$ of P_t

Note:

let

$P_t = P_c + P_{SB}$

$50W = 30W + 20W$ (60% 40%)

$60W = 30W + 30W$ (50% 50%)

$\mu \uparrow$

* P_c is independent of μ and share of carrier power in total power decreases as μ increases.

* An unmodulated AM transmitted power is given 500 w. Find AM transmitted power with 100% modulation?

Soln: Given $P_t = 500W$, $\mu = 0 \Rightarrow P_c = 500W$ & $P_{SB} = 0$

$\mu = 1$; $P_t = P_c + P_{SB}$

$= 500 +$

$P_t = \frac{3}{2} P_c$

$P_c = \frac{2}{3} \times 500$

$= 750$ watt.

$P_t = P_c \left\{1 + \frac{\mu^2}{2}\right\}$

$\boxed{P_t = 500 \left\{1 + \frac{1}{2}\right\} = 750W}$

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Note:

Q. For an AM signal total sideBand power is given by 100W with 50% of modulation. Find total AM transmitted power.

(26)

Soln:

$P_{SB} = 100W$; $\mu = 1/2$.

$P_t = ?$

$$P_{SB} = \frac{P_c \mu^2}{2}$$

$$P_c = \frac{P_{SB} \cdot 2}{\mu^2} = 100 \times 2 \times 4$$

$P_c = 800$.

So, $P_t = P_c + P_{SB}$

$$\boxed{P_t = 900W}$$

Q2. For an AM each of the S.B power is given by 2KW and carrier power is given by 8KW. Find % of modulation.

Soln:

$P_c = 8KW$

$P_{SB} = 4KW$ $\{P_{SB} = P_{USB} + P_{LSB} = 2+2\}$.

So,

$$P_{SB} = \frac{P_c \mu^2}{2}$$

$$\mu^2 = \frac{P_{SB} \times 2}{P_c}$$

$$\mu^2 = \frac{4 \times 2}{8}$$

$$\boxed{\mu = 1}$$ Ans

100% modulation

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Q3. A carrier of 10 Ws of 1 MHz is amplitude modulated by a message signal of $4\cos 4\pi \times 10^3 t$ with 50% of modulation. Antenna Resistance is given by 5Ω.

1. Find all the parameters of AM
2. Plot AM spectrum and Identify the spectral components.

(2)

Soln:- Given:

$$c(t) = 10\cos 2\pi \times 10^6 t = A_c \cos 2\pi f_c t.$$

$$m(t) = 4\cos 4\pi \times 10^3 t = A_m \cos 2\pi f_m t$$

So,

$A_m = 4\,V$; $f_m = 2\times 10^3 = 2\,KHz$

$A_c = 10\,V$; $f_c = 10^6 = 1\,MHz$

So, for single tone modulation

$$B.W = 2f_m$$

$$B.W = 2\times 2K$$

$$\boxed{B.W = 4KHz}$$ Ans.

And, $\mu = 0.5$; $R = 5\Omega$

So, $$\boxed{P_c = \frac{A_c^2}{2R} = \frac{10^2}{2\times 5} = 10\,W}$$ Ans.

$$P_t = P_c\left\{1+\frac{\mu^2}{2}\right\} = 10\left\{1+\frac{0.5^2}{2}\right\}$$

$$\boxed{P_t = 11.25\,W}$$ Ans.

So, $$\boxed{P_{SB} = P_t - P_c = 1.25\,W}$$ Ans.

$$\boxed{P_{USB} = P_{LSB} = P_{SB}/2 = 0.625\,W}$$ Ans.

And, $$\boxed{\eta = \frac{P_{SB}}{P_t} = \frac{\mu^2}{2+\mu^2} = 0.11 = 11\%}$$ Ans.

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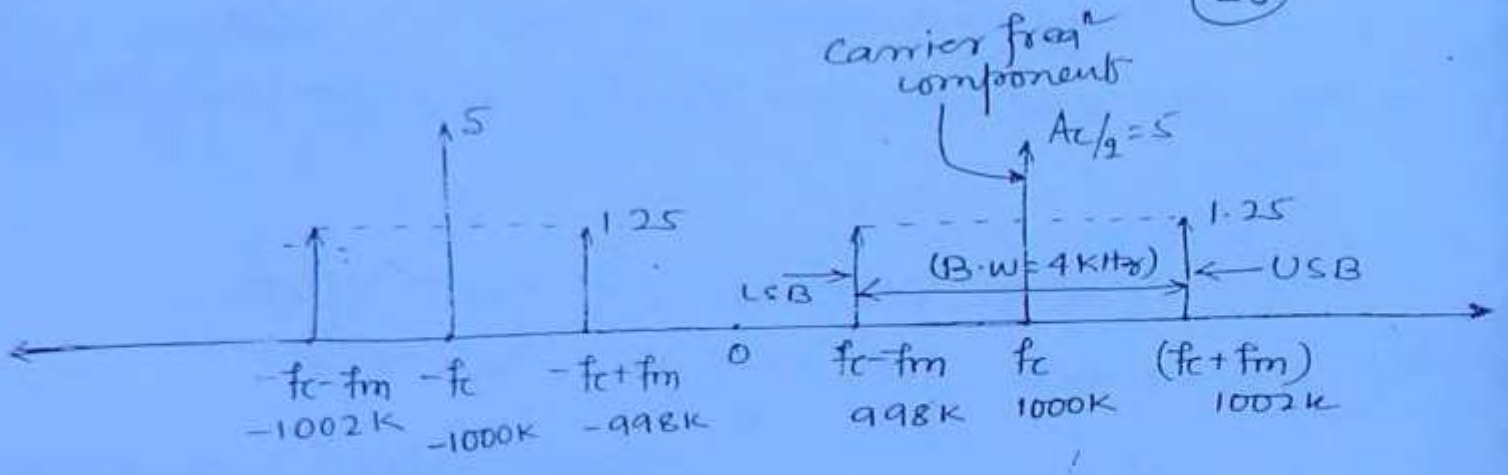

Q4. A carrier of $10\cos 8\pi \times 10^5 t$ is amplitude modulated by a msg signal of $6\cos \pi \times 10^4 t$. i) Find all the parameters of AM.
ii) Plot spectral components & spectrum.

Soln: Given:

$$c(t) = 10\cos 8\pi \times 10^5 t = A_c \cos 2\pi f_c t$$

$$A_c = 10 \; ; \; f_c = 4 \times 10^5 = 400\text{KHz}.$$

$$m(t) = 6\cos \pi \times 10^4 t = A_m \cos 2\pi f_m t$$

$$A_m = 6 \; ; \; f_m = 5\text{KHz}.$$

Now, As μ is not given. So, take :-

$$\boxed{\mu = \frac{A_m}{A_c} = \frac{6}{10} = 0.6}$$

Now, $P_c = \frac{A_c^2}{2R} = \frac{10^2}{2\times 1}$ { As Antenna Resistance is not given. Hence $R = 1\Omega$ }

$$\boxed{P_c = 50\text{W}.}$$

$$P_t = P_c\left\{1 + \frac{\mu^2}{2}\right\}$$

$$P_t = 50\left\{1 + \frac{0.6^2}{2}\right\}$$

$$\boxed{P_t = 59\text{W}}$$

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AM Bandwidth = α for

AM B.W = 2 × 5K

(29)

$\boxed{\text{AM B.W} = 10\text{K}}$ Ans

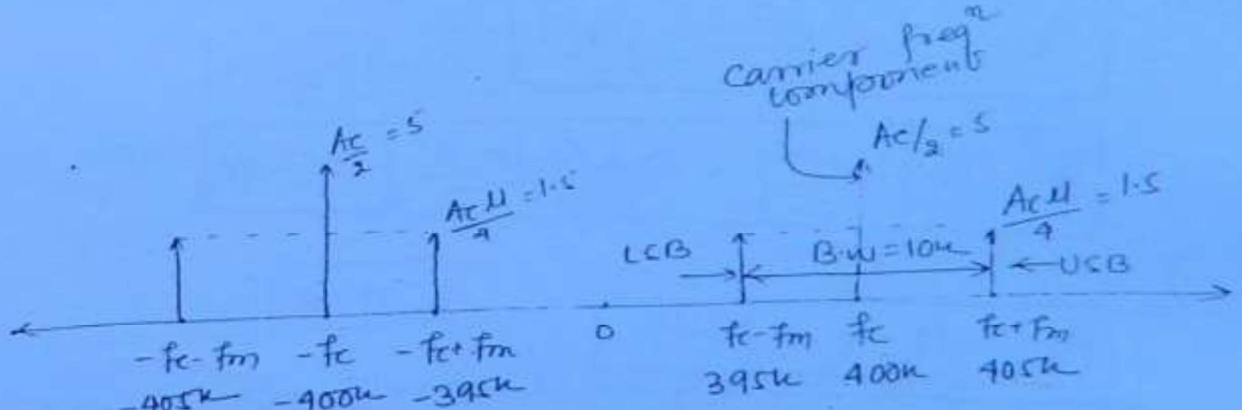
Now, $\boxed{P_{SB} = P_t - P_c = 9\text{W}}$ Ans

$\boxed{P_{USB} = P_{LSB} = 4.5\text{W}}$ Ans

And, $m = \frac{\mu^2}{2+\mu^2} = \frac{0.6^2}{2+0.6^2}$

$\boxed{m = 0.15}$ Ans

spectrum :-



Qs. An AM signal is given by

$s(t) = 4\cos 3200\pi t + 10\cos 4000\pi t + 4\cos 4800\pi t$

Find all the parameters of AM and plot the spectrum.

Soln:- As,

$$s_{AM}(t) = A_c \cos 2\pi f_c t + \frac{A_c\mu}{2}\cos 2\pi(f_c+f_m)t + \frac{A_c\mu}{2}\cos 2\pi(f_c-f_m)t$$

Comparing we get:-

$s_{AM}(t) = 10\cos 4000\pi t + 4\cos 4800\pi t + 4\cos 3200\pi t$

{As the value of magnitude remains same for USB & LSB}

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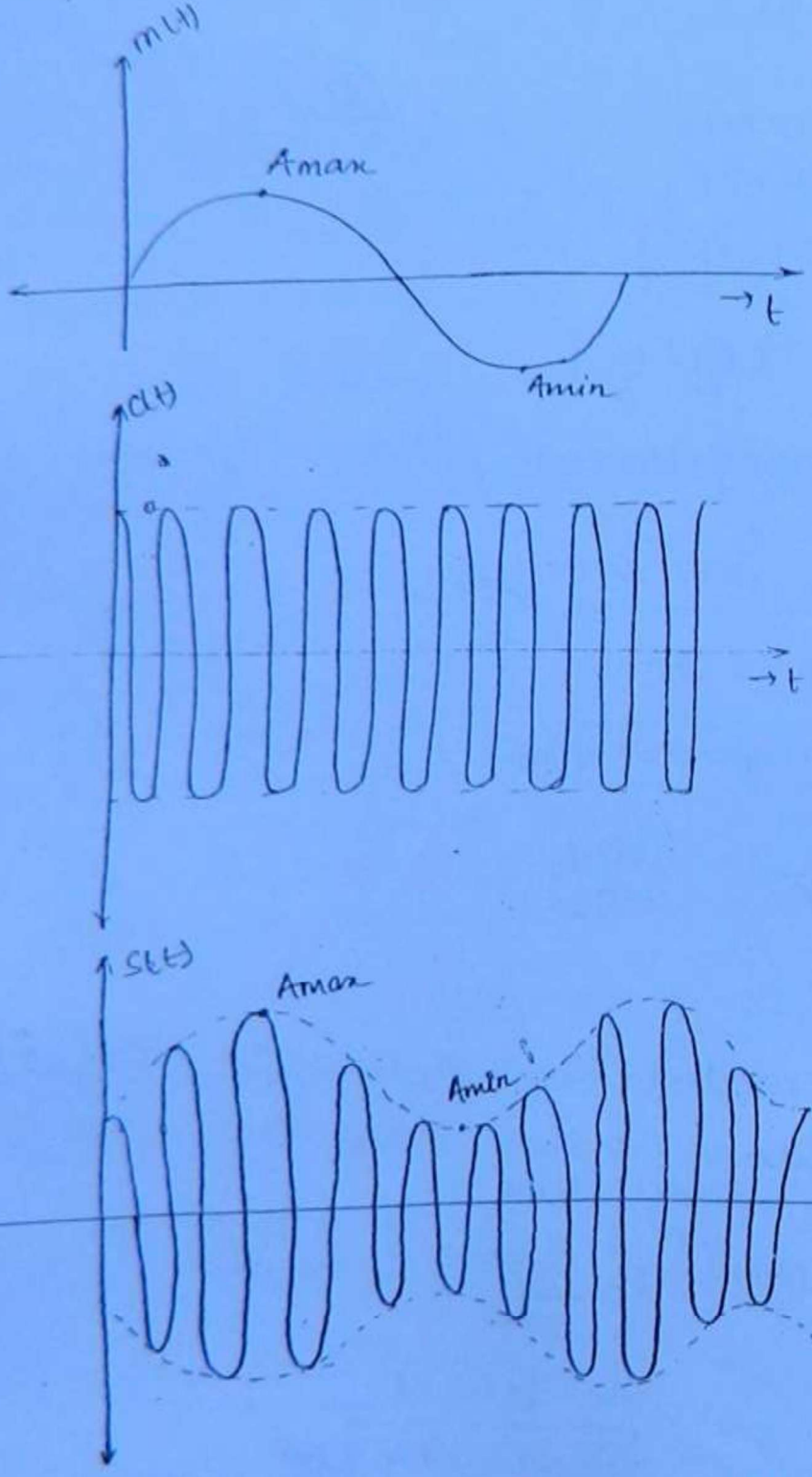
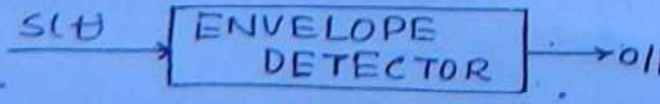
→ The peak amplitude of the carrier after modulation is not const but it varies in accordance to the msg signal.

Now,



Note :- If the above signal is fed to the input of the envelope detector for its demodulation as in the fig. shown below:-



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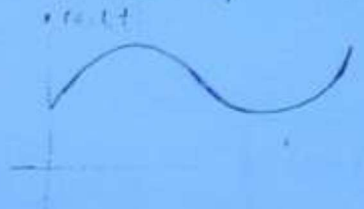
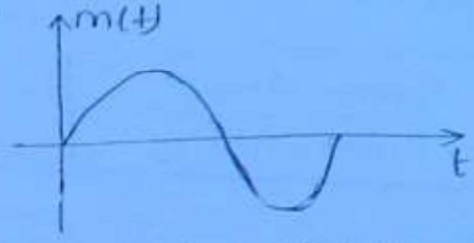
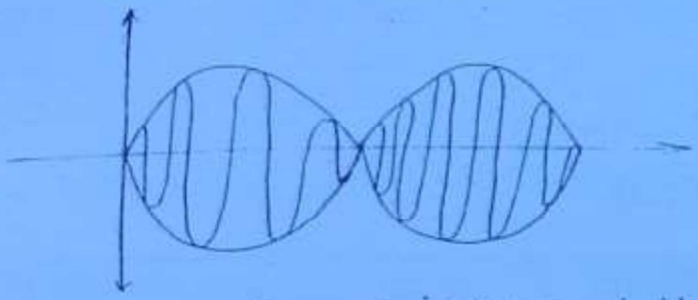
The O/P of the envelope detector is as shown:



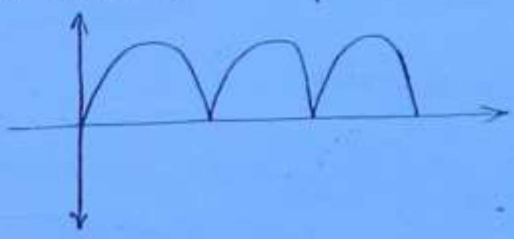
The above signal when shifted by some D.C value, the waveform obtained resembles to that of the original signal. Its waveform is as shown below:



Now, if the wave of the AM transmitted signal is as follows:



And if this above signal is passed through the ED the O/P obtained is as follows:-



← This signal doesn't correspond to the original message signal and by any means it cannot be converted to the original message signal. Hence the above waveform is not the AM transmitted wave.

Note:-

Max^m peak of AM signal, $A_{max} = A_c\{1+u\}$.

Min^m peak of AM signal; $A_{min} = A_c\{1-u\}$

∵ max^m value of the $\cos 2\pi f_m t = 1$

$-1 \leq \cos 2\pi f_m t \leq 1$

So, $A_{max} = A_c(1+u) = 12.64(1+0.707) = 21.5$ volts

$A_{min} = A_c(1-u) = 12.64(1-0.707) = 3.7$ volts

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Also,

$A_{max} + A_{min} = 2A_c$

So,

$$A_c = \frac{A_{max} + A_{min}}{2}$$ (34)

Substituting the value of A_c in above eqn of A_{max} & A_{min} we get:

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

7 The peak voltage of AM signal varies from 2 volts to 10 volts. Find μ, P_t & η?

Sol^n: Given:

$A_{max} = 10$ Volts

$A_{min} = 2$ Volts

So,

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}} = \frac{10-2}{10+2} = \frac{8}{12} = 0.66$$ Ans.

$$A_c = \frac{A_{max} + A_{min}}{2} = \frac{2+10}{2} = 6 \text{ volts}$$

$$P_c = \frac{A_c^2}{2R} = \frac{6^2}{2 \times 1} = 18 \text{ W}$$

$$P_t = P_c \left\{1 + \frac{\mu^2}{2}\right\}$$

$$P_t = 18 \left\{1 + \frac{0.66^2}{2}\right\} = 22 \text{ W}$$ Ans

$$\eta = \frac{\mu^2}{2+\mu^2} = \frac{0.66^2}{2+0.66^2}$$

$$\eta = 18.1$$ Ans

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× MULTI-TONE AMP. MODULATION.

Assume,

$$m(t) = A_{m1}\cos 2\pi f_{m1}t + A_{m2}\cos 2\pi f_{m2}t$$

and,

$$S_{AM}(t) = A_c\{1 + K_a m(t)\}\cos 2\pi f_c t \quad (3)$$

So,

$$S_{AM}(t) = A_c\{1 + K_a A_{m1}\cos 2\pi f_{m1}t + K_a A_{m2}\cos 2\pi f_{m2}t\}\cos 2\pi f_c t$$

Let:

$K_a A_{m1} = \mu_1$

$K_a A_{m2} = \mu_2$

So,

$$\boxed{S_{AM}(t) = A_c\{1 + \mu_1\cos 2\pi f_{m1}t + \mu_2\cos 2\pi f_{m2}t\}\cos 2\pi f_c t}$$

on expanding we get:

$$S_{AM}(t) = A_c\cos 2\pi f_c t + A_c\mu_1\cos 2\pi f_{m1}\cos 2\pi f_c t + A_c\mu_2\cos 2\pi f_{m2}t\cos 2\pi f_c$$

$$S_{AM}(t) = \underbrace{A_c\cos 2\pi f_c t}_{\text{carrier frequency component}} + \underbrace{\frac{A_c\mu_1}{2}\cos 2\pi (f_c + f_{m1})t}_{USB1} + \underbrace{\frac{A_c\mu_1}{2}\cos 2\pi (f_c - f_{m1})t}_{LSB1} + \underbrace{\frac{A_c\mu_2}{2}\cos 2\pi (f_c + f_{m2})t}_{USB2} + \underbrace{\frac{A_c\mu_2}{2}\cos 2\pi (f_c - f_{m2})t}_{LSB2}$$

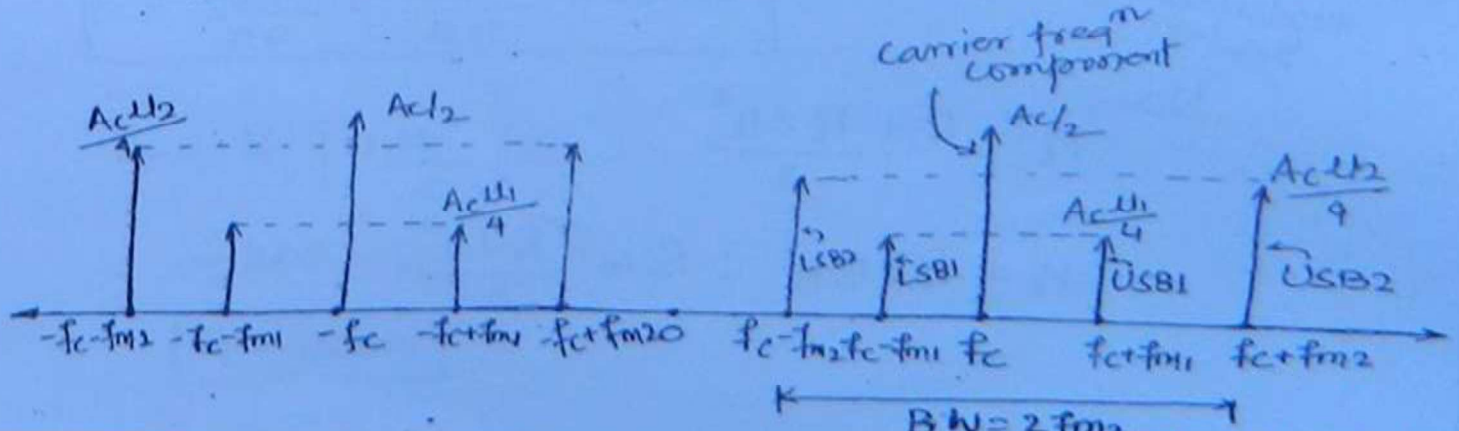
Spectrum: For plotting spectrum let us assume

$f_{m2} > f_{m1}$

&

$\mu_2 > \mu_1$

So,



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* TOTAL POWER OF AM (Multitone):

The total power (P_t) is given as:

$$P_t = P_c + P_{SB}$$

$$P_t = P_c + P_{USB}(\text{total}) + P_{LSB}(\text{total})$$

$$\boxed{P_t = P_c + P_{USB1} + P_{USB2} + P_{LSB1} + P_{LSB2}}$$

Now,

$$P_c = \frac{A_c^2}{2R} \; ; \; P_{USB1} = \frac{(A_c m_1)^2}{2R} = \frac{A_c^2 m_1^2}{8R} = P_{LSB1}$$

$$P_{USB2} = \frac{\left(\frac{A_c m_2}{2}\right)^2}{2R} = \frac{A_c^2 m_2^2}{8R} = P_{LSB2}$$

So,

$$P_t = \frac{A_c^2}{2R} + \frac{A_c^2 m_1^2}{4R} + \frac{A_c^2 m_2^2}{4R}$$

$$P_t = \frac{A_c^2}{2R}\left\{1 + \frac{m_1^2 + m_2^2}{2}\right\}$$

So,***

$$\boxed{P_t = P_c\left\{1 + \frac{m_t^2}{2}\right\}}$$

where,

$$m_t = \sqrt{m_1^2 + m_2^2}$$

m_t → Total modulation index.

Now,

$$P_t = P_c + \frac{P_c m_t^2}{2}$$

$$P_t = P_c + P_{SB} \; ; \; P_{SB} = \frac{P_c m_t^2}{2}$$

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Now,

$$\eta = \frac{P_c \cdot B}{P_t}$$

$$\eta = \frac{P_c \mu_t^2}{\frac{2}{P_c\left\{1+\frac{\mu_t^2}{2}\right\}}}$$

(37)

So, ***

$$\boxed{\eta = \frac{\mu_t^2}{2+\mu_t^2}}$$

Q1. A Carrier of $20\cos 4\pi \times 10^6 t$ is amp. modulated by a msg signal of having frequencies 10KHz, 20KHz with modulation indexes 0.6 & 0.8 respt. Find all the parameters and plot the spectrum?

Soln: Given:

$$A_c \cos 2\pi f_c t = 20\cos 4\pi \times 10^6 t$$

So, $A_c = 20$, $f_c = 2000$KHz

$f_{m1} = 10$KHz ; $f_{m2} = 20$KHz

$\mu_1 = 0.6$; $\mu_2 = 0.8$

So, $\mu_t = \sqrt{\mu_1^2+\mu_2^2} = \sqrt{0.36+0.64} = 1$

Now,

$$B.W = 2f_{max}$$

$$= 2 \times 20K$$

$$\boxed{BW = 40\text{KHz}}$$

And,

$$\boxed{P_c = \frac{A_c^2}{2R} = \frac{400}{2\times 1} = 200\text{W}}$$

$$P_t = P_c\left\{1+\frac{\mu_t^2}{2}\right\}$$

$$= 200 \times \frac{3}{2}$$

$$\boxed{P_t = 300\text{W}}$$

$P_{SB} = P_1 - P_c$

$= 100\ W$

$P_{USB} = P_{LSB} = 50W$

(38)

$P_{USB1} = \frac{A_c^2 \mu_1^2}{8R} = 18W = P_{LSB1}$

$P_{USB2} = 32W = P_{LSB2}$

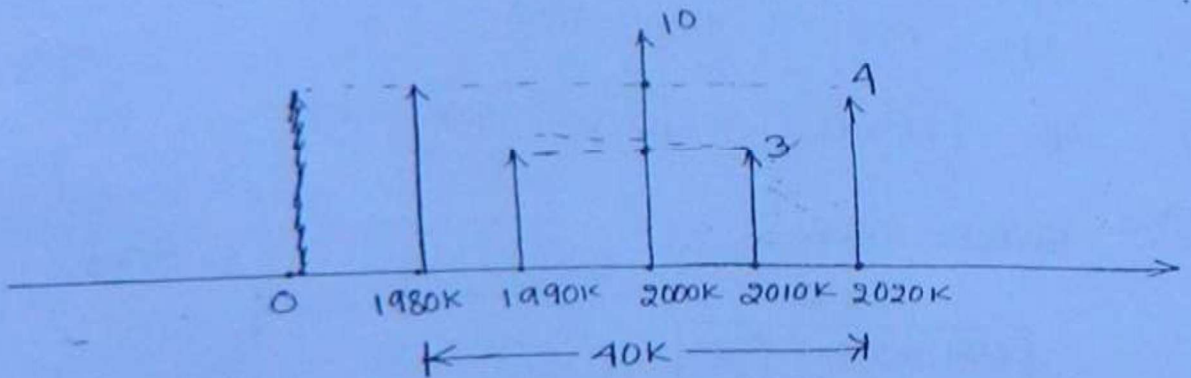
And

$\eta = \frac{\mu_t^2}{2+\mu_t^2}$

$\eta = \frac{1}{1+2}$

$\eta = 0.33 = 33.3\%$

Spectrum:



2. An AM signal is given by

$s(t) = \{20 + 4\cos 8\pi \times 10^4 t + 8\cos \pi \times 10^5 t\} \cos \omega_c t$

Find μ_t, P_t, B.W & η?

Sol:- As, $s(t) = 20\{1 + 4/20 \cos 8\pi \times 10^4 t + 8/20 \cos \pi \times 10^5 t\} \cos \omega_c t$.

$S_{AM}(t) = A_c\{1 + \mu_1 \cos 2\pi f_{m1} t + \mu_2 \cos 2\pi f_{m2} t\} \cos 2\pi f_c t$

$= A$

So, $A_c = 20V$, $\mu_1 = 4/20$, $\mu_2 = 8/20$.

$f_{m1} = 40K$; $f_{m2} = 50KHz$.

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So, $m_1 = 1/20 = 0.2$; $m_2 = 0.4$

$B.W = 2 f_{max}$

$\boxed{BW = 100K}$ Ans

(39)

$\boxed{m_t = \sqrt{0.2^2 + 0.4^2} = 0.44.}$ Ans

$\boxed{P_c = \frac{A_c^2}{2R} = \frac{400}{2 \times 1} = 200W}$ Ans

$P_t = P_c \left\{1 + \frac{m_t^2}{2}\right\}$

$= 200 \left\{1 + \frac{0.44^2}{2}\right\}$

$\boxed{P_t = 220W}$ Ans

And, $\eta = \frac{m_t^2}{2 + m_t^2} = \frac{0.44^2}{2 + 0.44^2}$

$\boxed{\eta = 0.09 = 9\%}$ Ans

CURRENT RELATIONS IN AM:-

As,

$P_t = P_c \left\{1 + \frac{m_t^2}{2}\right\}$.

$I_t^2 R = I_c^2 R \left\{1 + \frac{m_t^2}{2}\right\}$

PSU's ***

$\boxed{I_t^2 = I_c^2 \left\{1 + \frac{m^2}{2}\right\}}$

So, PSU's ***

$\boxed{I_t = I_c \sqrt{1 + \frac{m^2}{2}}}$

I_t = total Antenna Current
I_c = Carrier current.

→ I_c is independent of "m"

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VOLTAGE RELATIONS IN AM

As.

$$P_t = P_c\left\{1+\frac{\mu^2}{2}\right\}$$

(40)

$$\frac{V_t^2}{R} = \frac{V_c^2}{R}\left\{1+\frac{\mu^2}{2}\right\}$$

$$\boxed{V_t = V_c\sqrt{1+\frac{\mu^2}{2}}}$$

Q1. An unmodulated AM transmitter current is given by 5A. Find AM transmitter current with 50% of modulation.

Soln: Given:

$\mu = 0.5$

$I_c = 5A$

So,

$$I_t = I_c\sqrt{1+\frac{\mu^2}{2}}$$

{∵ $\mu = 0$, $I_t = I_c = 5A$ as modulation index changes, I_c doesn't changes}

$$I_t = 5\sqrt{1+\frac{0.5^2}{2}}$$

$$I_t = 5\sqrt{1+\frac{0.25}{2}}$$

$$\boxed{I_t = 5.32A}$$

Q2. An AM transmitter current is given by 10A with 40% of modulation. Find AM transmitter current with 80% of modulation.

Soln: Given, $I_t = 10A$

∵ as $\mu = 0.4$; $I_c = 9.6A$ $\left\{10 = I_c\sqrt{1+\frac{0.16}{2}},\ I_c = 9.6A\right\}$.

So, $$I_t = I_c\sqrt{1+\frac{\mu^2}{2}}$$

$$= 9.6\sqrt{1+\frac{0.8^2}{2}} = 9.6\sqrt{1+0.32}$$

$$\boxed{I_t = 11.05A}$$

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CURRENT RELATION FOR MULTI TONE AM

As,

$$P_t = P_c\left[1+\frac{\mu_t^2}{2}\right]$$

(41)

So,

$$\boxed{I_t = I_c\sqrt{1+\frac{\mu_t^2}{2}}} \quad ; \quad \boxed{\mu_t = \sqrt{\mu_1^2+\mu_2^2}}$$

Q1. An Unmodulated AM transmitter power is given by 10KW when the carrier is modulated by single sinusoidal message signal transmitter power becomes 13.5 KW. Find AM transmitter power if the carrier is simultaneously modulated by 2nd message signal with 60% of modulation.

a) 12 KW b) 15 KW c) 20 KW d) 22.5 KW

Sol^{n}: $\because \mu = 0$

So, $P_t = P_c = 10\text{KW}$

Now

$$P_t = P_c\left\{1+\frac{\mu_1^2}{2}\right\}.$$

$$\frac{13.5}{10} = 1+\frac{\mu_1^2}{2}$$

$$\frac{\mu_1^2}{2} = 0.35 \Rightarrow \mu_1^2 = 0.7$$

Now, $\mu_t = \sqrt{\mu_1^2+\mu_2^2} = \sqrt{0.7+0.36} \approx 1.$

So,

$$P_t = P_c\left(1+\frac{\mu_t^2}{2}\right\}.$$

$$P_t = 10\times\frac{3}{2}$$

$$\boxed{P_t = 15\text{KW}}$$ Ans

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OR An AM transmitter current is given by 10A with carrier is modulated by single sinusoidal message signal with 40% of modulation with the carrier is simultaneously modulated by 2nd message signal, AM transmitter current is increased to 10.5 A. Find % of modulation due to 2nd message signal? (42)

Soln:

$$I_t = I_c \sqrt{1 + \frac{\mu_1^2}{2}}$$

$$10 = I_c \sqrt{1 + 0.08}$$

$$I_c = \frac{10}{\sqrt{1+0.08}}$$

$$I_c = 9.624\ A$$

NOW,

$$10.5 = 9.624 \sqrt{1 + \frac{\mu_t^2}{2}}$$

$$\left(\frac{10.5}{9.624}\right)^2 = 1 + \frac{\mu_t^2}{2}$$

$$\frac{\mu_t^2}{2} = 0.1903$$

$$\mu_t = \sqrt{0.3806} = \sqrt{\mu_1^2 + \mu_2^2}$$

$$\mu_1^2 + \mu_2^2 = 0.3806.$$

$$\mu_2^2 = 0.3806 - 0.16 \Rightarrow \mu_2 = \sqrt{0.2206}$$

$$\boxed{\mu_2 = 0.47 = 47\%.}$$

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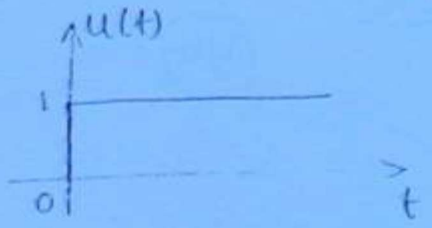
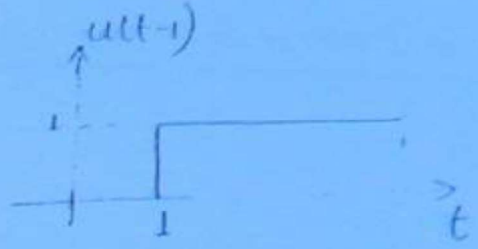
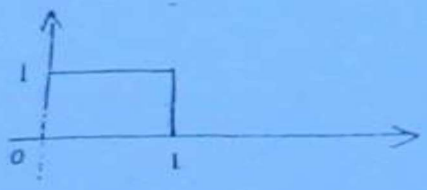
Note :-

$x(t) = u(t) - u(t-1)$

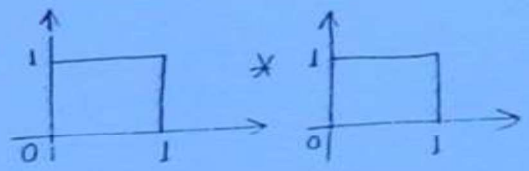


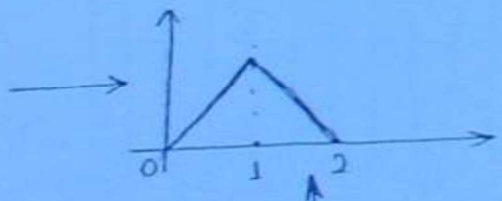




So,

$x(t) * x(t)$





Now,

$$\{u(t) - u(t-1)\} * \{u(t) - u(t-1)\}$$

$$= r(t) - r(t-1) - r(t-1) + r(t-2)$$

$$= r(t) - 2r(t-1) + r(t-2)$$

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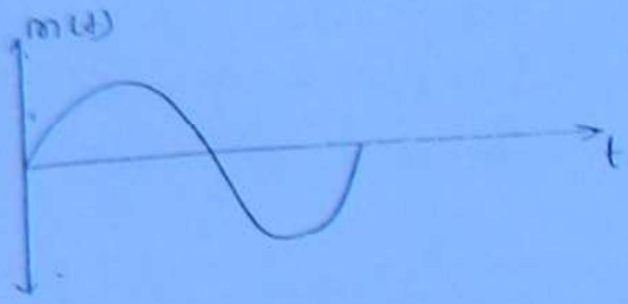
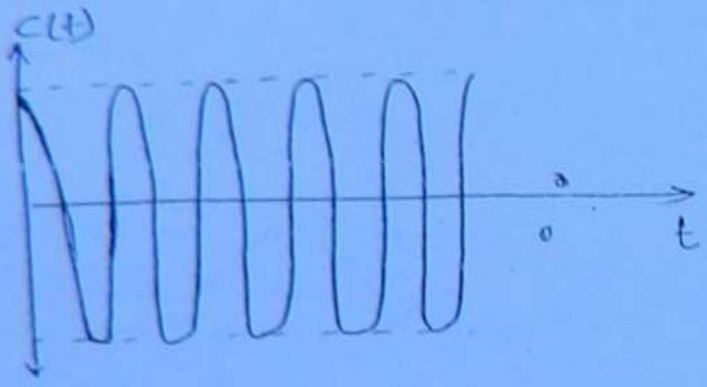
Under, Critical and over modulation:-

let,





i) Under modulation:-

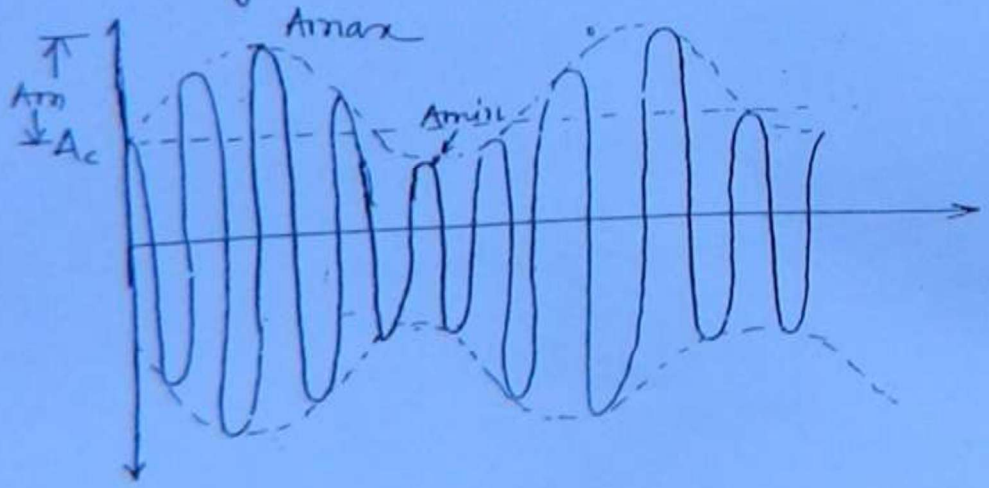
For under modulation

$$u < 1$$

$$u = \frac{Am}{Ac} < 1 \Rightarrow Am < Ac$$

$$Amin = Ac\{1-u\} = +ve$$

So, the waveform is:-



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ii) Critical modulation:

$u = 1$

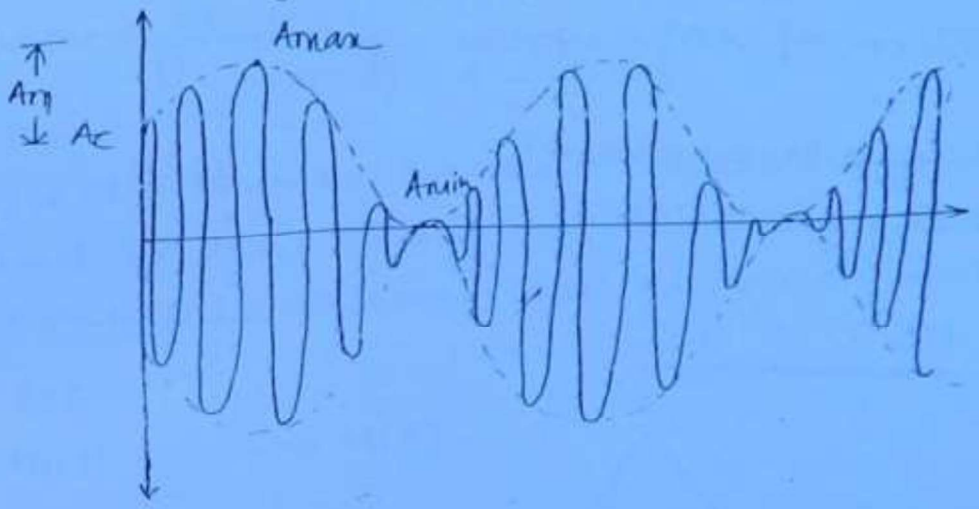
$\frac{Am}{Ac} = 1$

45

$Am = Ac$

So, $Amin = Ac\{1-u\} = 0$

So, the waveform is given as:-



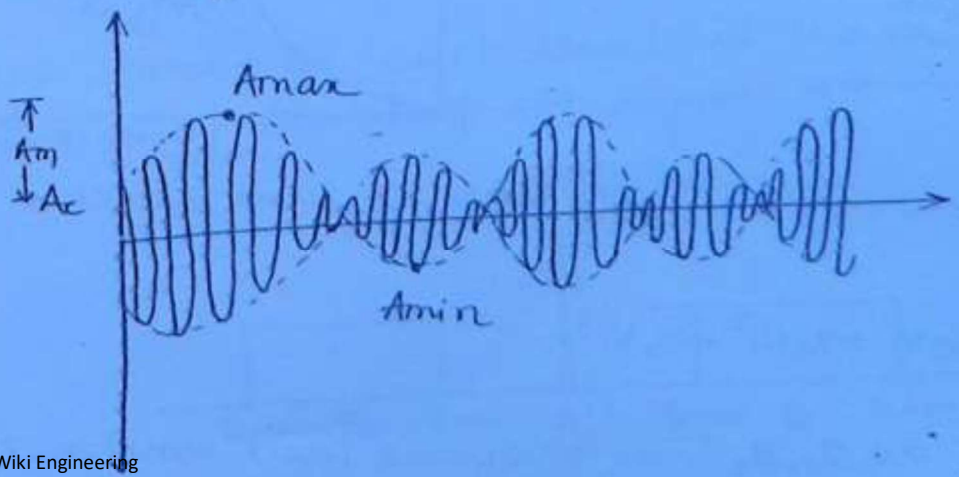
iii) Over modulation:-

$u > 1$

$\frac{Am}{Ac} > 1 \Rightarrow Am > Ac$

So, $Amin = Ac\{1-u\} = -ve$

So, the waveform is given as:-



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Note:

* In under and critical modulation, message signal is preserved in the form of +ve envelope, so demodulation becomes simple

* In over modulation, the message signal is not stored in the form of +ve envelope, so demodulation becomes complex.

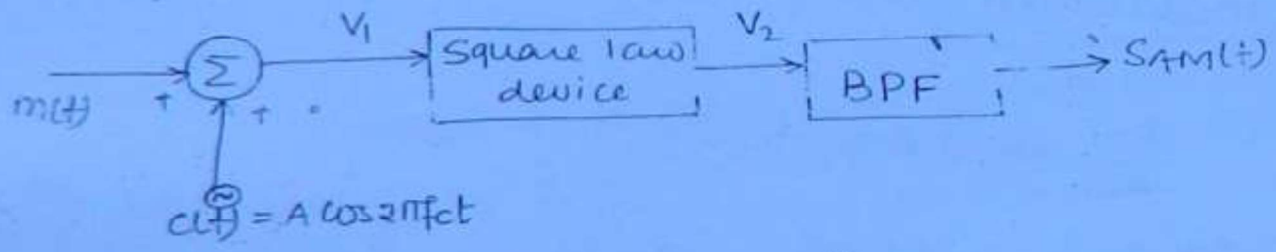
* Generation of AM Signal :-

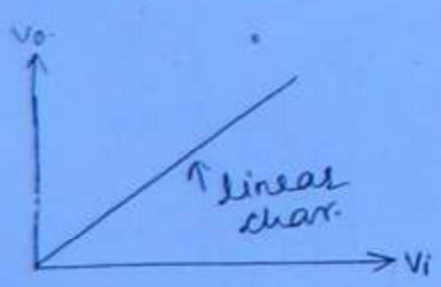
For the generation of AM signal, following modulators are used:

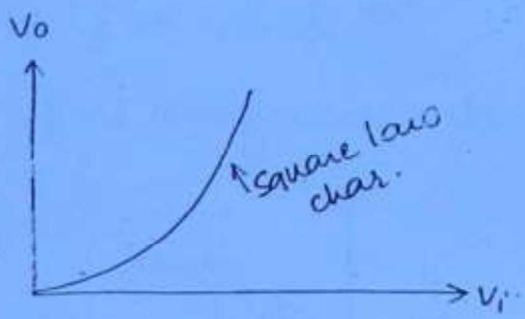
1) Square law modulator.
2) Switching modulator

* SQUARE-LAW MODULATOR:

Block diag^m:







* The Relation b/w Vo & Vi for square law device is given as:

$$V_o = a_0 V_i + a_1 V_i^2 + a_2 V_i^3 + \cdots$$

where, $a_0, a_1, a_2 \cdots \rightarrow$ Square law constants

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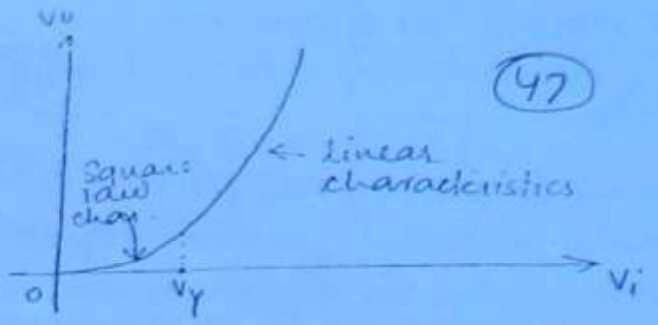
Note:



when the applied voltage $V_i < V_\gamma$ (diode), then the diode exhibits square law characteristics and if $V_i > V_\gamma$ diode) it then exhibits linear characteristics.

So,

Input of the diode (Square law device) is:-

$$V_I = m(t) + C(t)$$
$$= m(t) + A_c \cos 2\pi f_c t$$

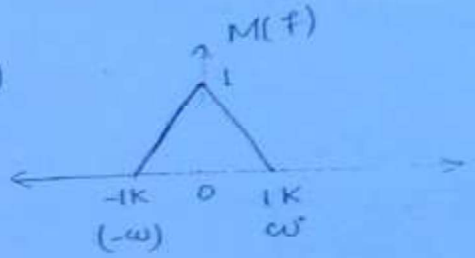
*$m(t)$ & $C(t)$ should be such that the peak voltage of V_I must be less than cutin voltage of diode, so that diode exhibits Square law characteristics

Now let

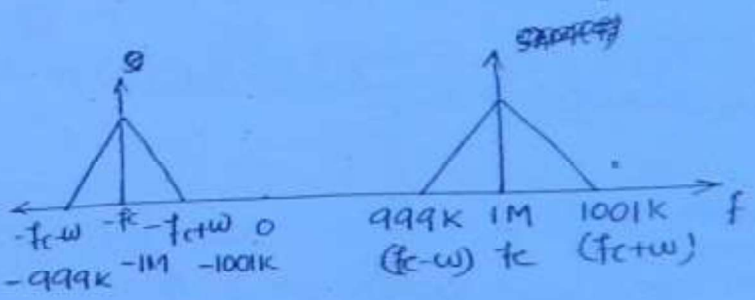
$m(t) \longleftrightarrow M(f)$

$f_c = 1MHz$

$= 1000K$

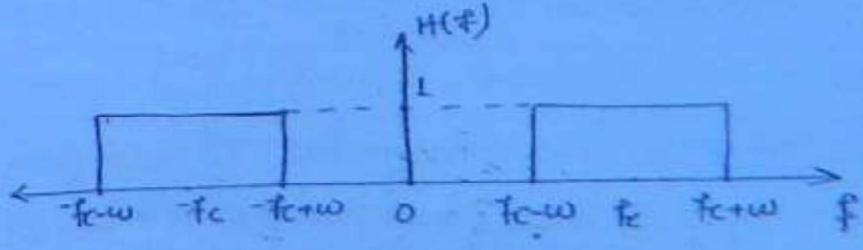


So, $S_{AM}(t)$



The BPF should be such that it has to allow only the frequency Band of AM signal

So,



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The o/p of square law device is given as

$$(SLD)_{o/p} = V_2 = a_1 V_1 + a_2 V_1^2$$

$$= a_1\{m(t) + A_c\cos 2\pi f_c t\} + a_2\{m^2(t) + A_c^2\cos^2 2\pi f_c t + 2A_c m(t)\cos 2\pi f_c t\}$$

Now,

$$(BPF)_{o/p} = a_1 A_c \cos 2\pi f_c t + 2a_2 A_c m(t)\cos 2\pi f_c t$$

So

$$(BPF)_{o/p} = a_1 A_c \cos 2\pi f_c t + 2a_2 A_c m(t)\cos 2\pi f_c t$$

$m^2(t) = m(t) \times m(t)$

$\updownarrow$

$M(f) * M(f)$

$\cos^2 2\pi f_c t = \frac{1+\cos 4\pi f_c t}{2}$ (d.c)

$$\boxed{(BPF)_{o/p} = a_1 A_c\left[1 + \frac{2a_2}{a_1} m(t)\right]\cos 2\pi f_c t = S_{AM}(t)}$$

Comparing with standard AM signal.

$$A_c'\{1 + K_a m(t)\}\cos 2\pi f_c t$$

where,

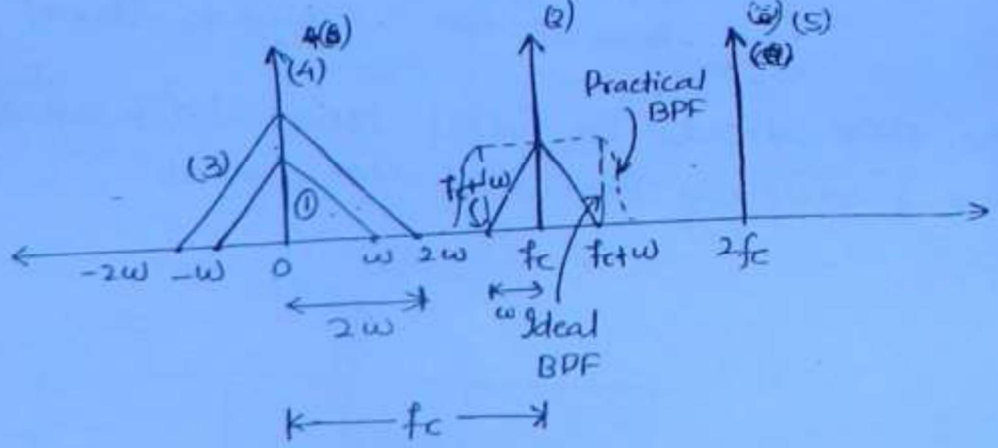
$$A_c' = a_1 A_c$$

$$\boxed{K_a = \frac{2a_2}{a_1}}$$

Now,

$$V_2 = a_1\{\overset{①}{m(t)} + \overset{②}{A_c\cos 2\pi f_c t}\} + a_2\{\overset{④}{A_c^2\cos^2 2\pi f_c t} + \overset{③}{m^2(t)} + 2A_c \overset{⑤}{m(t)}\cos 2\pi f_c t\}$$

$V_2(t) \longleftrightarrow$



$\boxed{f_c >>> 3\omega}$ ← To avoid overlapping of desired & undesired signal.

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Note :-

To avoid undesired frequencies at BPF output

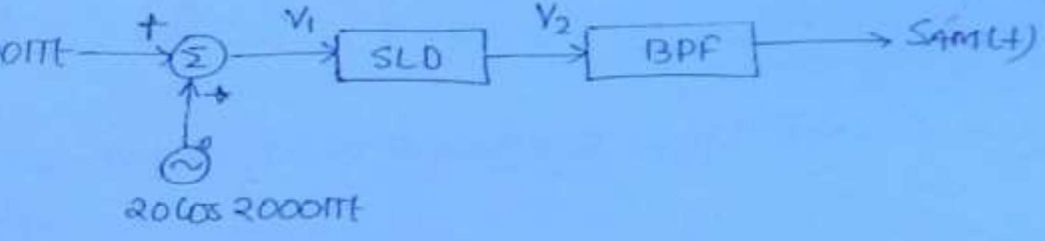
$f_c >>> 3W$

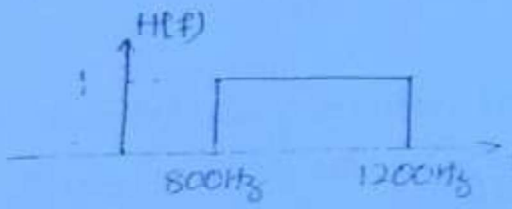
Q) For the following Square law modulator, where the square law device is characterised by

$$V_2 = V_1 + 0.1V_1^2$$

and pass Band of BPF will be from 800Hz to 1200Hz find all the parameters of Resulting AM signal

Sol^n:





Sol^n:

$$V_2 = (2\cos 2000\pi t + 20\cos 2000\pi t) + 0.1(2\cos 2000\pi t + 20\cos 2000\pi t)^2$$

$$= 2.0\cos 2000\pi t + 20\cos 2000\pi t + 0.1\{4\cos^2 2000\pi t + 400\cos^2 2000\pi t + 80\cos 2000\pi t \cos 2000\pi t\}$$

$$V_2 = (2\cos 2000\pi t + 20\cos 2000\pi t + 0.4\underbrace{\cos^2 2000\pi t}_{(1+\cos 4000\pi t/2)} + 40\cos^2 2000\pi t + 8\cos 2000\pi t \cos 2000\pi t)$$

$$\cos(2200\pi t) + \cos(1800\pi t)$$

So,

$$(BPF)_{o/p} = SAM(t) = 20\cos 2000\pi t + 8\cos 2000\pi t \cos 2000\pi t$$

$$\boxed{SAM(t) = 20\{1 + 0.4\cos 2000\pi t\}\cos 2000\pi t}$$

above signal is single tone AM. So

$$SAM(t) = A_c\{1 + \mu\cos 2\pi f_m t\}\cos 2\pi f_c t$$

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So,
$A_c = 20\,V$; $\mu = 0.4$; $f_m = 100\,Hz$, $f_c = 1000\,Hz$

(50)

objective $\mu = ?$

as, $\mu = K_a A_m$

and $K_a = \frac{2a_2}{a_1}$

by compaing with $V_2 = V_1 + 0.1V_1^2$

$V_2 = a_1 V_1 + a_2 V_1^2$

$a_1 = 1$; $a_2 = 0.1$

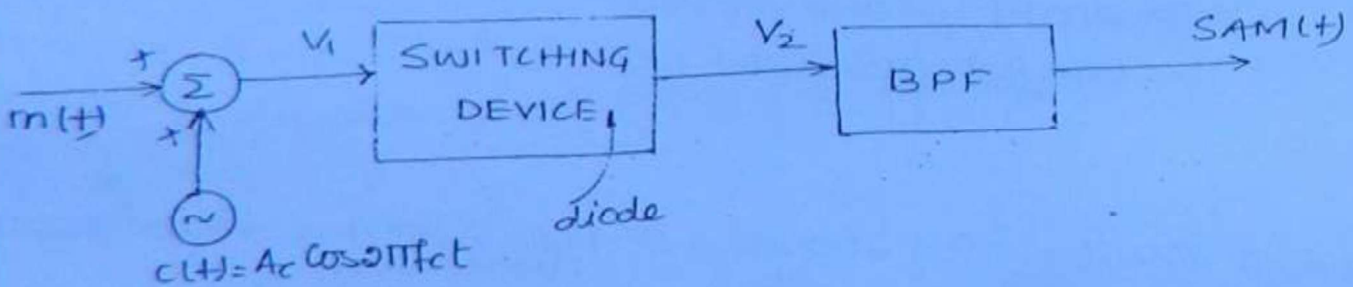
So, $K_a = \frac{2 \times 0.1}{1} = 0.2$

$\mu = 0.2 \times 20$

$\boxed{\mu = 0.4}$

* SWITCHING MODULATOR:

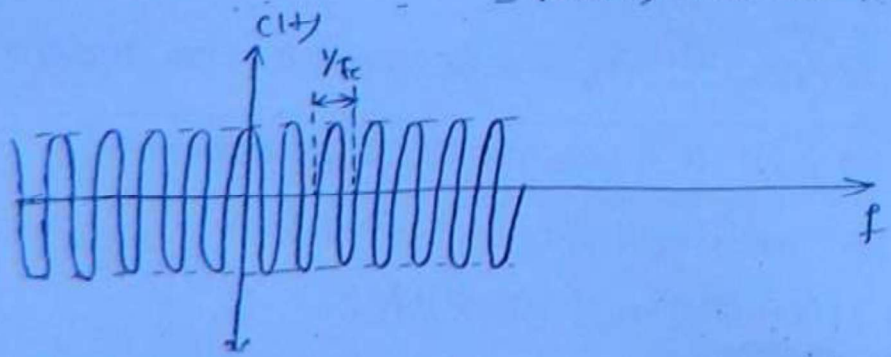
BLOCK diagm:



Analysis :-

I/P of diode $\Rightarrow V_1 = m(t) + C(t)$

$= m(t) + A_c \cos 2\pi f_c t$



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* The strength of Practical message signal will be generally of having less strength and carrier signal is generated with high strength. So the operation of the diode is mainly controlled by carrier signal. (57)

1. when c(t) is +ve, diode is forward Biased ie S.C then $\boxed{V_2 = V_1}$

2. when c(t) is -ve, diode is Reverse Biased ie O.C then $\boxed{V_2 = 0}$

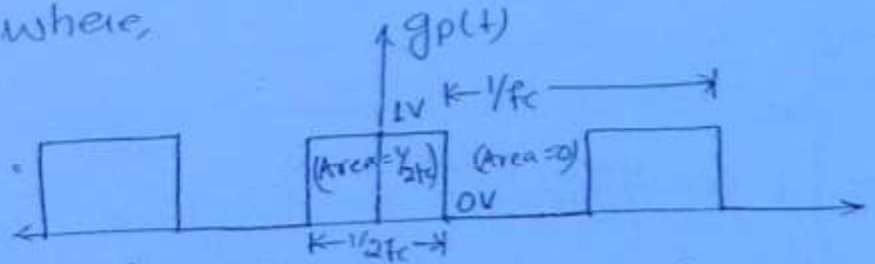
* O/P of diode switches b/w V_1 and 0 with the time interval of $1/f_c$.

Note:

∵ The O/P of switching device, ie V_2 is switching continuously b/w V_1 & 0, so plotting its spectrum is not easy. Hence to plot its spectrum, let

$$V_2 = V_1 \times g_p(t) \quad \text{---} \quad (1)$$

where,



The analysis of Periodic signal is done by Fourier series. Hence by Trigonometric FS we have:

$$f(t) = a_0 + \sum_{n=1}^{\infty} \{a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\} \; ; \omega_0 = 2\pi/T \quad \text{---} \quad (2)$$

∵ $g_p(t)$ is even, hence $b_n = 0$.

So, $$g_p(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$$

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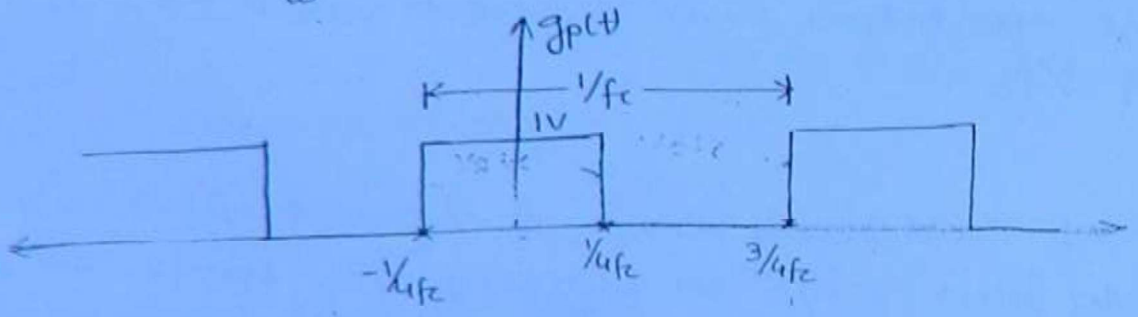
Now, $a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)\,dt = \frac{\int_{t_0}^{t_0+T} f(t)\,dt \text{ (Area)}}{T}$ (52)

$$a_0 = \frac{\frac{1}{2}f_c}{1/f_c}$$

$$\boxed{a_0 = 1/2}$$

&

$$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\cos n\omega_0 T\,dt \quad ; \omega_0 = 2\pi/T = 2\pi f_c$$



So,

$$a_n = \frac{2}{1/f_c}\int_{-1/4f_c}^{3/4f_c} g_p(t)\cos n 2\pi f_c t\,dt$$

$$a_n = \frac{2}{1/f_c}\int_{-1/4f_c}^{1/4f_c} 1\cdot\cos 2\pi n f_c t\,dt + 0$$

$$= 2f_c \times \frac{\sin 2\pi n f_c t}{2\pi n f_c}\Bigg|_{-1/4f_c}^{1/4f_c}$$

$$= \frac{1}{n\pi}\left\{\sin 2\pi n f_c \times \tfrac{1}{2}\tfrac{1}{4f_c} - \sin 2\pi n f_c \times -\tfrac{1}{4f_c}\right\}$$

$$= \frac{1}{n\pi}\left\{\sin\frac{n\pi}{2} + \sin\frac{n\pi}{2}\right\}$$

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$$a_n = \frac{2}{n\pi} \sin n\pi/2$$

So, from equation (1) we get:

$$g_p(t) = 1/2 + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi/2 \cos n 2\pi f_c t$$

and from equation (1) we get

$$V_2 = V_1 \cdot g_p(t)$$

$$= (m(t) + A_c \cos 2\pi f_c t)\left\{ \frac{1}{2} + \frac{2}{\pi} \cos 2\pi f_c t - \frac{2}{3\pi} \cos 2\pi (3f_c)t + \frac{2}{5\pi} \cos 2\pi (5f_c)t + \cdots \right\}$$

* V_2 is then passed through a BPF. So its O/P is given as:

$$(BPF)_{O/P} = S_{AM}(t) = \left(\frac{2}{\pi} m(t) \cos 2\pi f_c t + \frac{A_c}{2} \cos 2\pi f_c t\right)$$

$$S_{AM}(t) = \frac{A_c}{2}\left\{1 + \frac{4}{\pi A_c} m(t)\right\} \cos 2\pi f_c t$$

Comparing with standard equation of AM we get:-

$$A_c'\{1 + K_a m(t)\} \cos 2\pi f_c t$$

So,

$$A_c' = \frac{A_c}{2}$$

*** & $K_a = \frac{4}{\pi A_c}$

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DEMODULATION OF AM SIGNAL:

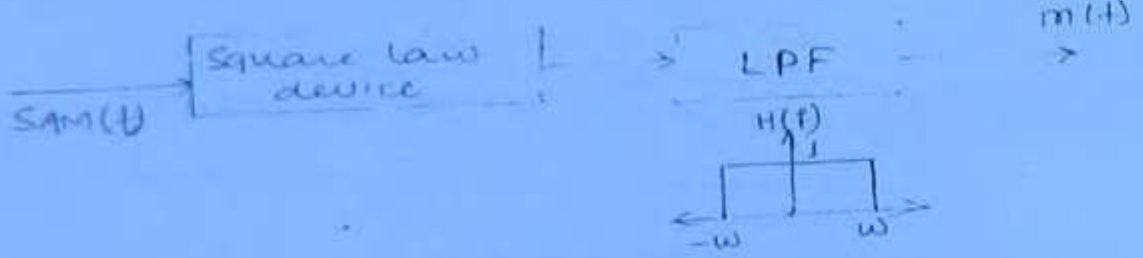
The demodulators that are used for demodulation of AM are:

1) Square law demodulator
2) Envelope detector (diode detector) } $\mu \leq 1$
3) Synchronous detector } any value of μ

* Synchronous detector is complex and square law demodulator will have some drawbacks. So generally for demodulation of AM; envelope detector will be used.

SQUARE LAW DEMODULATOR:

B.Diag^M



As,

$$S_{AM}(t) = A_c\{1 + K_a m(t)\}\cos 2\pi f_c t$$

$$= A_c \cos 2\pi f_c t + K_a m(t)\cos 2\pi f_c t \cdot A_c$$

The o/p of the square law device may be given as:

$$(SLD)_{o/p} \Rightarrow V_2 = a_1 S_{AM}(t) + a_2\{S_{AM}(t)\}^2$$

$$= a_1\{A_c \cos 2\pi f_c t + A_c K_a m(t)\cos 2\pi f_c t\}$$

$$+ a_2\{A_c^2 \cos^2 2\pi f_c t + K_a^2 m^2(t)\cos^2 2\pi f_c t \cdot A_c^2 + 2K_a A_c^2 m(t)\cos^2 2\pi f_c t\}.$$

So,

$$V_2 = \left[a_1 A_c \cos 2\pi f_c t + a_1 A_c K_a m(t)\cos 2\pi f_c t + \frac{a_2 A_c^2}{2}(1+\cos 4\pi f_c t) + \frac{a_2 A_c^2 K_a^2 m^2(t)}{2}(1+\cos 4\pi f_c t) + a_2 A_c^2 K_a m(t)(1+\cos 4\pi f_c t)\right].$$

✓ (but only Part of)
✓ (but blocked by blocking capacitor)

the signal v_2 is given to LPF and the LPF is given as following:-

$$(LPF)_{OIP} = \underbrace{\frac{a_2 A_c^2 K_a^2 m^2(t)}{2}}_{\text{unwanted signal or Noise}} + \underbrace{\frac{a_2 \cdot 2A_c^2 K_a m(t)}{2}}_{\text{expected Signal}} \quad (5)$$

If

$\frac{S}{N} >>>> 1$; $m(t)$ can be Reconstructed (almost in the Range of 100 or 1000)

$\frac{S}{N} < 1$; $m(t)$ can't be Reconstructed

Now,

$$\frac{S}{N} = \frac{a_2 A_c^2 K_a m(t)}{\frac{a_2 A_c^2 K_a^2 m^2(t)}{2}}$$

$$\frac{S}{N} = \frac{2}{K_a m(t)}$$

Now let $m(t) = A_m \cos 2\pi f_m t$

So,

$$\frac{S}{N} = \frac{2}{K_a \cdot A_m \cos 2\pi f_m t}$$

$$\frac{S}{N} = \frac{2}{u \cos 2\pi f_m t}$$

As, $-1 \leq \cos 2\pi f_m t \leq 1$

So, $\boxed{\frac{S}{N} = \frac{2}{u \times 1}}$ $\{\cos 2\pi f_m t = 1\}$.

Now $\frac{S}{N} \propto \frac{1}{u} \Rightarrow$ To get $\frac{S}{N} = 100 \Rightarrow u = 0.02$

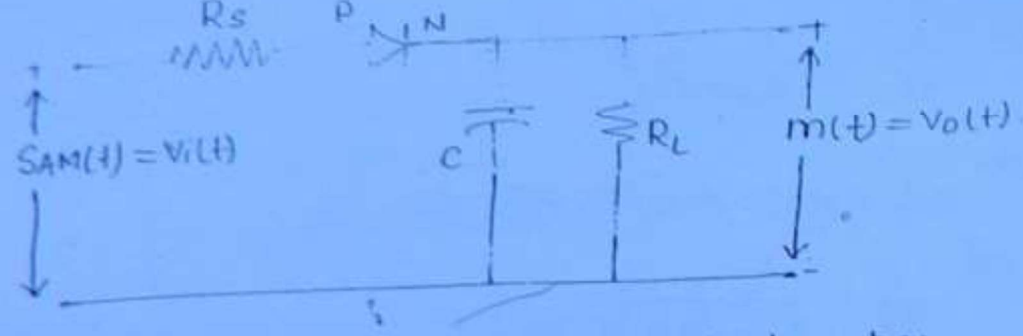
For such small value of u, η will be worstly affected ie low share of sideband power in AM signal.

* Hence in the square law demodulator either of the two 'E' or 'η' has to be sacrificed. It means that if 'E' is increased P_{SB} is increased but S/N is reduced and vice versa.

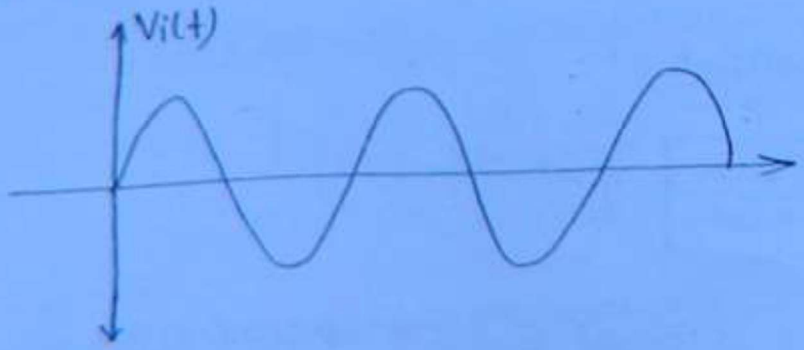
* To get high value of S/N, 'E' should be very small. For small values of 'E', modulation efficiency will be very much low (worstly affected). But for efficient power distribution, 'η' should be max^m. So this method is practically not preferred for AM demodulation.

* ENVELOPE DETECTOR:

CKt diagram:



→ Envelope detector is very simple and cheaper.
→ The +ve envelope of applied signal will be produced at the o/p so it is called as ENVELOPE DETECTOR.



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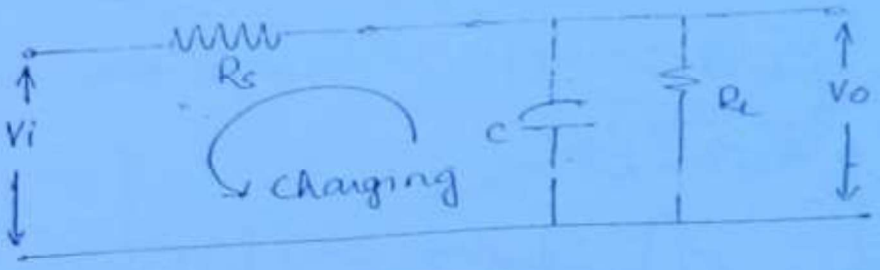
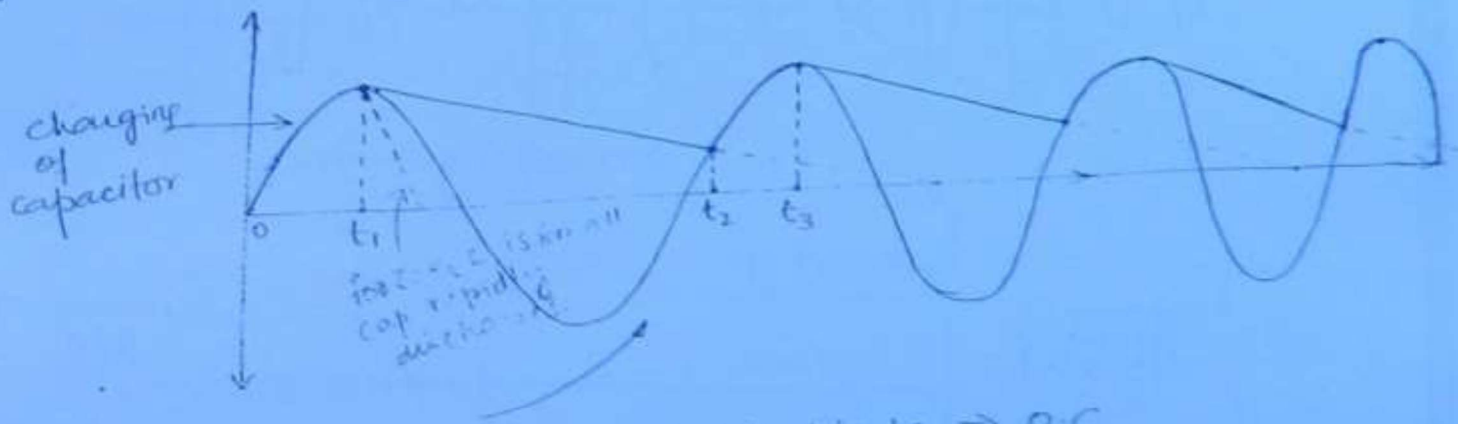
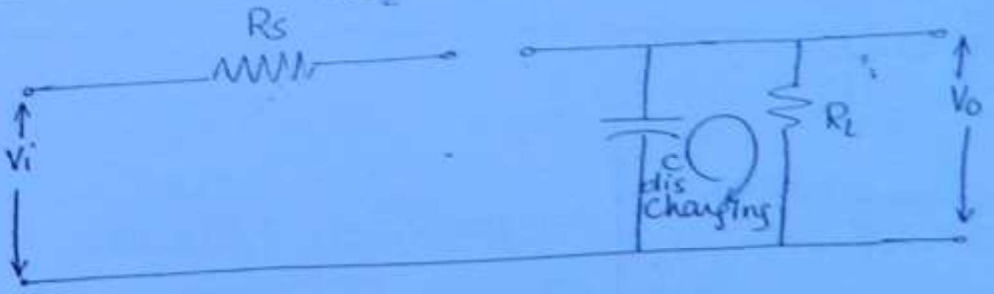


Note :-
The time const = $R_s \cdot C$ should be very small. Hence capacitor rapidly changes to input voltage.



Case 2 : For $t = t_1^+$ to t_2 ; P < N ⇒ RB diode ⇒ O.C



If $\tau = R_L \cdot C$ is very small, the capacitor rapidly discharges to zero.

If $\tau = R_L \cdot C$ is very high, the capacitor discharges slowly

Note :-
$\tau = R_L \cdot C$; should be very high then capacitor discharges very slowly.

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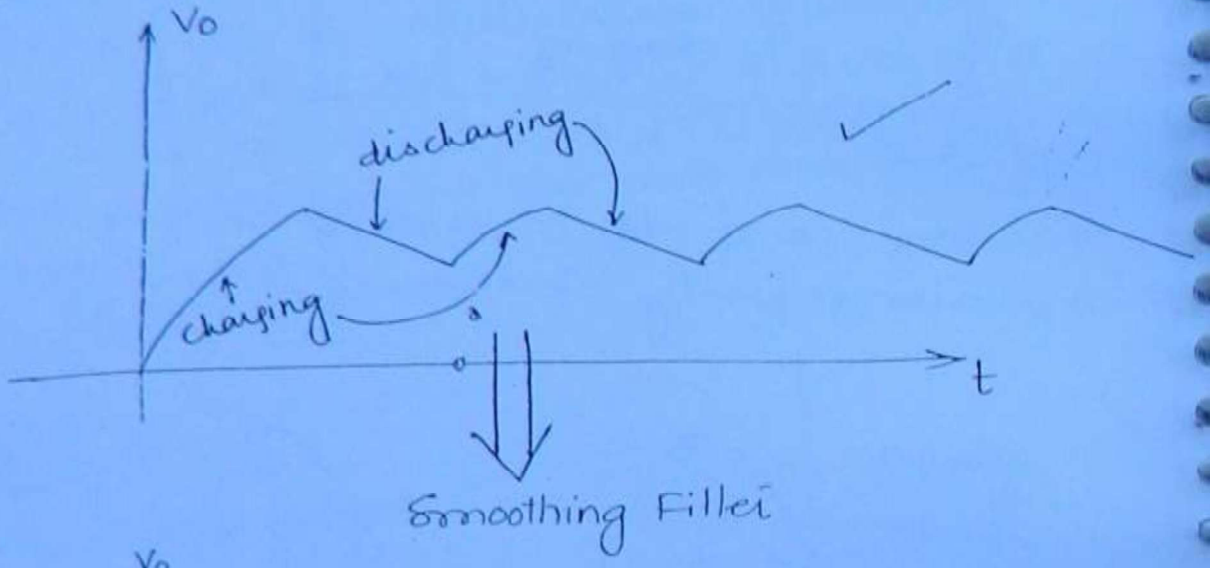
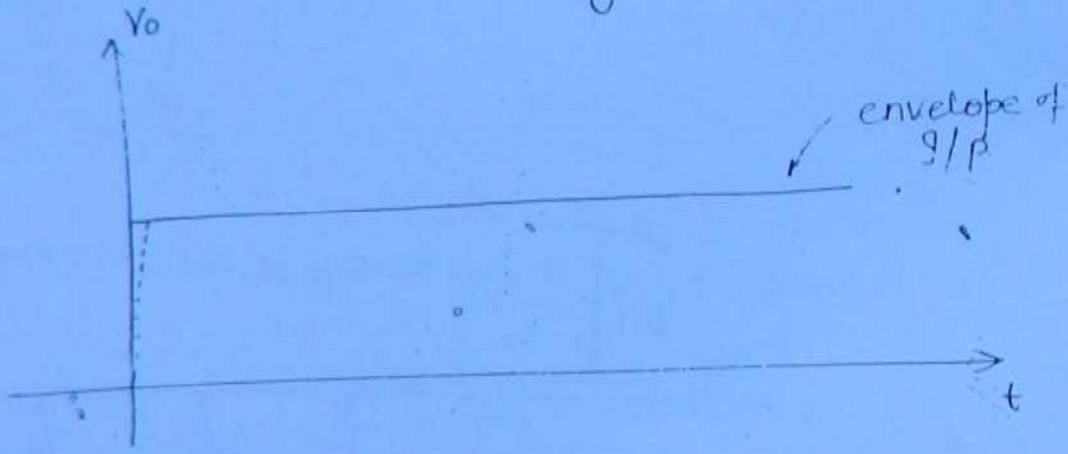
The curve shown at back

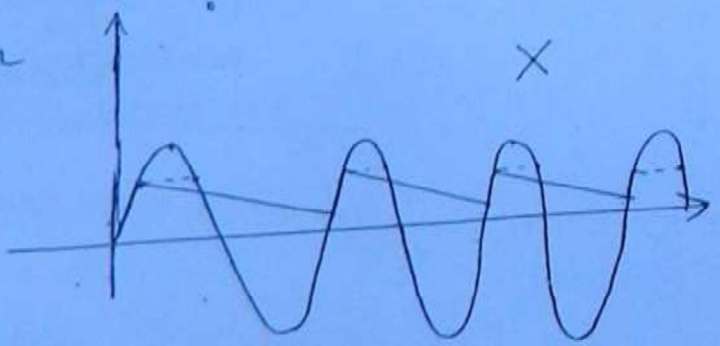
Voltage across capacitor:



Smoothing Filter



Note:- If $R_S \cdot C$ = very high



If $R_S \cdot C$ is very high, even before capacitor voltage reaches to peak voltage of input, diode becomes R and capacitor will be discharged.

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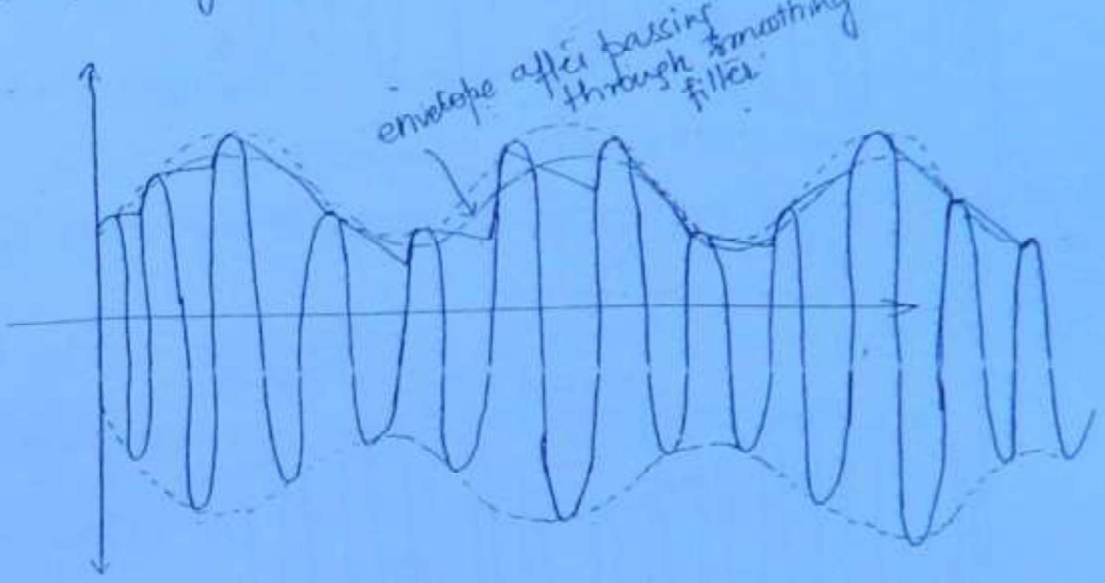
→ To make capacitor voltage to follow the envelope of input RsC should be very small and RLC should be high.

Note:

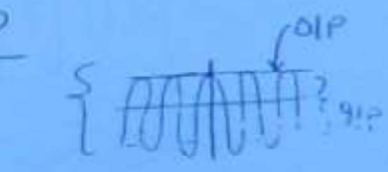
If the AM signal is applied to the diode detector then:



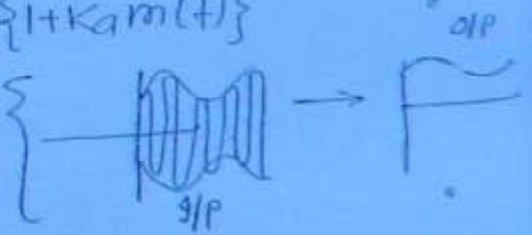
Analysis:

ED I/P → ED O/P

1. $V_m \cos 2\pi f_m t \longrightarrow V_m$



2. $S(t) = A_c\{1 + K_a m(t)\}\cos 2\pi f_c t \longrightarrow A_c\{1 + K_a m(t)\}$



3. $A\cos 2\pi f_0 t + B\sin 2\pi f_0 t \longrightarrow \sqrt{A^2+B^2}$

$= \left\{\sqrt{A^2+B^2} \times \cos(2\pi f_c t + \phi)\right\}$

$\phi = \tan^{-1} B/A$

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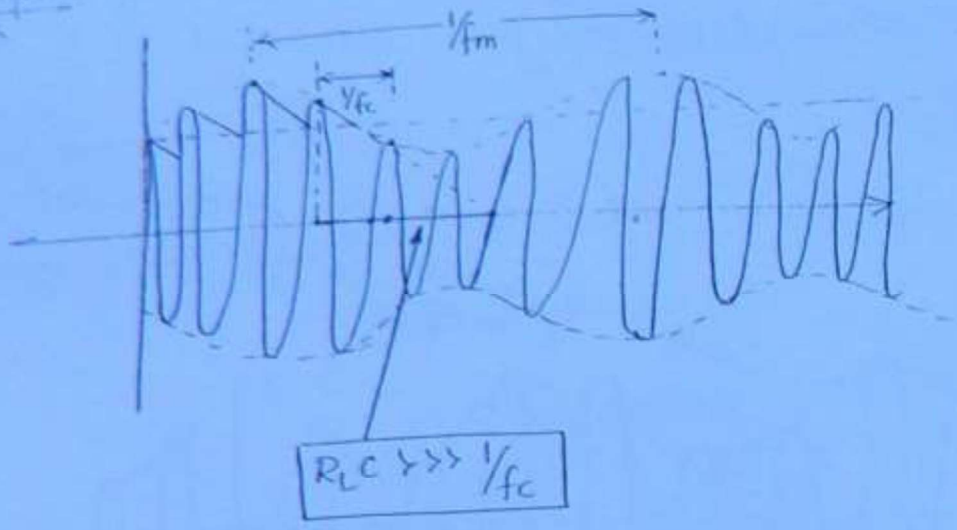
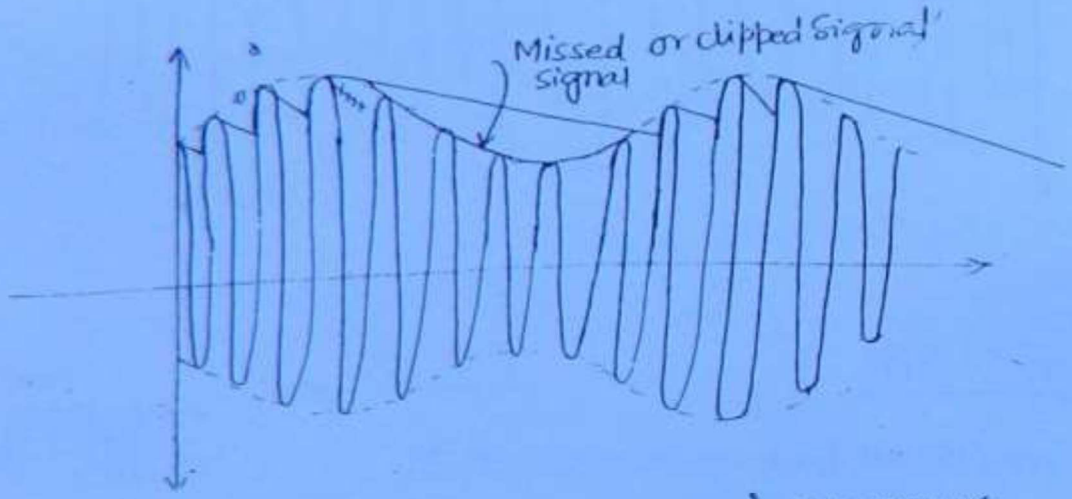
Analysis:





CASE OF DIAGONAL CLIPPING

To avoid diagonal clipping

$R_L C >>> \frac{1}{f_m}$ $\quad$ $R_L C <<< \frac{1}{f_m}$

* Proper choice of $R_L C \Rightarrow \frac{1}{f_c} <<< R_L \cdot C <<< \frac{1}{f_m}$

Q1. The carrier signal of 1 MHz is amplitude modulated by message signal of frequency 2 KHz. The proper choice of $R_L \cdot C$ for envelope detection should be

a) 1 μsec b) 100 μsec c) 200 μsec d) 0.5 μsec

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Soln: $f_c = 1\,MHz \Rightarrow T_c = ?$

$f_m = 2K \Rightarrow \frac{1}{f_m} = 0.5\,msec = 500\,\mu sec.$

(6)

$\frac{1}{f_c} << R_L C << \frac{1}{f_m}$

Q2. An AM signal is given by

$$S(t) = A_c \cos\omega_c t + 2\cos\omega_c t \cos\omega_m t$$

For proper envelope detection, the min^m value of A_c shou'd be:

a) 0.5 volts b) 2 volts c) 1 volts d) 2.5 volts

Soln: Comparing with AM signal:

$$S(t) = A_c\{1 + \frac{2}{A_c}\cos\omega_m t\}\cos 2\pi f_c t$$

$$S_{AM}(t) = A_c\{1 + \mu\cos\omega_m t\}\cos 2\pi f_c t$$

So, $\mu = \frac{2}{A_m}$ $\quad \mu = \frac{2}{A_c}$

$\frac{A_m}{A_c} = \frac{2}{A_m}$. for envelope detection $\mu \le 1$.

$\Rightarrow \frac{2}{A_c} \le 1$

$\boxed{A_c \ge 2}$ Ans.

Hence, the minm value of A_c is 2 volts.

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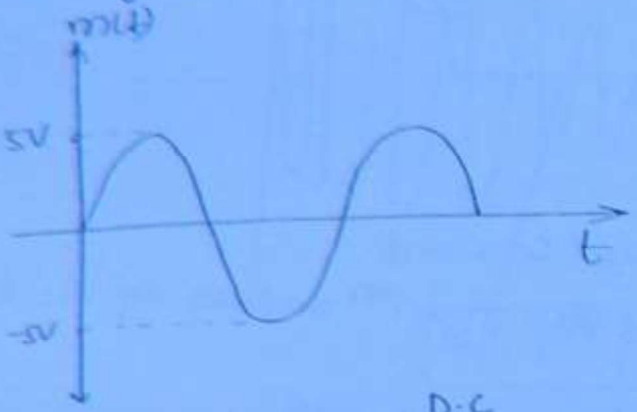
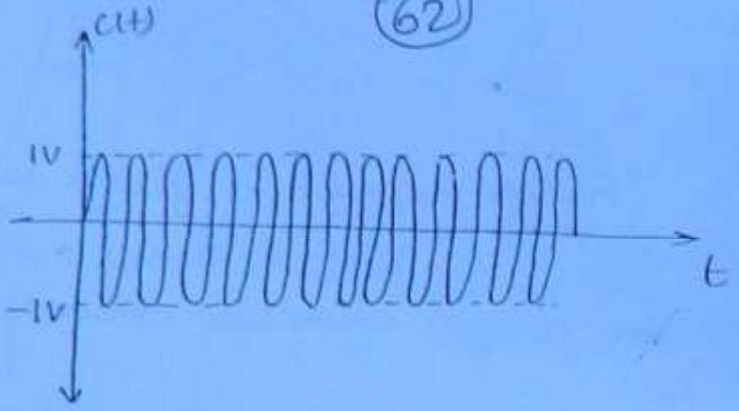
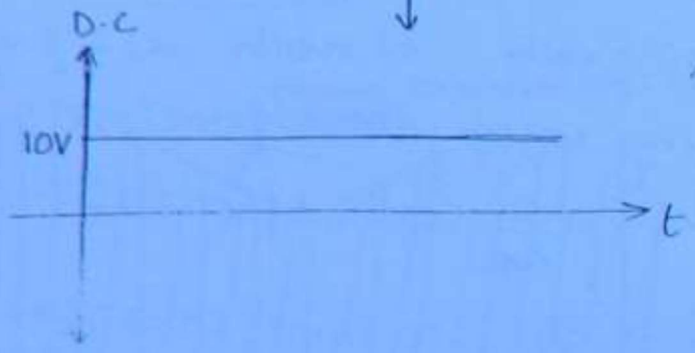
Analysis:

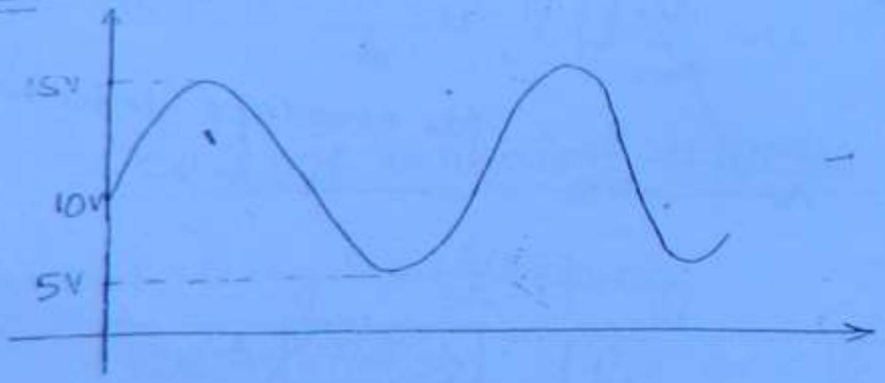
i.



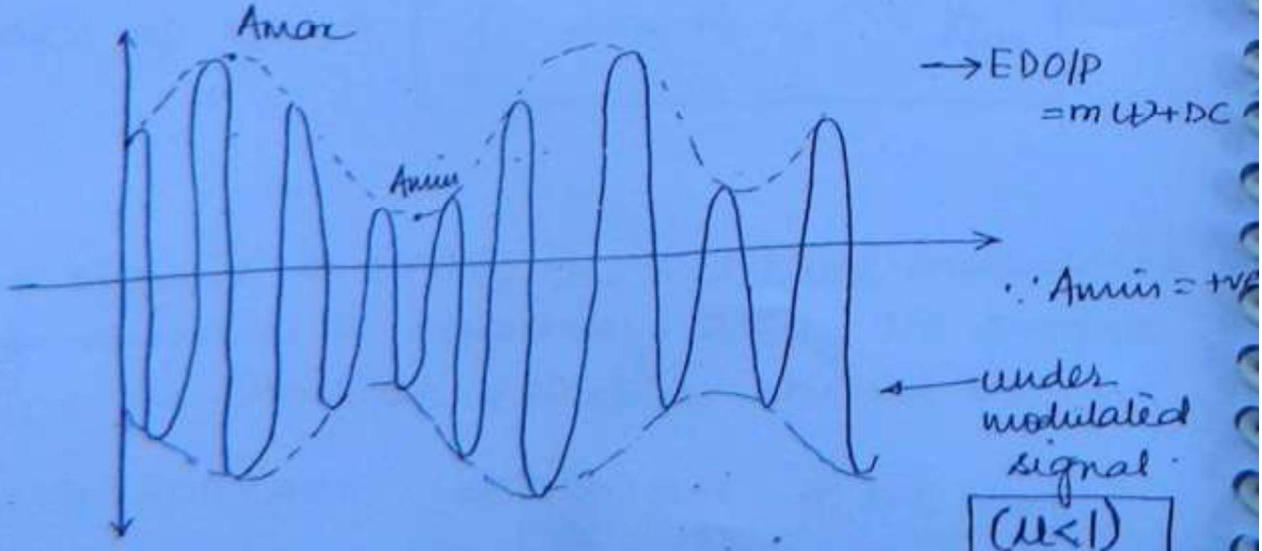




m(t) + DC :-



Now, $SAM(t) = \{m(t) + DC\} c(t) = \{m(t) + DC\} \cos 2\pi f_c t$



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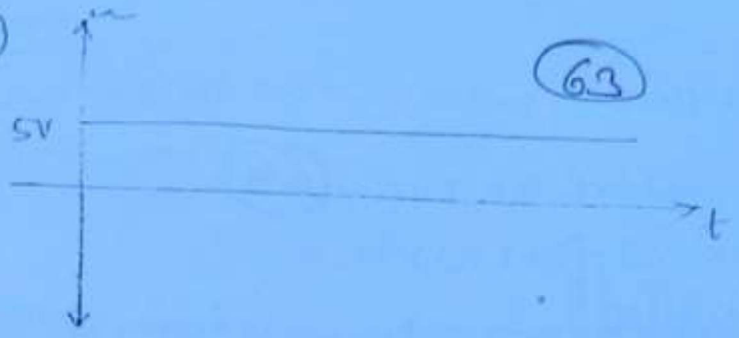
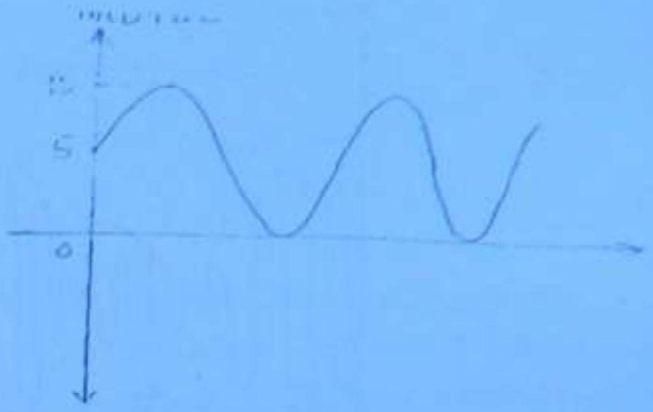
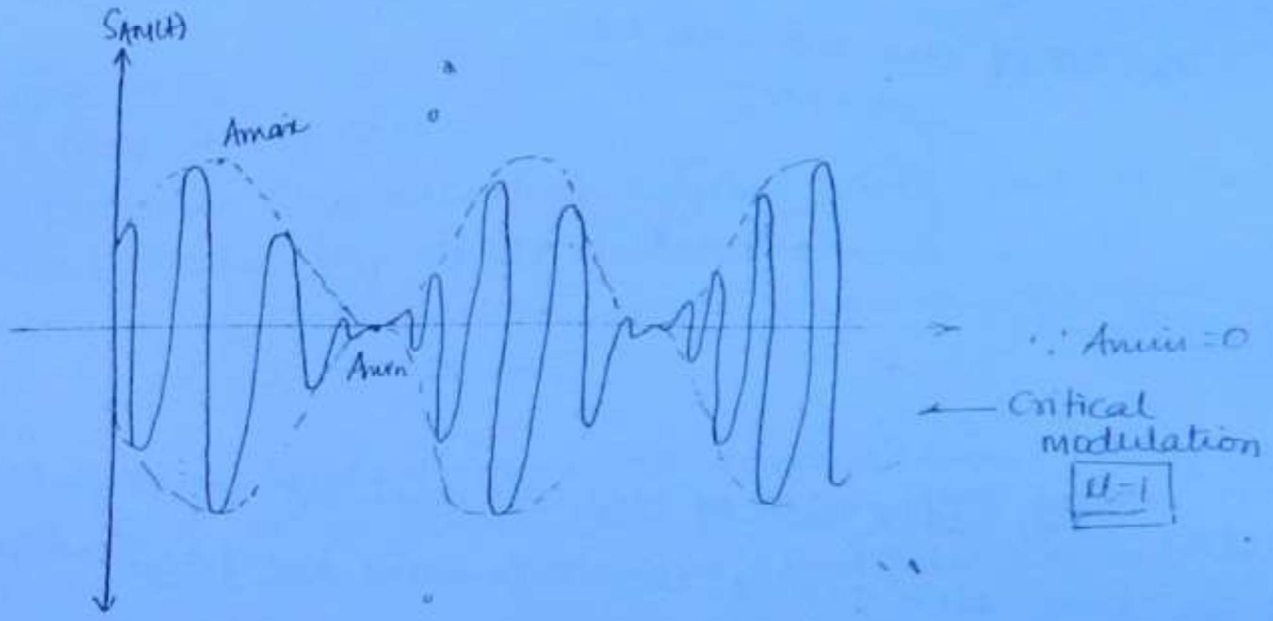
(ii)

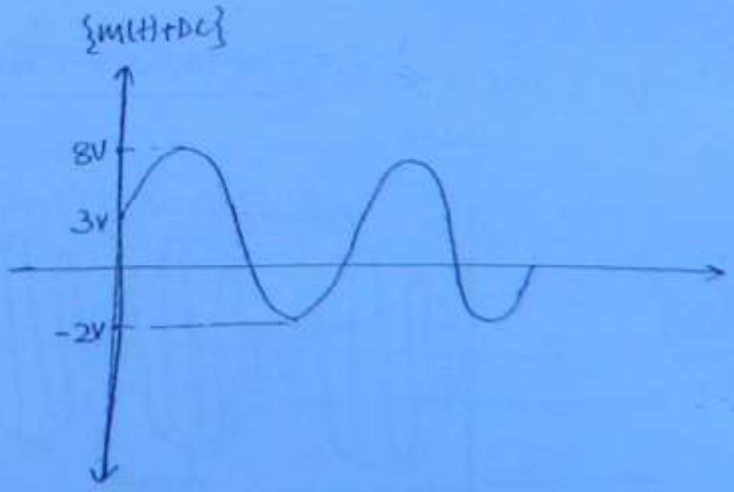




So,

$$S_{AM}(t) = \{m(t) + DC\} \cos \omega_c t$$





So,

$$S_{AM}(t) = \{m(t) + DC\} \cos \omega_c t$$

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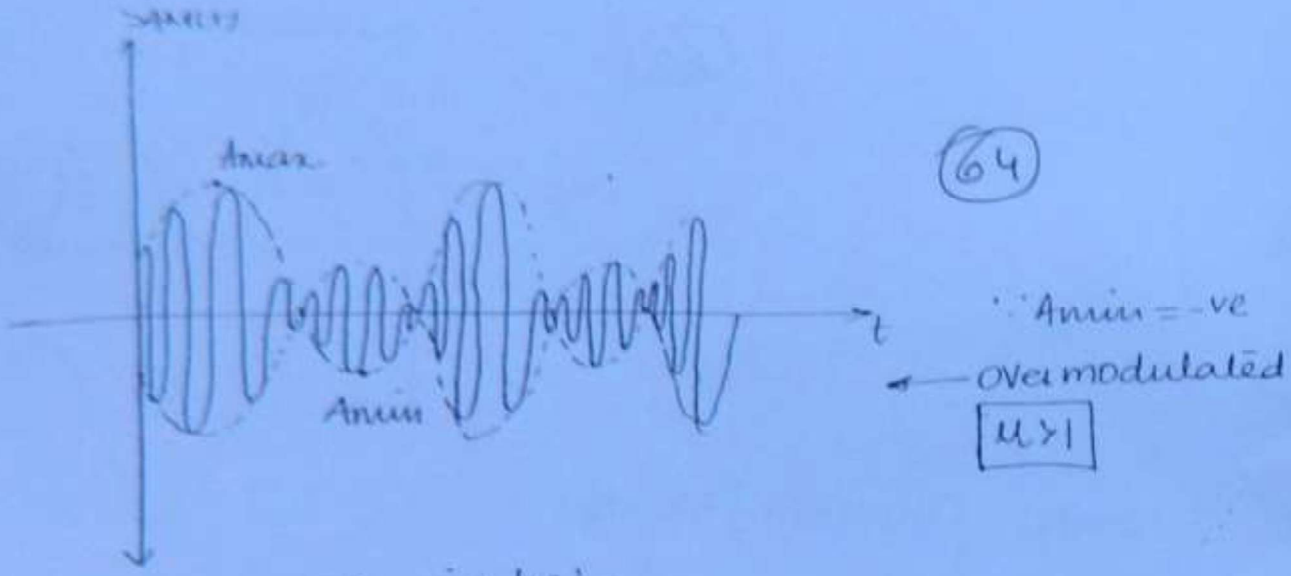
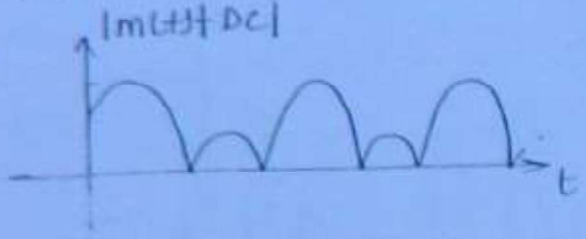


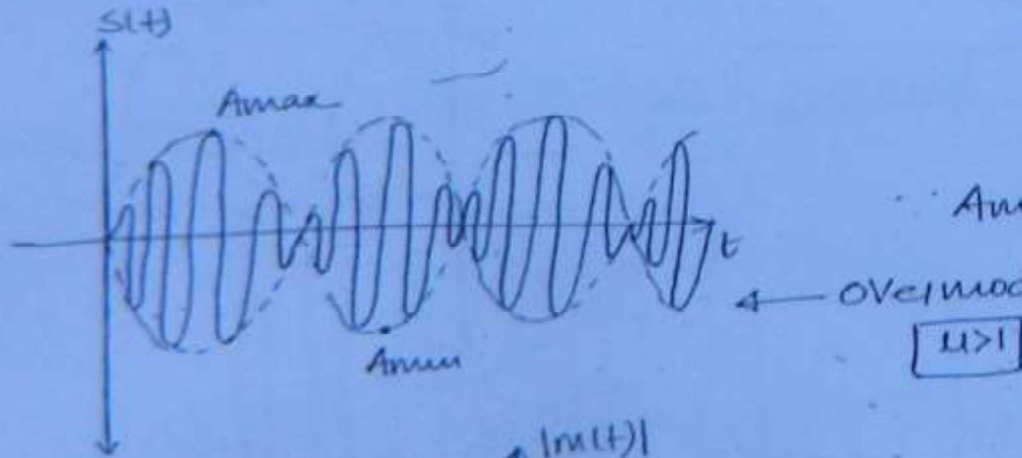
The O/P of the ED will be:-



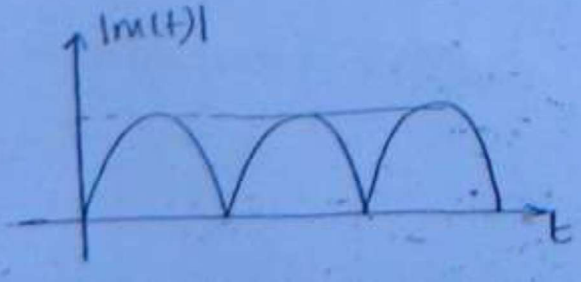
Note:

1. For $\mu \leq 1$, the O/P of the ED will be $A_c\{1+k_a m(t)\}$
2. For $\mu > 1$, the O/P of the ED will be $|A_c\{1+k_a m(t)\}|$

IV $s(t) = m(t) \cdot c(t) = m(t)\cos 2\pi f_c t$



O/P of ED is given as:-



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NOTE :-

1. The O/P of ED in case (i) & (iii) is totally +ve.

and, $SAM(t) = \{m(t) + DC\}c(t)$ (65)

$= m(t)c(t) + D.C \times c(t)$

If the DC term is not added in the modulation of the signal the term $D.C \times c(t)$ would not be obtained Hence the AM signal obtained would be over modulated and it can't be demodulated back to obtain $m(t)$ by ED. Hence the ED fails in its demodulation.

2. For case iv ie

$s(t) = m(t) \cdot c(t)$

it contains the message signal $m(t)$, but it can't be demodulated by diode detector.

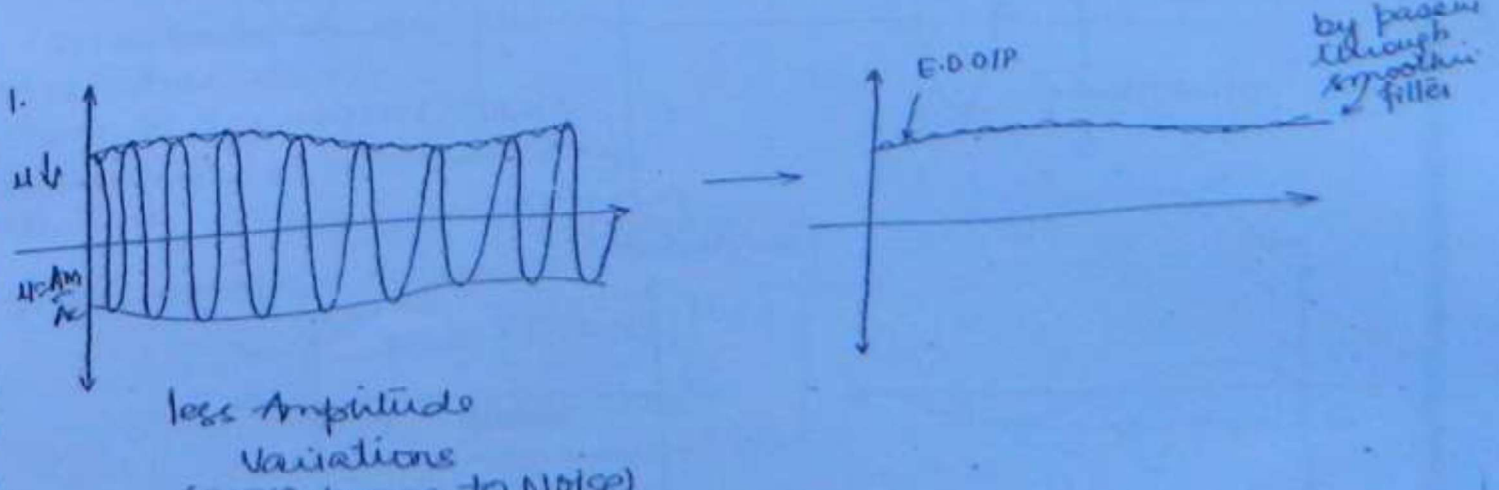
This scheme of $s(t) = m(t) \cdot c(t)$ is given a specific name called as DSB-SC

3. Case IV corresponds to DSB-SC modulation. The advantage of DSB-SC is that the Transmitter power will be saved but its drawback is that its Demodulation becomes complex

* IMPORTANCE OF Ka (amp. Sensitivity) :-

As, we know that :-

$SAM(t) = Ac\{1 + Kam(t)\}\cos 2\pi fct$



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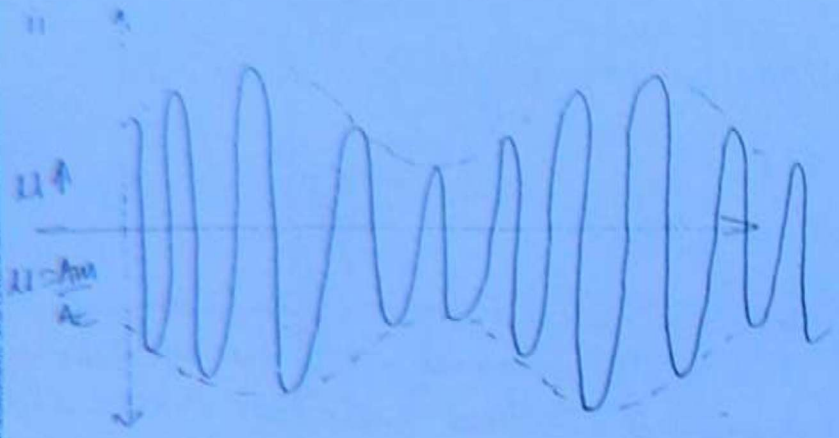
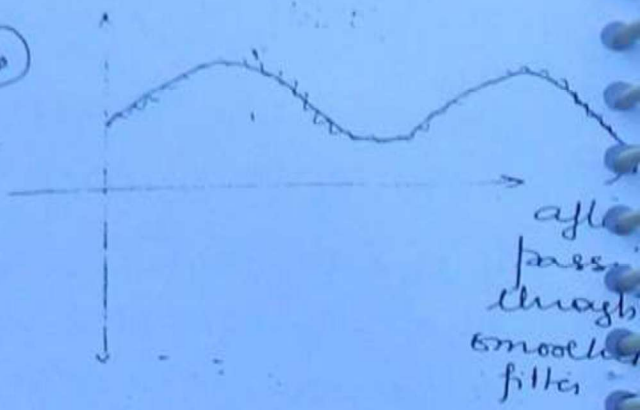




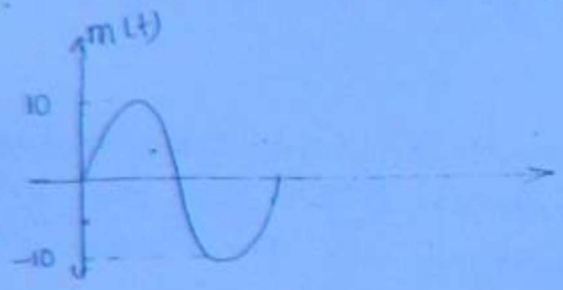
Note:

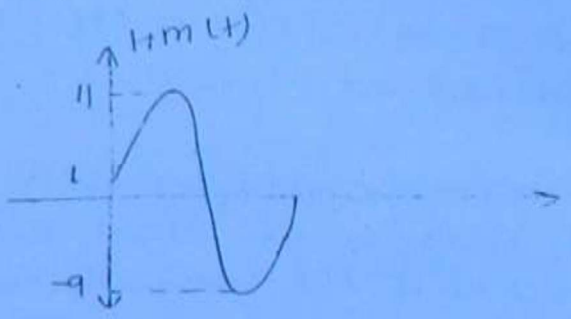
1. u is high the effect of Noise is less on the reconstructed message signal
2. u is less, the effect of Noise is dominant

Now,

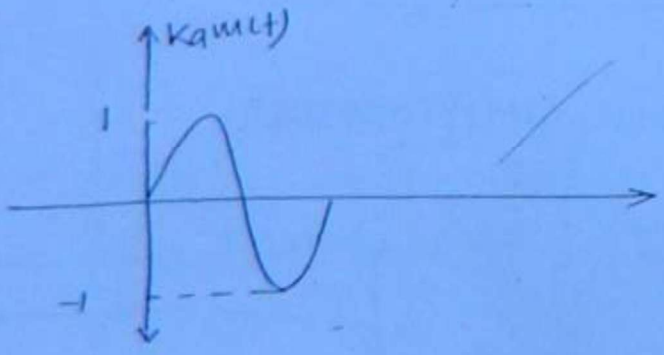
$S_{AM}(t) = A_c\{1+m(t)\}\cos 2\pi f_c t$

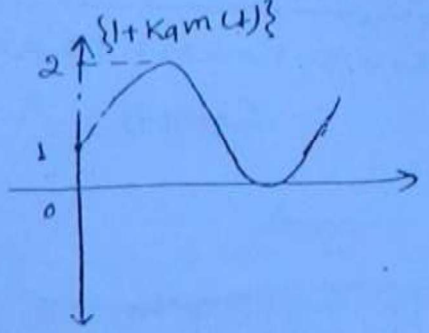
1.



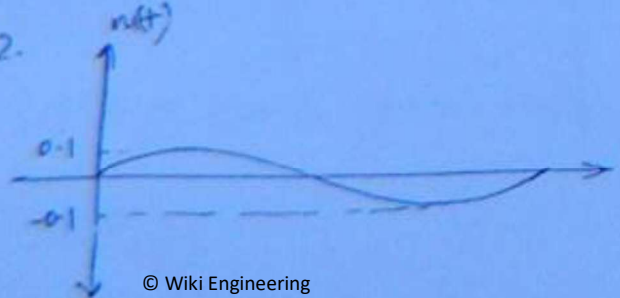


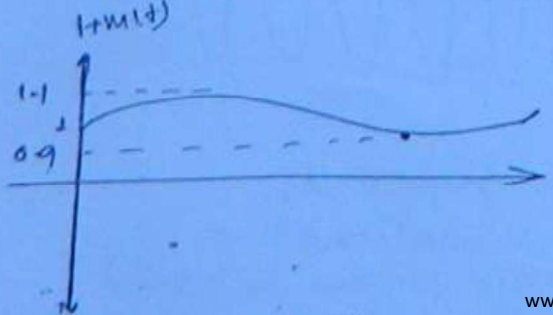
Now $K_a = 1/10 \Rightarrow K_a m(t)$ → attenuating message signal





2.





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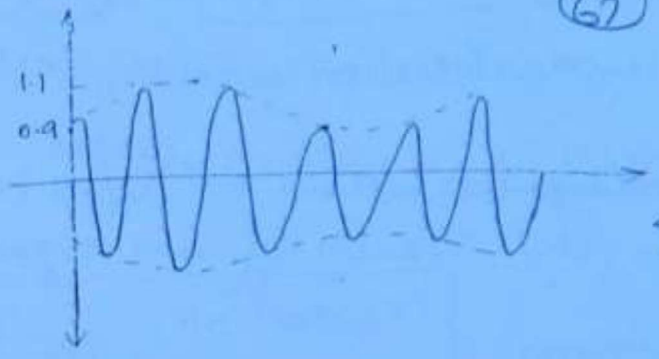
The AM signal has little amp variations and hence effect of Noise is high

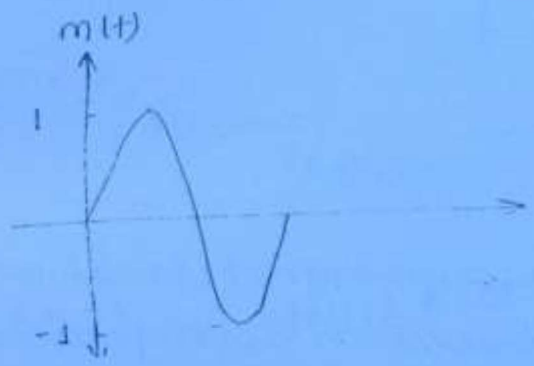


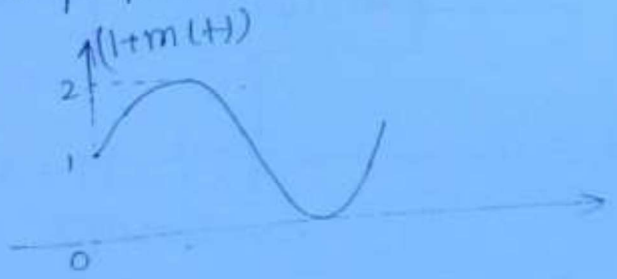
X ← As effect of Noise is high

← If dc value of 10 is also added hence, Amax = 10.1 ; Amin = 9.9
Then also amp variations are small and effect of Noise is present.

3. As effect of Noise was high, since for small amp. variations. So, the message needs to be amplified.

So, $K_a = 10 \Rightarrow K_a m(t)$ = amplified.





Note:-

1. Ka specifies, normalisation of message signal for proper envelope detection and according to SIMON-HAYKINS,

$$u = K_a \cdot A_m$$ ← For Square law & Switching modulator

Note:- originally, $u = \frac{A_m}{A_c}$, Corresponds to the value of u when Ka = 1.
Now, If the signal needs to amplified or attenuated by Ka. Then, $u = K_a \cdot A_m$ ← value of Ac is included in Ka or taken as 1

where, $K_a = \frac{2a_2}{a_1}$ {Square law modulator}

$K_a = \frac{1}{\pi A_c}$ {Switching modulator}

2. For envelope detection,

$$\max\{K_a m(t)\} \leq 1$$

or $u \leq 1$

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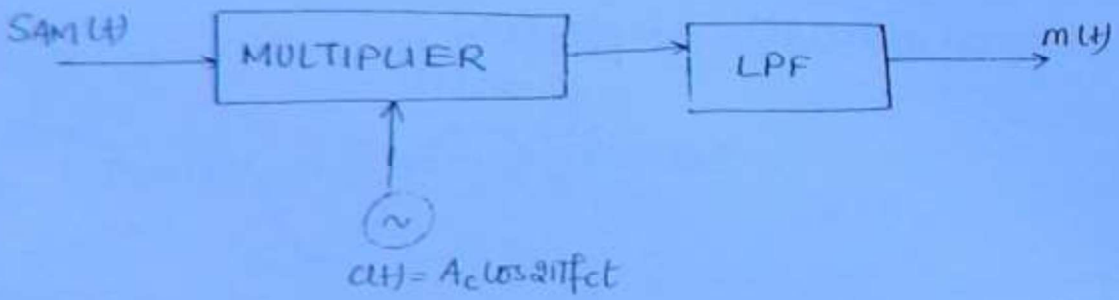
3. For $\max\{K_a m(t)\} > 1$ → over modulation

the O/P of the diode detection will be $|m(t)|$.

4. It specifies the voltage variations of super imposed message signal in the Resulting AM signal.

* SYNCHRONOUS DETECTION:

Block-diagram:



* For perfect Reconstruction of message signal, Local oscillator O/P should be perfectly synchronised in both frequency and phase
* Frequency synchronisation can be easily achieved but achieving phase synchronisation is very complex & difficult
* To achieve phase synchronisation additional complex ckts has to be used which make the Synchronous detector very complex.

Analysis:

Case1:-

1. Assume L.O(O/P) = $A_c \cos 2\pi f_c t$ {perfect synchronisation}.

& $SAM(t) = A_c\{1 + K_a m(t)\}\cos 2\pi f_c t$

$= A_c \cos 2\pi f_c t + A_c K_a m(t) \cos 2\pi f_c t$

The multiplier O/P is given as:

$SAM(t) \times c(t)$.

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So,

$(Mul)_{o/p} = SAM(t) \times c(t)$

$= \{A_c \cos 2\pi f_c t + A_c K a m(t) \cos 2\pi f_c t\} \cdot A_c \cos 2\pi f_c t$

$= A_c^2 \cos^2 2\pi f_c t + A_c^2 K a m(t) \cos^2 2\pi f_c t$ (69)

$= \frac{A_c^2}{2}(1 + \cos 4\pi f_c t) + \frac{A_c^2 K a m(t)}{2}(1 + \cos 4\pi f_c t)$

So, $\boxed{(LPF)_{o/p} = \frac{A_c^2 \cdot K a m(t)}{2}}$

→ As the AM signal is transmitted for long distances, its amp gets reduced.

→ So, for its Reconstruction, it is passed through Ampl^r

Case 2 :-

Assume, $(L.O)_{o/p} = \cos\{2\pi f_c t + \Phi\}$ ← {NO phase synchronisation}

So,

$(Mul)_{o/p} = SAM(t) \times \cos\{2\pi f_c t + \Phi\}$

$= \{A_c \cos 2\pi f_c t + A_c K a m(t) \cos 2\pi f_c t\} \cos\{2\pi f_c t + \Phi\}$

$= \frac{A_c^2}{2}\cos(4\pi f_c t + \Phi) + \frac{A_c^2}{2}\cos\Phi + \frac{A_c^2 K a m(t)}{2}\cos(4\pi f_c t + \Phi)$ ✓ (But blocked by block. cap. of Ampl^r)

$+ \frac{A_c^2 K a m(t)}{2}\cos\Phi.$

So, $\boxed{(LPF)_{o/p} = \frac{A_c^2 \cdot K a \cdot m(t)}{2}\cos\Phi}$

Note :-

1. The message signal can be Reconstruction back, if the value of $\cos\Phi$ ie Φ remains const.
2. But if the value of Φ changes continuously with time, we have to implement an Ampl^r whose gain changes with Φ to maintain $m(t)$ at const value, practically it is impossible to construct.

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* Also if $\phi = 90^\circ$

so $\cos\phi = \cos 90^\circ = 0$ (70)

So the O/P of the demodulator is 0, ie no message can be extracted. This is called as "Quadrature NULL EFFECT" (QNE).

Note:

1. For perfect Reconstruction of message signal $\phi = const$
2. To maintain, $\phi = const$ additional complex ckt is has to be used which makes SD very complex.
3. The demodulation of AM by SD is affected by "QNE"

* ADVANTAGES OF AM:

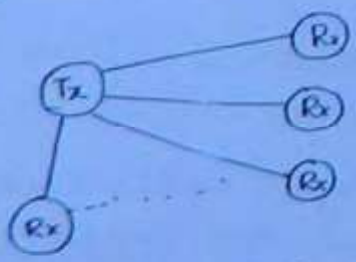
1. Primary Advantage is that the demodulation is simple
2. used for long distance comm^n

* DRAWBACKS OF AM:

1. Transmitter power is wasted. (max^m η = 33% of $P_t = P_{SB}$)
2. It needs high transmission Bandwidth
3. Highly affected by Noise
4. QNE.

* APPLICATIONS OF AM:

1. It is foreferred to be used in Broadcasting. (Point to Multi-point)



Since AM Reciever is simple and cheaper, it is highly foreferred in Broadcasting.

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× DOUBLE SIDE BAND - SUPPRESSED CARRIER (DSB-SC):

Assume, message signal = $m(t)$

Carrier signal = $c(t) = A_c \cos 2\pi f_c t$ (1)

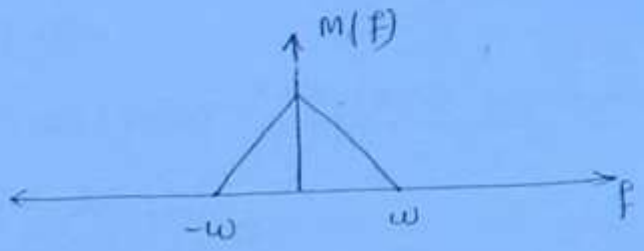
×General eqn of DSB-SC signal:

$$S_{DSB}(t) = m(t)\,c(t)$$

$$\boxed{S_{DSB}(t) = A_c\, m(t) \cos 2\pi f_c t}$$

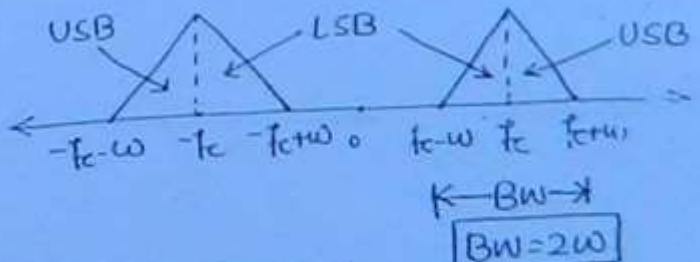
×Frequency spectrum:

$m(t) \longleftrightarrow$



$c(t) = A_c \cos 2\pi f_c t \longleftrightarrow$

$S_{DSB}(t) \longleftrightarrow$



$$\boxed{\text{Bandwidth} = 2 \times \text{message signal B.W}}$$

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* SINGLE TONE DSB:

Assume,

$$m(t) = A_m \cos 2\pi f_m t$$

So,

$$S_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$$

$$= A_c A_m \cos 2\pi f_m t \cdot \cos 2\pi f_c t$$

$$S_{DSB}(t) = \frac{A_c \cdot A_m}{2} \cos 2\pi (f_c + f_m)t + \frac{A_c A_m}{2} \cos 2\pi (f_c - f_m)t$$

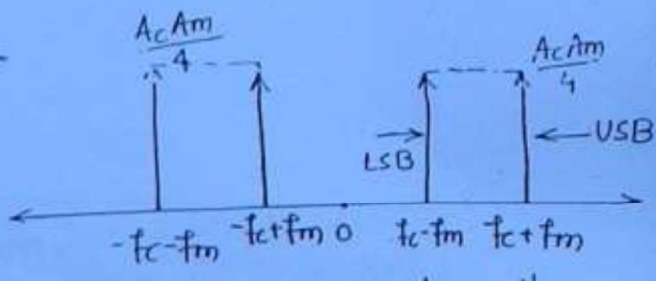
"USB" "LSB"

Frequency spectrum Analysis :-

$m(t)$ ⟷

$= A_c \cos 2\pi f_c t$ ⟷

$S_{DSB}(t)$ ⟷



Bandwidth = 2x message signal frequency

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Note:

let $m(t) = 5\cos 1\times10^3 t$

$c(t) = 10\cos 20\times10^6 t$

Then Peak amp of the DSB signal $= \frac{A_c A_m}{2}$

(same for AM also)

Peak amp of the USB of DSB in spectrum $= \frac{A_c A_m}{4}$

Total Power of DSB-SC signal:

The total power is given as:

$$P_t = P_{SB}$$

$$= P_{USB} + P_{LSB}$$

$$P_{USB} = \left(\frac{A_c A_m}{2}\right)^2 / 2R$$

$$P_{USB} = \frac{A_c^2 A_m^2}{8R}$$

So, $P_t = 2 \cdot \frac{A_c^2 A_m^2}{8R}$

$$\boxed{P_t = \frac{A_c^2 A_m^2}{4R}}$$

Modulation efficiency (η):

As, $\eta = \frac{P_{SB}}{P_t}$

$\eta = \frac{P_t}{P_t}$

$$\boxed{\eta = 1 = 100\%}$$

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Q1. A Carrier of $20\sqrt{2}\cos\pi\times10^5 t$ is DSB modulated by a msg signal of $2\sqrt{2}\cos\pi\times10^3 t$. Find all the parameters and plot the spectrum?

(74)

Soln: Given:

$$m(t) = 2\sqrt{2}\cos\pi\times10^3 t$$

$$A_m = 2\sqrt{2} \; ; \; f_m = \frac{10\times10^9}{2} = 0.5\ KHz$$

$$c(t) = 20\sqrt{2}\cos\pi\times10^5 t$$

$$A_c = 20\sqrt{2} \; ; \; f_c = \frac{10\times10^4}{2} = 50\ KHz$$

Now,

$$P_t = \frac{A_c^2 A_m^2}{4R} = \frac{800\times8^2}{4\times1}$$

$$\boxed{P_t = 1600\ W}$$ {Normalised power, R=1Ω} Ans.

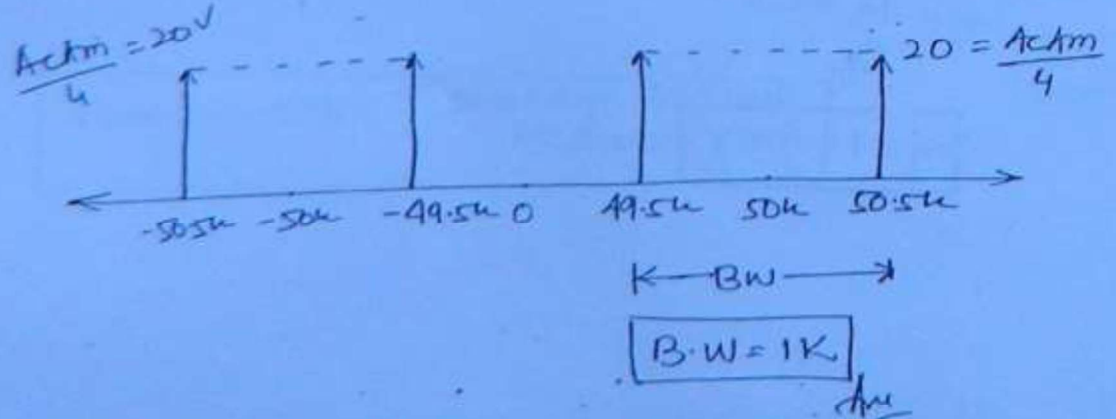
Band width = $2f_m = 2\times0.5K$

$$\boxed{B.W = 1\ KHz}$$ Ans.

2. $$\boxed{P_{USB} = P_{LSB} = \frac{P_t}{2} = 800\ W}$$ Ans.

3. $$\boxed{\eta = 100\%}$$ Ans.

Spectrum:



$$\boxed{B.W = 1K}$$ Ans.

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* MULTI-TONE DSB:

Assume:

$$m(t) = A_{m1}\cos 2\pi f_{m1}t + A_{m2}\cos 2\pi f_{m2}t$$

$$S_{DSB}(t) = A_c m(t)\cos 2\pi f_c t$$

$$= A_c\{A_{m1}\cos 2\pi f_{m1}t + A_{m2}\cos 2\pi f_{m2}t\}\cos 2\pi f_c t$$

$$= A_c A_{m1}\cos 2\pi f_{m1}t\cos 2\pi f_c t + A_c A_{m2}\cos 2\pi f_{m2}t\cos 2\pi f_c t$$

$$S_{DSB}(t) = \frac{A_c A_{m1}}{2}\cos 2\pi(f_c + f_{m1})t + \frac{A_c A_{m1}}{2}\cos 2\pi(f_c - f_{m1})t$$

(USB1, LSB1)

$$+ \frac{A_c A_{m2}}{2}\cos 2\pi(f_c + f_{m2})t + \frac{A_c A_{m2}}{2}\cos 2\pi(f_c - f_{m2})t$$

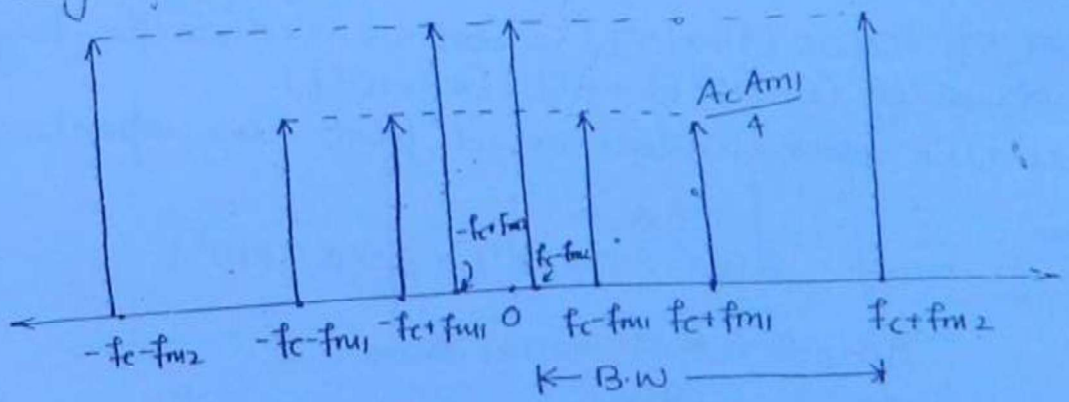
(USB2, LSB2)

* For plotting of spectrum.

let $f_{m2} > f_{m1}$

$A_{m2} > A_{m1}$.

* Frequency Spectrum:



B.W = 2 x Max^m frequency of msg. signal.

$$\boxed{B.W = 2 \times f_{max2}}$$

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* Total Power of Multitone DSB:

As,

$$P_t = P_{SB}$$

where,

$$P_{SB} = P_{USB(total)} + P_{LSB(total)}$$

$$= P_{USB1} + P_{USB2} + P_{LSB1} + P_{LSB2}$$

Now,

$$P_{USB1} = \left(\frac{A_c A_{m1}}{2}\right)^2 / 2R = \frac{A_c^2 A_{m1}^2}{8R} = P_{LSB1}$$

$$P_{USB2} = \left(\frac{A_c A_{m2}}{2}\right)^2 / 2R = \frac{A_c^2 A_{m2}^2}{8R} = P_{LSB2}$$

So,

$$P_t = \frac{A_c^2 A_{m1}^2}{4R} + \frac{A_c^2 A_{m2}^2}{4R}$$

$$\boxed{P_t = \frac{A_c^2 [A_{m1}^2 + A_{m2}^2]}{4R}}$$

Q1. A carrier of $10\cos(4\pi \times 10^6 t)$ is DSB modulated by a msg signal of $6\cos(6\pi \times 10^4 t) + 8\cos(8\pi \times 10^5 t)$. Find all the parameters and plot the spectrum.

Soln: Given:

$$m(t) = 6\cos 6\pi \times 10^4 t + 8\cos 8\pi \times 10^5 t$$

$A_{m1} = 6$; $A_{m2} = 8$

$f_{m1} = 3 \times 10^4$ Hz $= 30$K ; $f_{m2} = \frac{10 \times 10^4}{2} = 50$K

$$c(t) = 10\cos(4\pi \times 10^6 t)$$

$A_c = 10$; $f_c = 2$MHz $= 2000$K

So, $\boxed{B.W = 2.f_{m(max)} = 2 \times 50\text{K} = 100\text{K}}$

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2. $P_t = \frac{A_c^2 A_{m1}^2}{4R} + \frac{A_c^2 A_{m2}^2}{4R}$ (72)

$$P_t = \frac{A_c^2}{4R}\{A_{m1}^2 + A_{m2}^2\}$$

$$= \frac{100}{4}\{36+64\}$$

$$\boxed{P_t = 2500\,W}$$

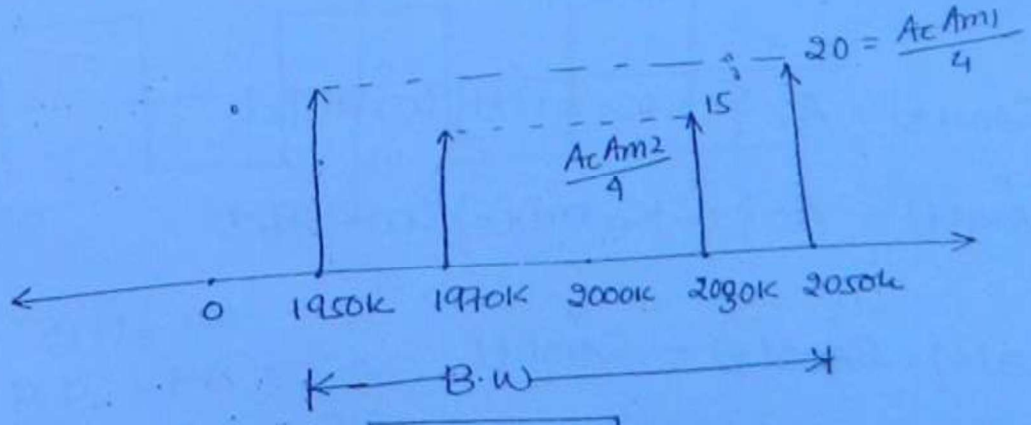
3. $P_{USB\,(total)} = P_{LSB\,(total)} = 1250\,W$

Now, $P_{USB1} = \frac{A_c^2 A_{m1}^2}{8R} = P_{LSB1}$

$$\boxed{P_{USB1} = \frac{\frac{25}{100} \times 36^{18}}{8_4} = 450\,W \quad = P_{LSB1}}$$

$$\boxed{P_{USB2} = 800 = P_{LSB2}}$$

Frequency spectrum:



$$\boxed{B.W = 100K}$$

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* GENERATION OF DSB-SC SIGNAL:

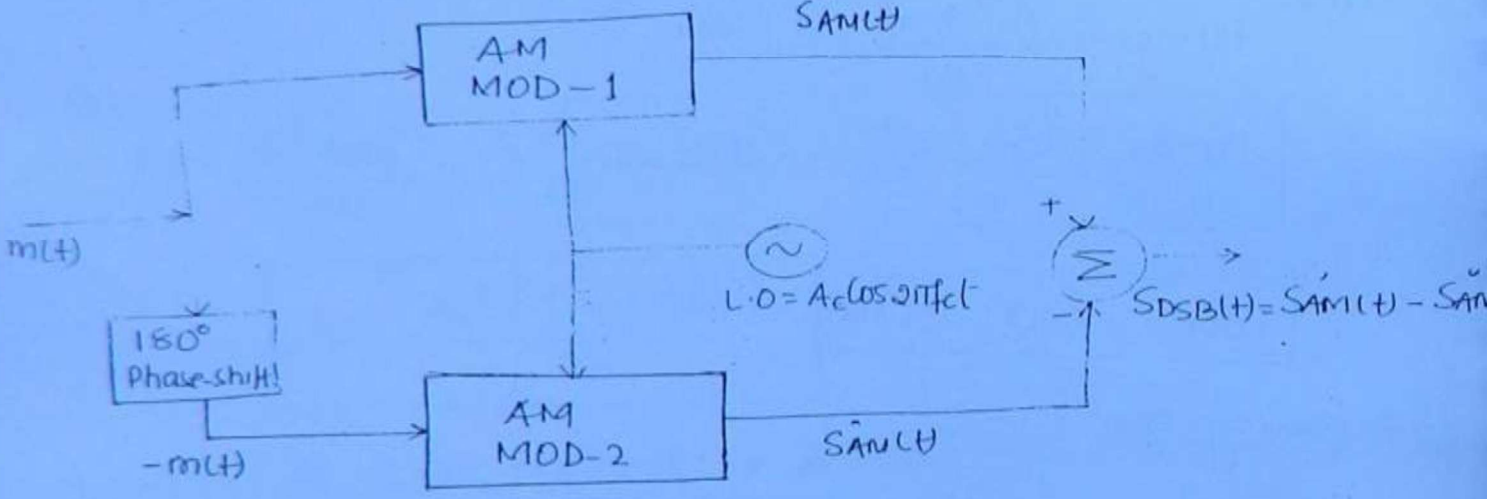
The Generation methods are:

1) Balanced Modulator.
2) Ring modulator.

* BALANCED MODULATOR:

1. In this two AM Modulators are connected in Balanced, to generate DSB signal.

Block diagram:



Now,

$$S'_{AM}(t) = A_c\{1 + K_a m(t)\}\cos 2\pi f_c t$$

&

$$S''_{AM}(t) = A_c\{1 - K_a m(t)\}\cos 2\pi f_c t$$

So,

$$S_{DSB}(t) = S'_{AM}(t) - S''_{AM}(t)$$

$$= 2A_c K_a m(t)\cos 2\pi f_c t$$

$$\boxed{S_{DSB}(t) = A_c' m(t)\cos 2\pi f_c t}$$

where $\boxed{A_c' = 2A_c K_a}$.

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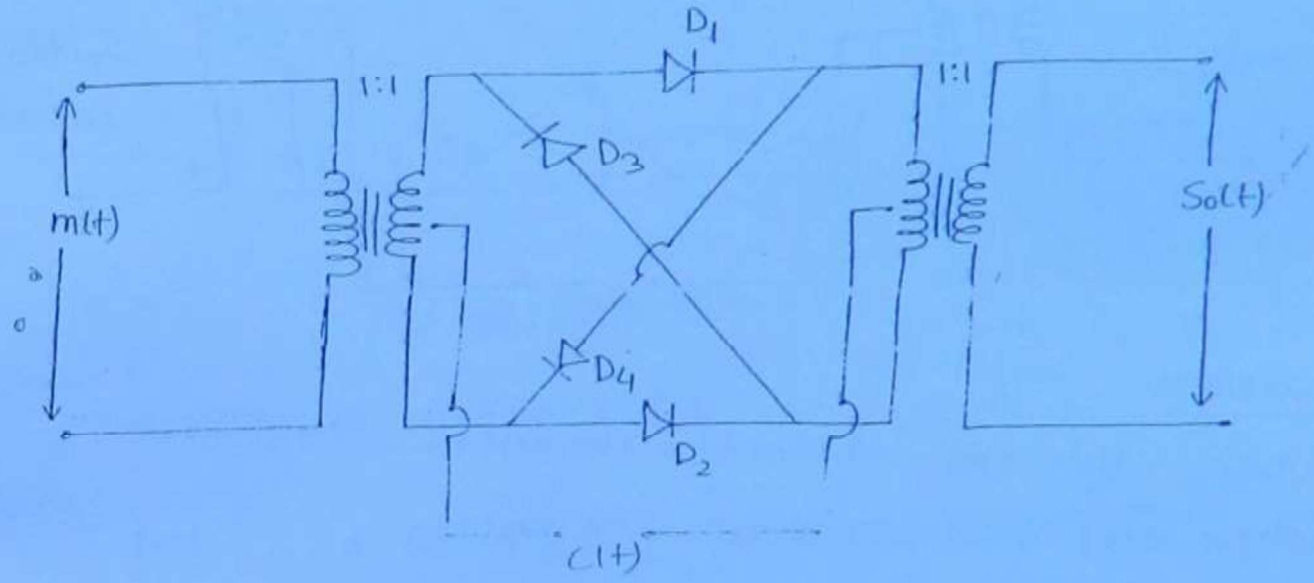
* In this 4 diodes are connected in the form of RING to generate DSB signal

(79)

* Block Diagram:

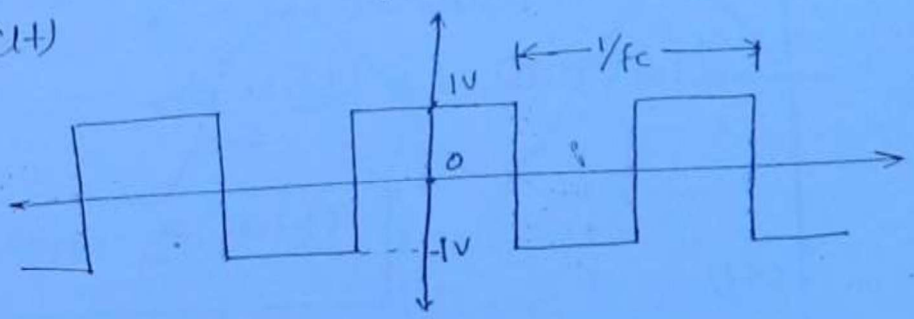


Note:

→ Assume the diode are ideal.

→ xmers are Centre-Tapped and of 1:1 type.

let, C(t)

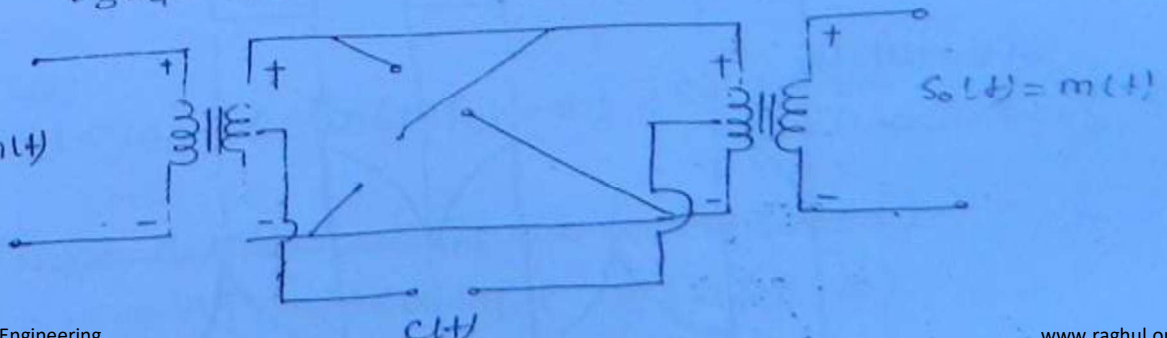


Opn:

1. when C(t) = +ve

$D_1 D_2$ = F.B = S.C

$D_3 D_4$ = R.B = O.C



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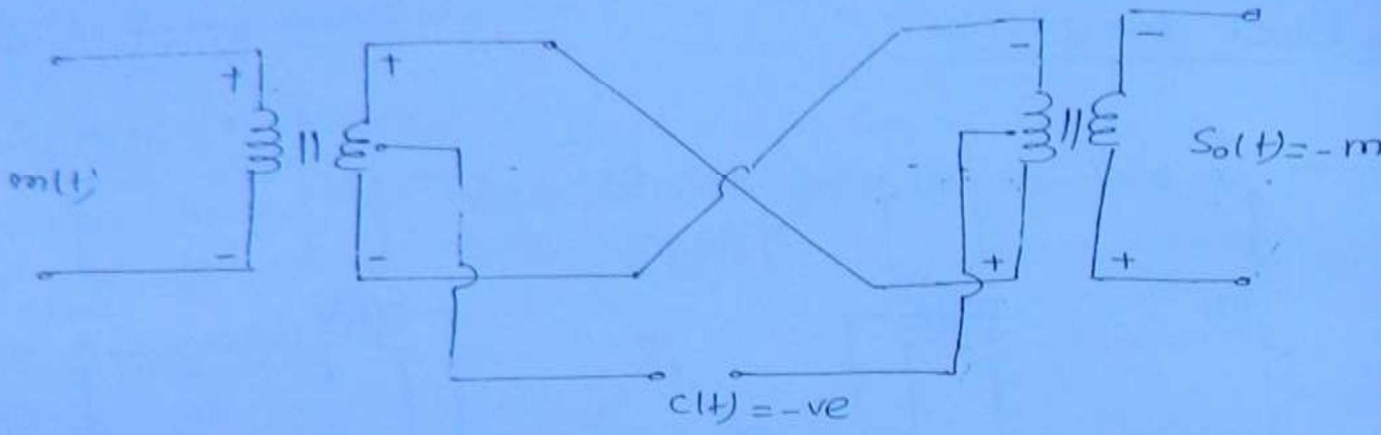
2. c(t) = -ve : then

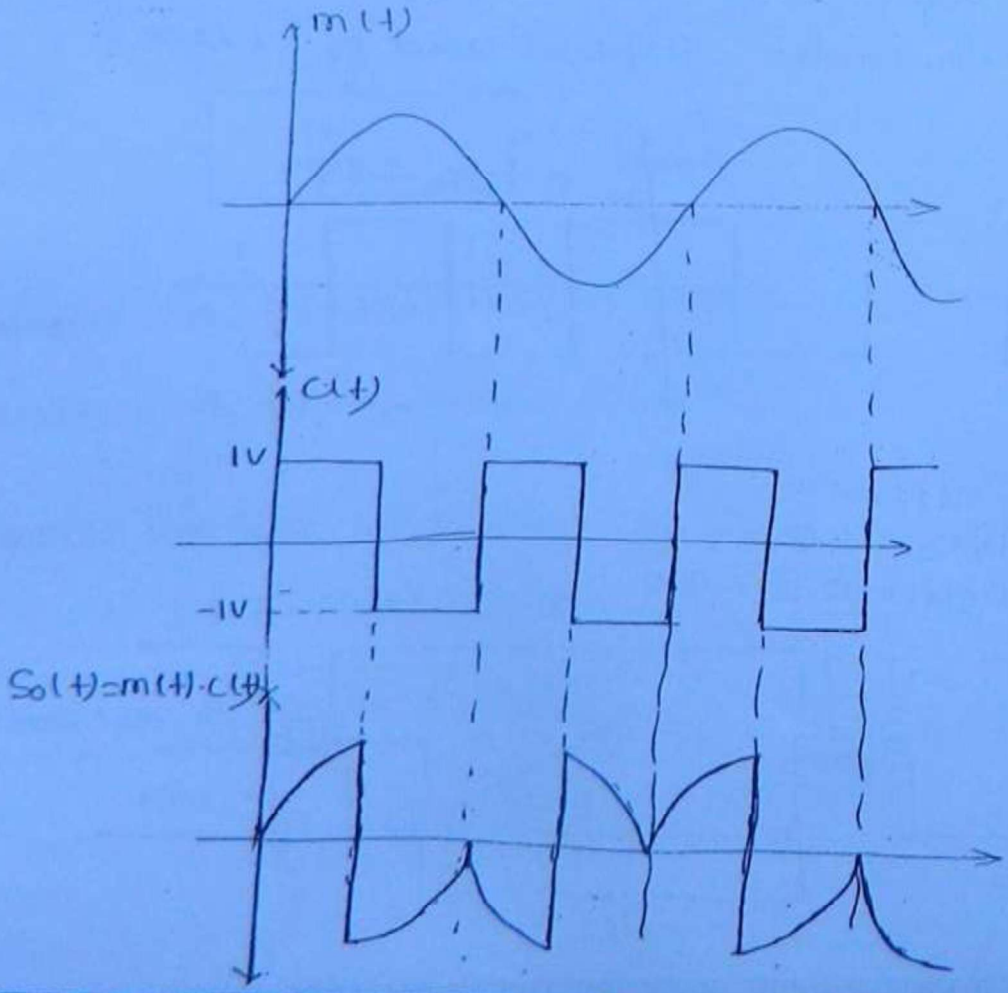
$D_3 D_4$ = FB = S C

$D_1 D_2$ = RB = OFF CKT



Conclusion:

1. when c(t) = +ve $\Rightarrow$ $S_o(t)$ = +ve m(t).
2. when c(t) = -ve $\Rightarrow$ $S_o(t)$ = -ve m(t).



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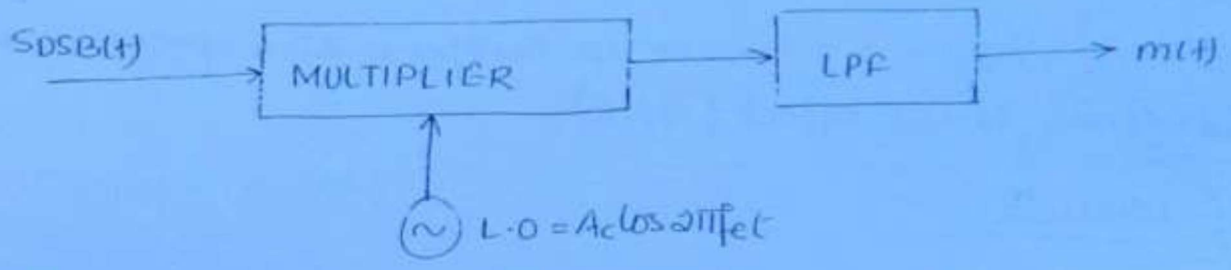
* DEMODULATION OF DSB-SIGNAL :-

* The demodulation of DSB is done by Synchronous detector

* SYNCHRONOUS - DETECTOR :

(81)

B. Diagm



Now:

$$S_{DSB}(t) = A_c\, m(t) \cos 2\pi f_c t$$

Case 1: let $L.O = A_c \cos 2\pi f_c t$ (perfect synchronization)

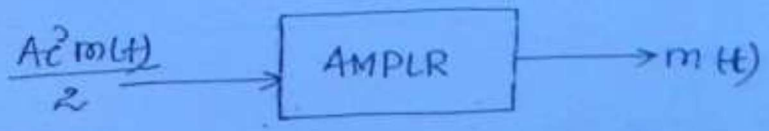
$$(Mul)_{O/P} = S_{DSB}(t) \times (L.O)_{O/P}$$

$$= A_c\, m(t) \cos 2\pi f_c t \times A_c \cos 2\pi f_c t$$

$$= A_c^2\, m(t) \cos^2 2\pi f_c t$$

$$= \frac{A_c^2\, m(t)}{2} \{1 + \cos 4\pi f_c t\}$$

$$(L.P.F)_{O/P} = \frac{A_c^2\, m(t)}{2}$$



Case 2:

let, $(L.O)_{O/P} = A_c \cos(2\pi f_c t + \Phi)$ {No phase synchronisation}

$$(Mul)_{O/P} = A_c\, m(t) \cos 2\pi f_c t \times A_c \cos(2\pi f_c t + \Phi)$$

$$= A_c^2\, m(t) \cos 2\pi f_c t \cdot \cos(2\pi f_c t + \Phi).$$

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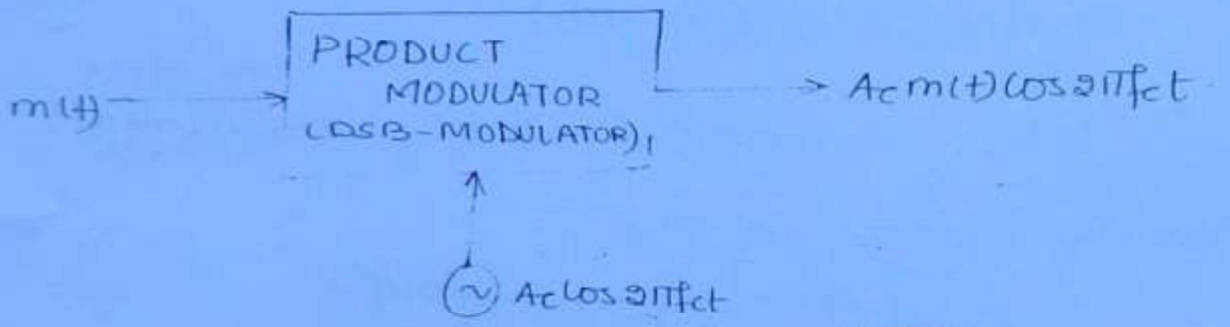
$$(Mul)_{o/p} = \frac{A_c^2 m(t)}{2}\cos(4\pi f_c t + \Phi) + \frac{A_c^2 m(t)}{2}\cos\Phi$$

(82)

$$*(LPF)_{o/p} = \frac{A_c^2 m(t)}{2}\cos\Phi$$

* To maintain the Φ const additional ckt is has to be utilised.
* when $\Phi = 90^\circ$, $o/p = 0$, hence it suffers the problem of Quadrature Null effect (QNE).

* Product Modulator :-



Note :-
* Product modulator is the alternate name given to a DSB modulator.

* ADVANTAGES OF DSB :-

1. Transmitter power will be saved ($\eta\% = 100\%$).
2. used for long distance comm^n.

* DRAWBACKS OF DSB :-

1. Demodulation is complex.
2. It needs high xmission Bandwidth.
3. affected by QNE.

* APPLICATION :-

1. It is used in Quadrature carrier Multiplexing

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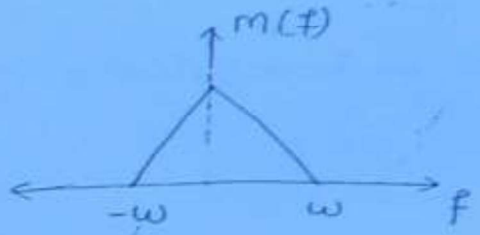
* SINGLE SIDE BAND - SUPPRESSED CARRIER (SSB):-

* The advantage of SSB over AM and DSB is both the xmitter power and B.w will be saved.

let

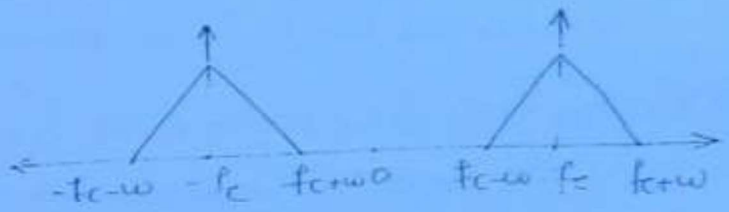
$m(t) \longleftrightarrow$



$c(t) = A_c \cos 2\pi f_c t$

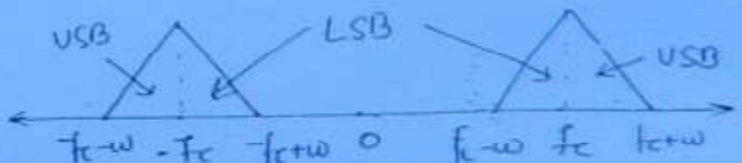
$SAM(t) \longleftrightarrow$

AM B.w = 2w

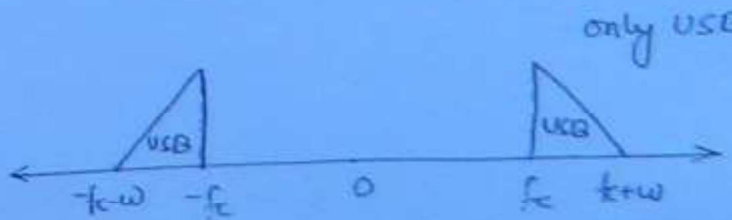


$SDSB(t) \longleftrightarrow$

DSB B.w = 2w



$SSSB(t) \longleftrightarrow$



SSB Bw = w

message signal BW

OR

only LSB.

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Note:

1. The demodulation of SSB is also done by the SD. The SD gives the message signal as O/P whenever the 2 sideBands (as in DSB) are given as I/P. If the SD gives m(t) whenever only 1 sideBand (as in SSB) is given. Hence SSB is preferred over DSB.

2. By doing this:
   1) Power is saved (for xmitting 2 sideBand as in DSB)
   2) Band width is saved

Note:

1. In SSB, Compared to DSB 50% of the transmitter power and 50% of transmission B.W will be saved.

2. The % of power saved in DSB and SSB compared to AM depends on the modulation index (μ).

Eg.

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* SINGLE TONE SSB:

Let

$$m(t) = A_m \cos 2\pi f_m t$$

$$SAM(t) = A_c\{1 + \mu \cos 2\pi f_m t\}\cos 2\pi f_c t$$

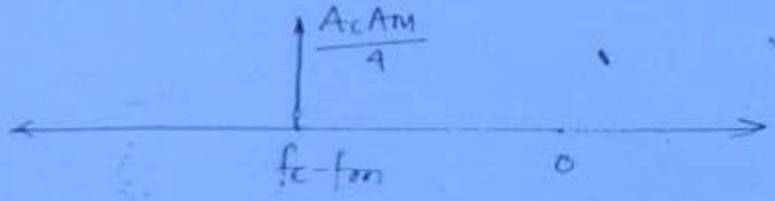
$$SDSB(t) = \frac{A_c A_m}{2}\cos 2\pi (f_c + f_m)t + \frac{A_c A_m}{2}\cos 2\pi (f_c - f_m)t$$

Now, the corresponding SSB expression is given as:-

$$SSSB(t) = \frac{A_c A_m}{2}\cos 2\pi [f_c \pm f_m]t$$

+ → USB
− → LSB.

Spectrum:-



Bandwidth of single tone SSB:-

$$B.W = 0$$

* Total Power:-

$$P_t = P_{SB}$$

$$= P_{USB} \text{ or } P_{LSB}$$

Now,

$$P_{USB} = \left(\frac{A_c A_m}{2}\right)^2 / 2R$$

$$P_{USB} = \frac{A_c^2 A_m^2}{8R}$$

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$$P_t = \frac{A_c^2 A_m^2}{8R}$$

* Modulation efficiency (η):

$$\%\eta = \frac{P_{SB}}{P_t} = \frac{P_{SB}}{P_{SB}} \times 100$$

$$\eta = 100\%$$

* General expression of SSB:-

The General expression of AM is:

$$S_{AM}(t) = A_c\{1 + K a m(t)\}\cos 2\pi f_c t$$

& for DSB is:

$$S_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$$

The General expression of SSB is given as:

$$S_{SSB}(t) = \frac{A_c A_m}{2} \cos 2\pi (f_c \mp f_m) t$$

$+ \rightarrow$ USB
$- \rightarrow$ LSB

$$S_{SSB}(t) = \frac{A_c A_m}{2} \cos 2\pi f_c t \cos 2\pi f_m t \mp \frac{A_c A_m}{2} \sin 2\pi f_c t \sin 2\pi f_m t$$

$- \rightarrow$ USB
$+ \rightarrow$ LSB

we have;

$$m(t) = A_m \cos 2\pi f_m t$$

and 90° phase shift of $m(t)$ we get $\{\hat{m}(t) = 90°$ phase shift of $m(t)\}$

$$\hat{m}(t) = A_m \sin 2\pi f_m t$$

where, $\hat{m}(t)$ = HILBERT xmation of $m(t)$

So, substituting in the above equation we get:

$$S_{SSB}(t) = \frac{A_c m(t)}{2} \cos 2\pi f_c t \mp \frac{A_c \hat{m}(t)}{2} \sin 2\pi f_c t$$

$- \rightarrow$ USB
$+ \rightarrow$ LSB

← single tone SSB signal

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% of power saved in DSB and SSB V/s AM

If, $\mu = 0.707$

Then, after modulation, for AM

$P_t = 100W$

$P_c = 80W$ $P_{SB} = 20W$

(88)

For AM:

P_t	P_c	P_{USB}	P_{LSB}
100W	80W	10W	10W

For DSB:

20W	–	10W	10W

For SSB:

10W	–	10W or 10W

Note:

1. In SSB as compared to AM, 90% of transmitter power is saved;
2. In DSB as compared to AM, 80% of transmitter power is saved.
3. In SSB as compared to DSB: 50% of xmitter power is saved.

Note:

1. The amount of power saved in DSB & SSB compared to AM depends on μ.
2. Whatever be the value of μ the amount of power saved in SSB compared to DSB is always 50%.

$$P_t = P_c \left\{1 + \frac{\mu^2}{2}\right\}$$

$$P_{SB} = \frac{P_c \mu^2}{2} \; ; \; P_{USB} = P_{LSB} = \frac{P_c \mu^2}{4}$$

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Now;

% of power saved in DSB compared to AM = Power saved / total power

$= \frac{P_c}{P_t} = \frac{P_c}{(1+\frac{\mu^2}{2})P_c}$

***

% of power saved in DSB V/S AM $= \frac{2}{2+\mu^2} = 1-\eta = 1-\frac{P_{SB}}{P_t}$

% of power saved in SSB compared to AM = Power saved / total power

$= \frac{P_c + \frac{P_c\mu^2}{4}}{P_c\{1+\frac{\mu^2}{2}\}}$

***

% of power saved in SSB V/S AM $= \frac{4+\mu^2}{4+2\mu^2}$

Q1. An AM transmitter power is given by 500W. Find the amount of power saved if carrier and one of the SB is suppressed with i) $\mu = 0.5$? ii) $\mu = 0.8$

Soln: Given:

$P_t = 500W$

for $\mu = 0$; $P_t = P_c = 500W$

So, $P_t = P_c\{1+\frac{\mu^2}{2}\}$

$= 500\{1+\frac{0.25}{2}\}$

$= 501.125$

So, % of power saved in SSB $= \frac{4+\mu^2}{4+2\mu^2}$

i) $\mu = 0.5$

% of power saved = 94.5%

Amount of power saved = 94.5% of 500

= 472.5 W Ans

ii) $\mu = 0.8$

% of power saved = 0.878 × 100

Amount of power saved = 87.8% of 500

= 439W Ans

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*Generation of SSB:

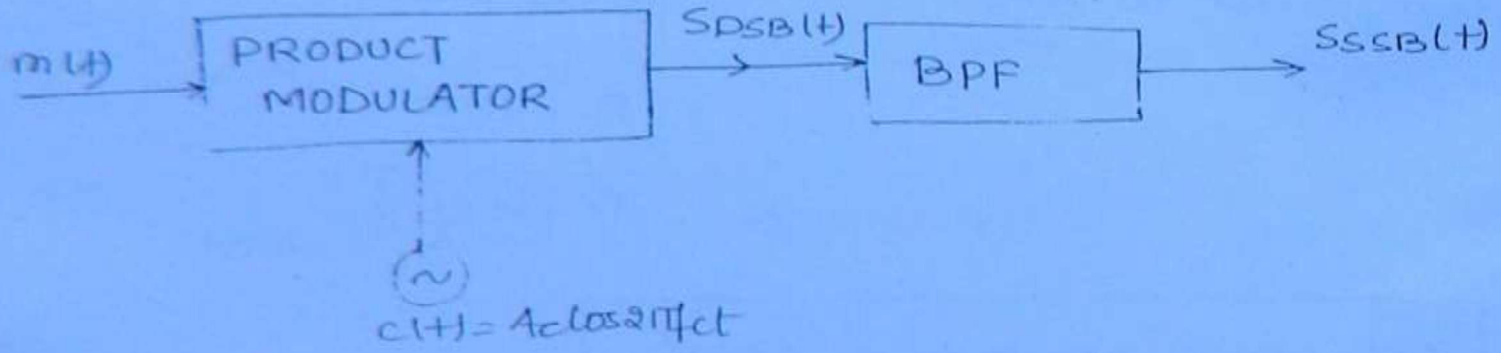
The Generation methods of SSB are:

i) Frequency discrimination method
ii) phase discrimination method

*FREQUENCY DISCRIMINATION METHOD:

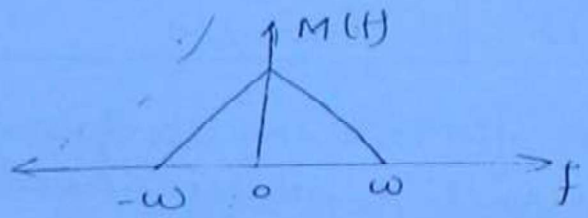
*In this DSB signal is transmitted through proper Band pass filter to generate SSB.

*Block Diagram:



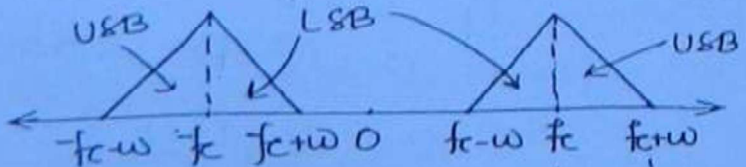
let

m(t) ⟷

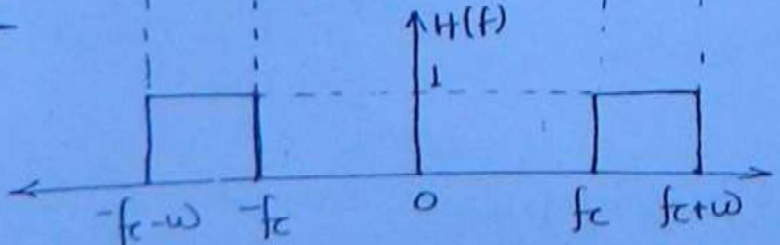


c(t) = Ac cos 2πfct

SDSB(t) ⟷



(BPF) ⟷



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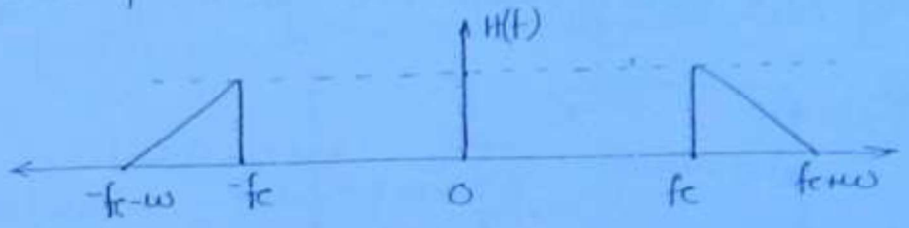
So the BPF; O/P is given as

$$y(f) = H(f)\,x(f)$$

(91)

$(BPF)_{O/P}$ = SSB spectrum



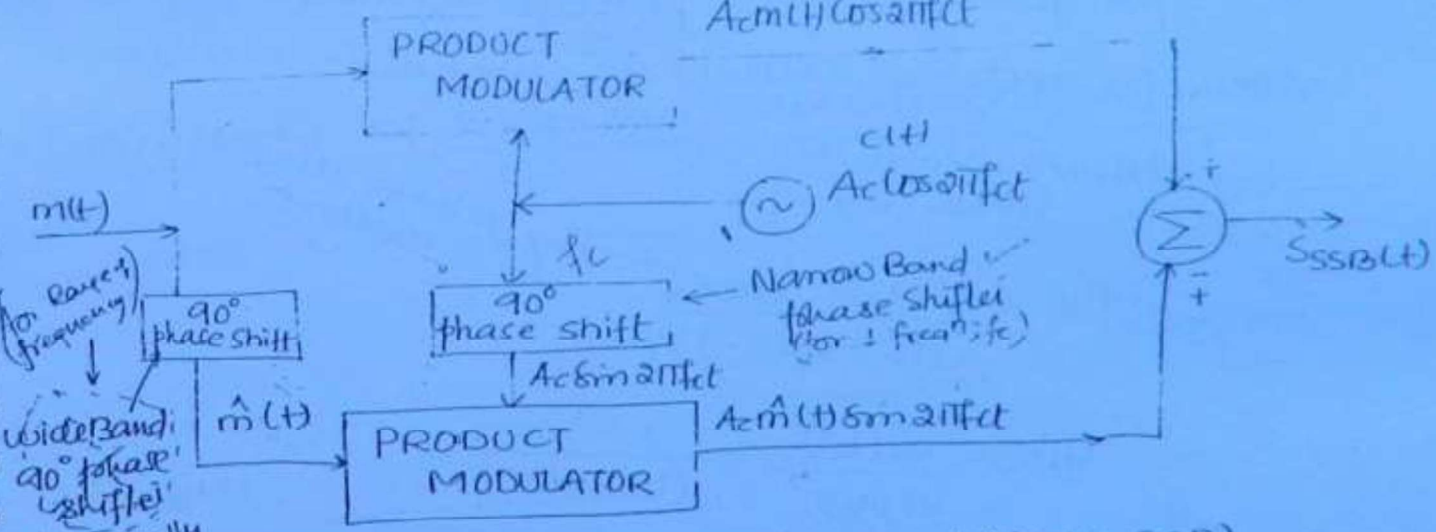
× PHASE DISCRIMINATION METHOD:

The General expression of SSB signal is given as:

$$S_{SSB}(t) = \frac{A_c m(t)\cos 2\pi f_c t}{2} \mp \frac{A_c \hat{m}(t)\sin 2\pi f_c t}{2}$$

(USB / LSB)

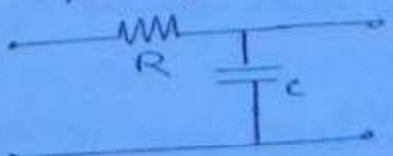
Block diagram:



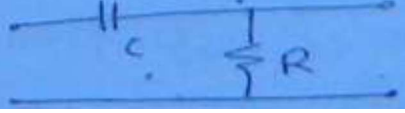
SSB Generation (for only multitone SSB)

Note:

1. Phase shift n/w is nothing but Integrator or Differentiator
2. Integrator → low pass filter with high RC



3. Differentiator →



→ High pass filter with low value of RC

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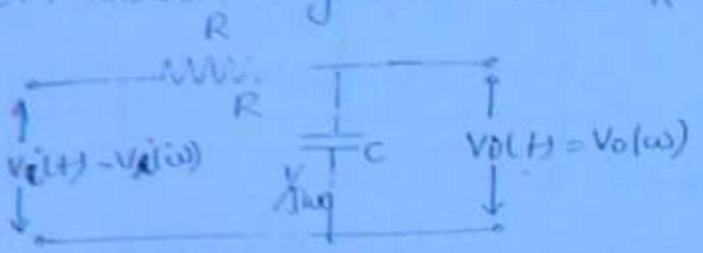
1. Generally for Realization of 90° phase shifter differentiator & Integrator will be used as

2. The following LPF with high RC works as Integrator.



So, $V_o(\omega) = V_i(\omega) \cdot H(\omega)$

$|V_o(\omega)| = |V_i(\omega)|\,|H(\omega)|$ ← Magnitude

$\angle V_o(\omega) = \angle V_i(\omega) + \angle H(\omega)$ ← phase

For getting phase shifted by 90°, $\angle H(\omega) = 90°$

Now, for getting $\angle H(\omega) = 90°$, we have:

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)}$$

← By voltage division method

$$H(\omega) = \frac{V_i(\omega) \cdot 1/j\omega C}{(R + 1/j\omega C)\,V_i(\omega)}$$

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = \frac{1}{(1 + j\omega RC)}$$

So, $\angle H(\omega) = 0 - \tan^{-1}\left(\frac{\omega RC}{1}\right)$

$$\boxed{\angle H(\omega) = -\tan^{-1}(\omega RC)}$$

→ To obtain $\angle H(\omega)$ the value of $\omega RC \approx \infty$, but practically it is not possible.

→ If the $v_i(t)$ be multitone ie contain range of frequency. Hence to obtain 90° phase shift for Range of frequency we need to vary R & C with frequency, which can't be implemented.

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Note

* The practical significance of Single tone SSB generation ckt is not there; as for transmission, multi tone SSB takes place. Hence there is no practical significance of Single tone SSB / Phase discrimination method

Note :-

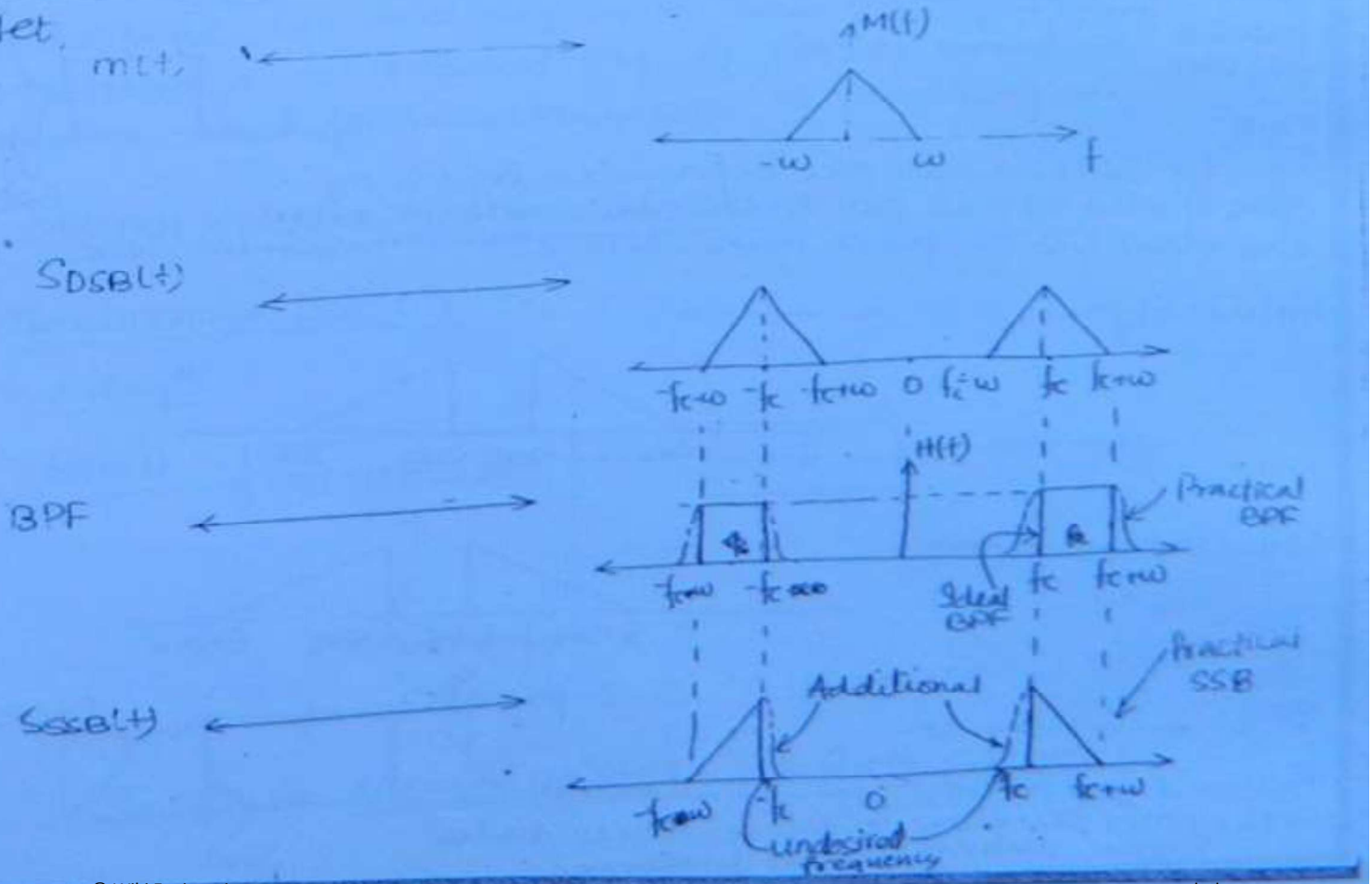
$$\angle H(j\omega) = -\tan^{-1}(\omega RC)$$

from above: diff. frequency components of Input are experiencing diff phase shifts. So, it is practically not possible to construct wide Band 90° phase shifter. So the above method is failed for the generation of Multi tone SSB signal

Note :

For Generation of Multitone SSB frequency discrimination method will be used.

* Drawbacks of Frequency discrimination method:



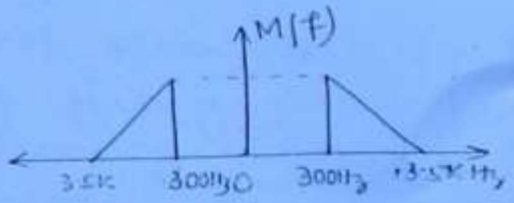
Note:

1. Since Ideal BPF cannot be constructed, the resulting SSB signal contains undesired frequencies in addition to actual side Bands.

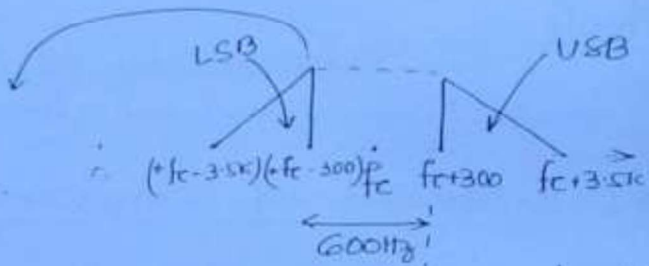
2. Because of above drawback SSB is limited only for voice signal transmission.

Analysis:

1. Voice Signal: $m(t)$ ⟷

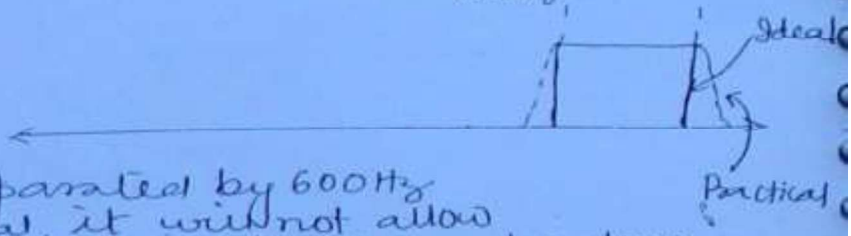


SDSB(t) ⟷



(BPF) ⟷

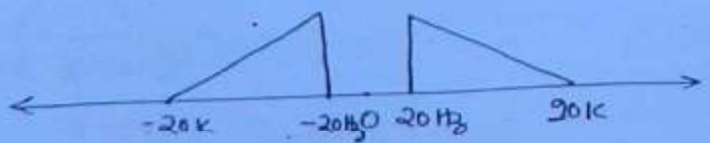
passing only USB



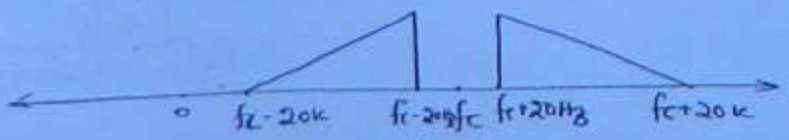
Note:

The SideBands are separated by 600Hz. Also, if the BPF is not ideal, it will not allow the other SB to pass. Hence the xmission can be done

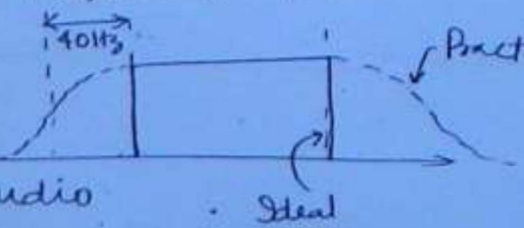
2. Audio Signal; $m(t)$ ⟷



SDSB(t) ⟷



BPF) ⟷



Band gap is 40Hz. So, the other SB may interfere. Hence Audio signal xmission is not preferred.

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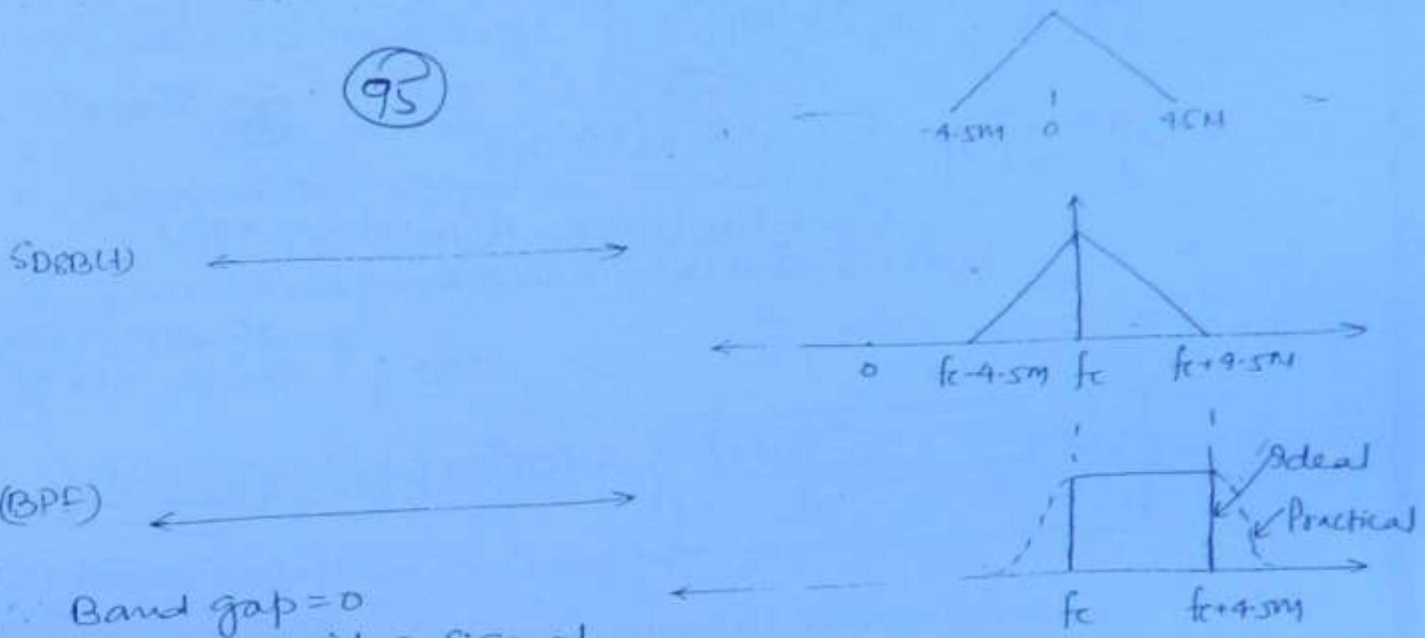


Band gap = 0
Hence the video signal can't be xmitted thro' SSB scheme as other SB may interfere.

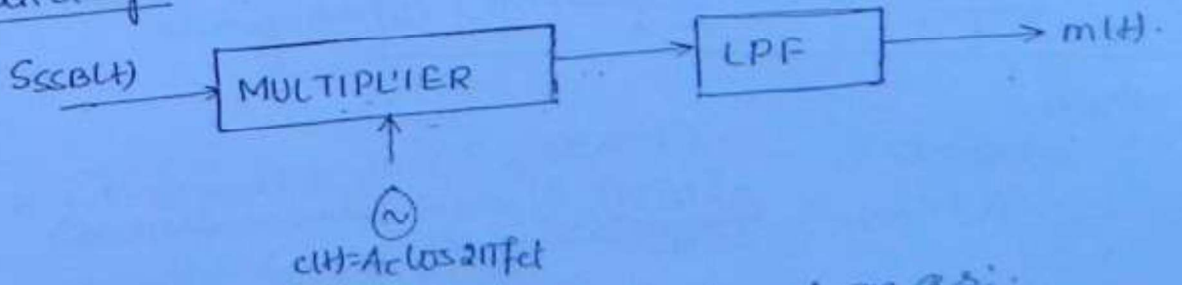
Conclusion:

1. For transmission of a message signal, in the Respective DSB spectrum, a wide Band gap should be existing blw side Bands; For voice signal in the corresponding DSB spectrum a wide Band gap of 600Hz exist so SSB can be used for transmission of voice signals.

* DEMODULATION OF SSB SIGNAL :-

* Synchronous Detector:

Block diagm:



General expression of SSB is given as:

$$S_{SSB}(t) = \frac{A_c m(t)}{2} \cos 2\pi f_c t - \frac{A_c \hat{m}(t)}{2} \sin 2\pi f_c t$$

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Case 1:

i) $(LO)_{O/P} = A_c \cos 2\pi f_c t$ {Perfect Synchronisation}

ii) $(Mul)_{O/P} = S_{SSB}(t) \times (LO)_{O/P}$

$$= \frac{A_c^2 m(t)}{2} \cos^2 2\pi f_c t \mp \frac{A_c^2 \hat{m}(t)}{2} \frac{\sin 4\pi f_c t}{2}$$

{$\because 2\sin\theta\cos\theta = \sin 2\theta$}

iii) $\boxed{(LPF)_{O/P} = \frac{A_c^2 m(t)}{4}}$ {$\because \cos^2 2\pi f_c t = \frac{1 - \cos 4\pi f_c t}{2}$}.

Case 2:

let,

i) $(LO)_{O/P} = A_c \cos(2\pi f_c t + \Phi)$ {NO phase synchronisation}.

ii) $(Mul)_{O/P} = S_{SSB}(t) \cdot (LO)_{O/P}$

$$= \left\{\frac{A_c^2 m(t)}{2} \cos 2\pi f_c t \mp \frac{A_c \hat{m}(t)}{2} \sin 2\pi f_c t\right\} \{A_c \cos(2\pi f_c t + \Phi)$$

$$= \frac{A_c^2 m(t)}{4} \cos(4\pi f_c t + \Phi) + \frac{A_c^2 m(t)}{4} \cos\Phi$$

$$\mp \frac{A_c^2 \hat{m}(t)}{4} \sin(4\pi f_c t + \Phi) \pm \frac{A_c^2 \hat{m}(t)}{4} \sin\Phi.$$

iii) $\boxed{(LPF)_{O/P} = \frac{A_c^2 m(t)}{4} \cos\Phi \pm \frac{A_c^2 \hat{m}(t)}{4} \sin\Phi.}$

when,

a) $\Phi = 0$

$$(LPF)_{O/P} = \frac{A_c^2 m(t)}{4}$$

b) $\Phi = 90^\circ$

$$(LPF)_{O/P} = \pm \frac{A_c^2 \hat{m}(t)}{4}$$ {$\because$ O/P $\neq 0$; hence SSB is not affected by QNE}.

NO QNE

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Note:

* Demodulation of SSB is not affected by QNE

* Advantages of SSB: (97)

1) Transmitter power is saved (½ of η = 100%).
2) Transmission Bandwidth is saved.
3) NO affect of QNE.

* Drawbacks of SSB:

1) Demodulation is complex.
2) Limited only for Voice signal transmission.

* APPLICATION:

1) Preferred in Voice signal transmission.

Note:

* Earlier SSB was used for Voice signal xmission in local telephone comm^{n}.

* VESTIGIAL SIDEBAND MOD^{N} (VSB):-

Video Signal = 0 – 4.5 MHz
B.W = 4.5 MHz

Hence, using AM or DSB modulation = 2 × B.W
= 9 MHz

* T.V Signal = 10 MHz (each)
(Including Audio Signal)
* Co-axial Cable B.W = 600 MHz
(Broadly used)
→ 60 TV channels can be xmitted.

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let, SSB modulation scheme is used.

Hence, S.S.B. BW = 4.5MHz

Total B.W TV Signal ≈ 5MHz
(Including audio signal)

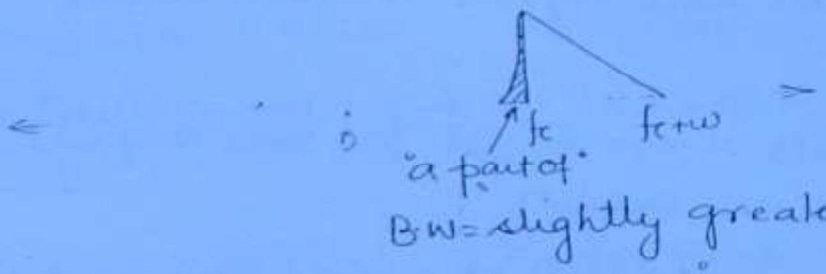
⇒ 120 TV channel

∵ SSB not used for Video Signal
Hence it fails. though it has advantage over AM or DSB (In B.W & power)

Note:-

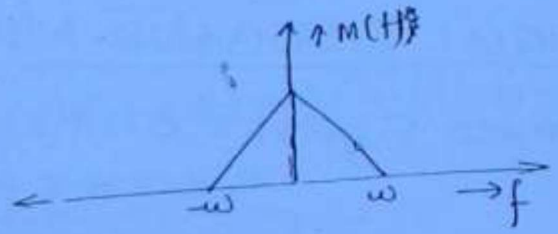
1. VSB provides almost of same Bandwidth as SSB and can be used for the xmission of Video signals

∴ 3 steps

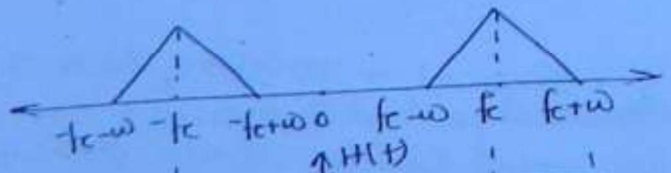


let, SSB

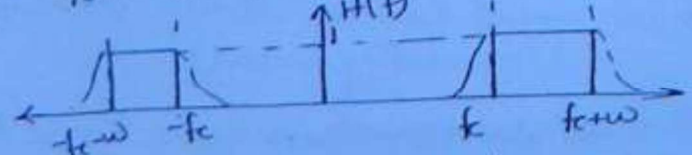
m(t) ⟷



SDSB(t) ⟷



BPF) ⟷



SSSB(t).

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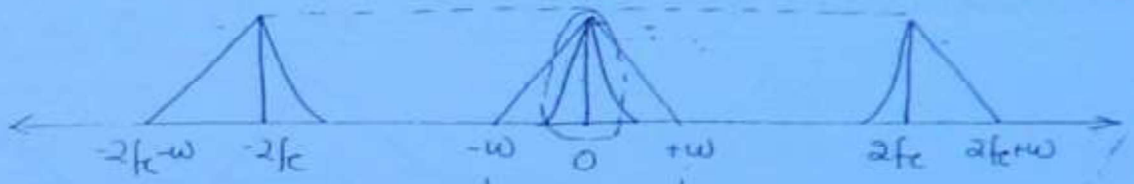
In the Recei (synchronous detector)
(Mul) o/p spectrum is given as

$$s(t)\cos 2\pi f_c t = \frac{S(f+f_c) + S(f-f_c)}{2}$$

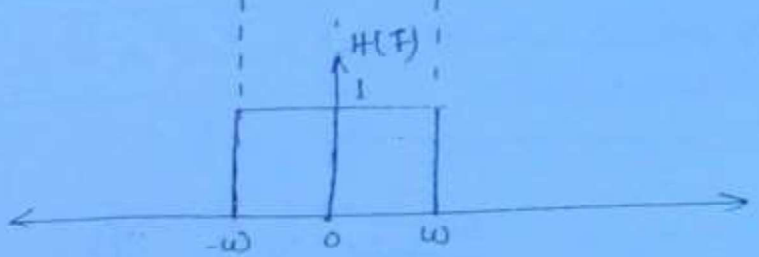
(99)

(SSB(+) shifted left & Right by f_c)

(Mul) o/p ←→

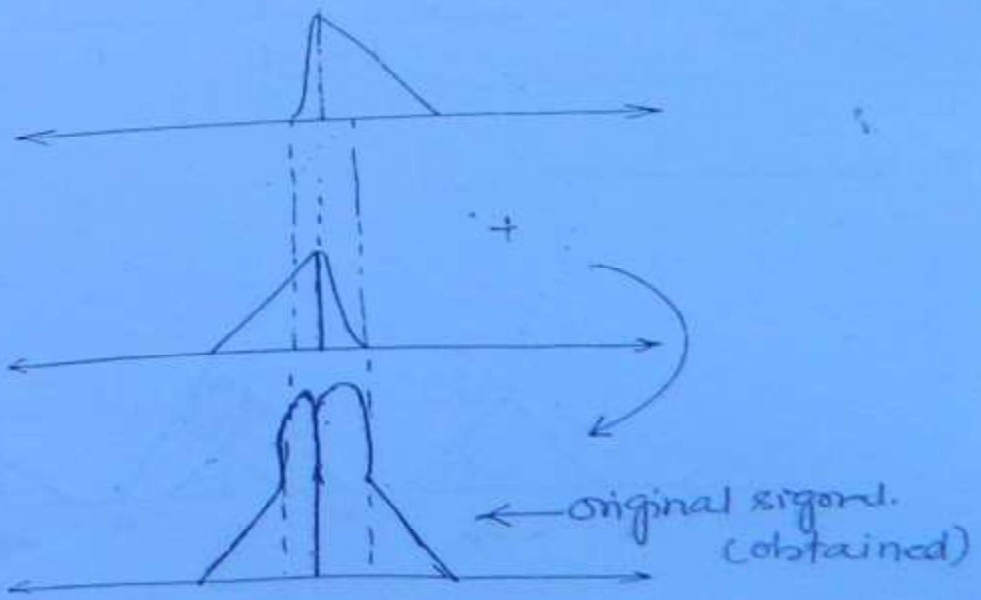


(LPF) ←→



Note:

* The Reconstructed message signal, the low frequencies near to origin will be interferenced by undesired frequencies so that message signal cannot be perfectly Reconstructed.



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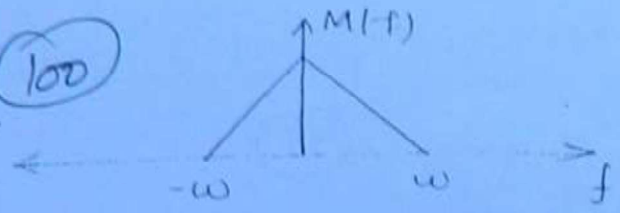
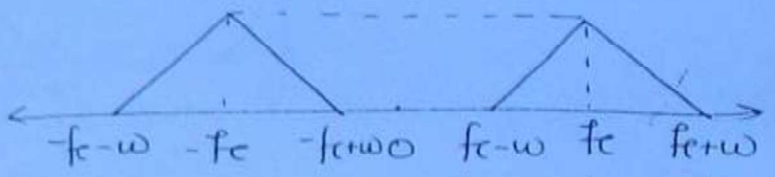
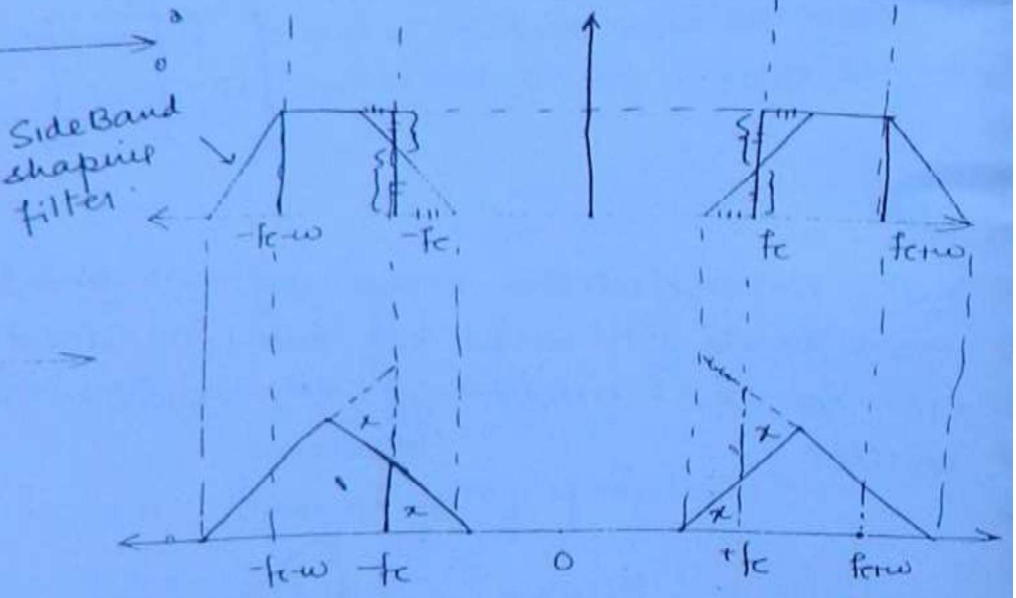
100



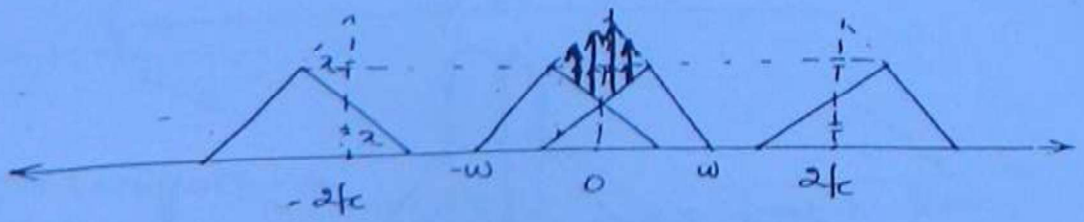
SDSB(t) ⟷



(BPF) ⟷



(Mul) o/p ⟷



Conclusion:

1. The ~~above~~ DSB signal is passed through sideband shaping filter to generate VSB.

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2 H(f) of SideBand shaping filter should be symmetrical about fc. (101)

3 For demodulation of VSB, synchronous Detector will be used.

Analysis of BW & Power with other Modulation schemes:

Bandwidth:-

AM & DSB > VSB > SSB ← Comparision of Bandwidth.

Powers:-

AM > DSB > VSB & SSB ← comparision of power

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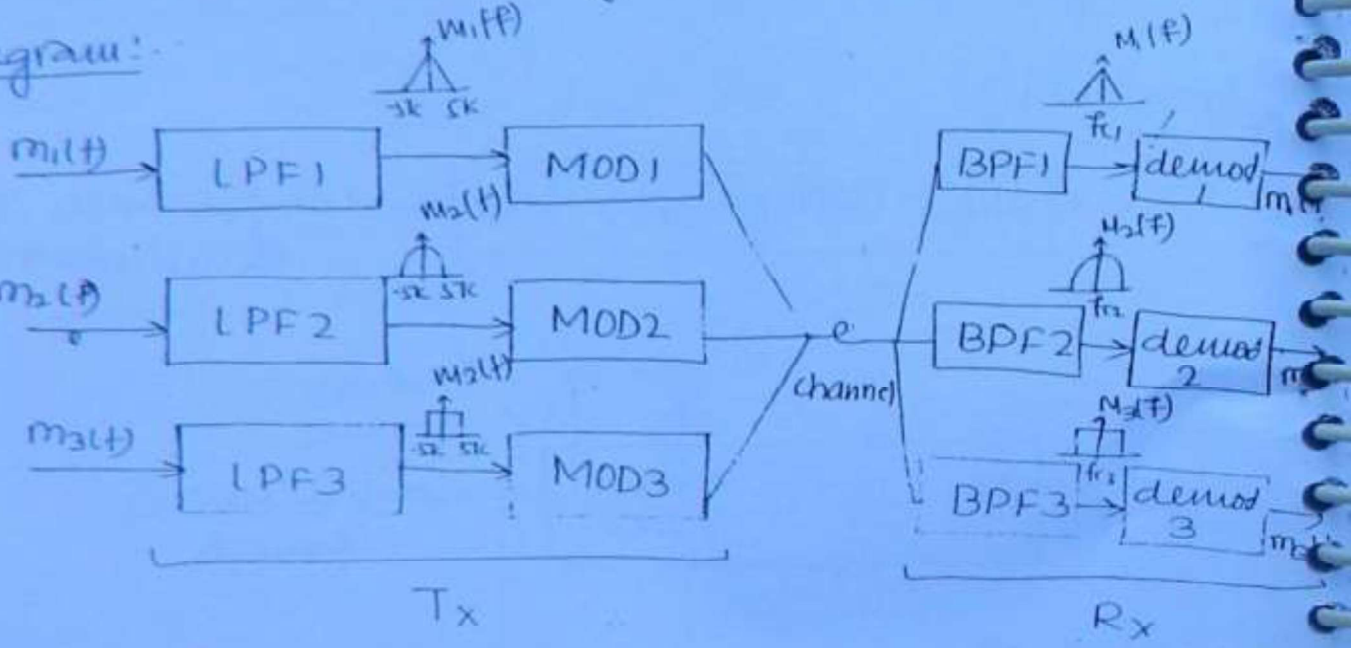
*FREQUENCY DIVISION MULTIPLEXING (FDM):

→used to multiplex continuous signal

*FDM is used for multiplexing continuous signals.

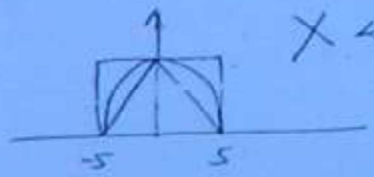
Block diagram:-



*LPF are used to Band limit the signals

Case 1 (no modulators):-

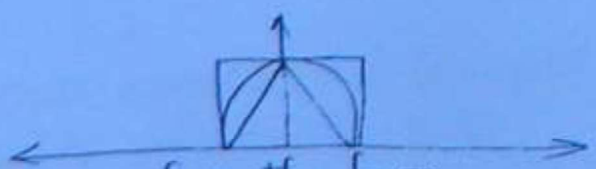
∵ All the messages are occupying same Band of frequencies. So, they all will get interfered with each other.



Case 2 (Same carrier frequency):-

let same carrier frequency, $f_c = A_c \cos 2\pi f_c t$

So,



← Here also the signal occupy same Band of frequency Hence interference is said to occur

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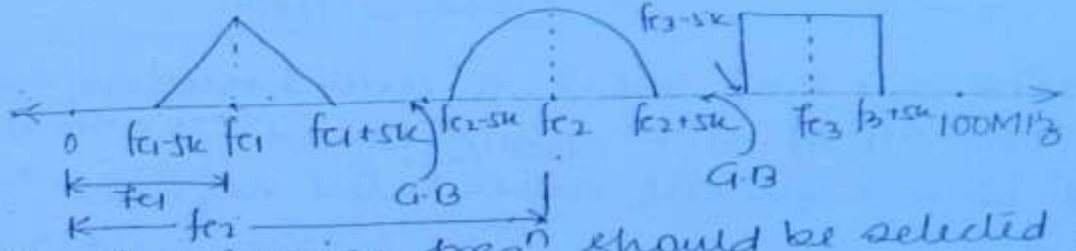
Case 3 (diff. Carrier frequency) :-

Let the channel B.W be 100MHz (0 to 100MHz)

The limitation is that the carrier frequency should be less than 100MHz

(103)



2. Also the carrier freqⁿ should be selected such that there exist some spacing b/w spectrums. This spacing is called as "GAURD BAND".

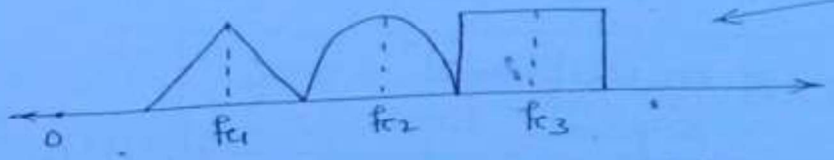
So, to avoid interference,

$$f_{c2} >> f_{c1} + 10K$$

$$f_{c3} >> f_{c2} + 10K$$

Case 4: $f_{c2} = f_{c1} + 10K$; $f_{c3} + f_{c2} + 10K$

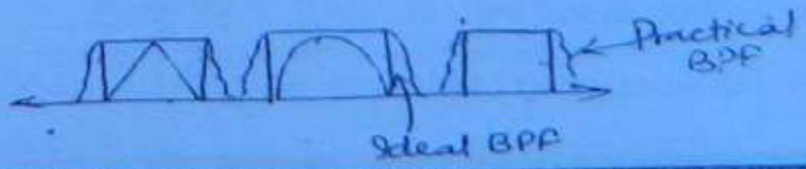
~~$f_{c2} = f_{c1} + 10K$~~



→ B.W is efficiently used, but practically not used as no Gaurd Band exists. Hence in the Rx section the part of the other msg signal will be extracted.

Note:

1. When the modulated signal as in case 3 is given to the BPF (practical) the message signal can be extracted efficiently



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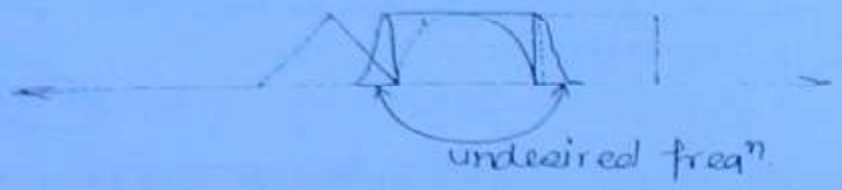
2. Whereas in case 1, when no G.B exists then.



Note :-

Since BPF are not ideal, to avoid undesired frequencies at the O/P of BPF; Gaurd Band has to be maintained b/w adjacent modulated signals.

Q1) Three msg signals each Band limited to 5KHz are multiplexed using FDM. Gaurd Band is 1 KHz. Find multiplexed signal BW, if the modulation schemes used are AM, DSB & SSB respt.

Soln: For 1st msg signal: 2nd msg signal 3rd

So, B.W of multiplexed signal = 27K

Q2. 10 msg signal, each Band limited to 10K are multiplexed using FDM. Gaurd Band is 0.5K. Find multiplexed signal B.W if modulation used is

a) AM b) DSB c) SSB

Soln: For AM

BW = 2W = 20K

So, B.W of 10 msg = 200K

So, B.W of multiplexed signal = 200K + 9x0.5K

= 204.5K (Same for DSB)

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For SSB:

B.W = 10K

So, B.W of 10 msg = 100K (05)

So, **B.W of multiplexed signal = 100 + 9x0.5 = 104.5K** Ans

Previous exam Questions:

01. An AM signal is given by

$$s(t) = \{1 + m(t)\}\cos\omega_c t$$

where, $m(t) = \frac{1}{2}\cos\omega m_1 t + \frac{1}{2}\sin\omega m_2 t$

Find modulation efficiency?

Soln: Given,

$$s(t) = \{1 + m(t)\}\cos\omega_c t$$

$$= \{1 + \frac{1}{2}\cos\omega m_1 t + \frac{1}{2}\sin\omega m_2 t\}\cos\omega_c t$$

Comparing with standard multitone AM:

$$s(t) = A_c\{1 + \mu_1\cos\omega m_1 t + \mu_2\cos\omega m_2 t\}\cos\omega_c t$$

for $A_{m_1}\cos\omega m_1 t + A_{m_2}\cos\omega m_2 t = m(t)$.

So, for if $m(t) = A_{m_1}\cos\omega m_1 t + A_{m_2}\sin\omega m_1 t$

then

$$s(t) = A_c\{1 + \mu_1\cos\omega m_1 t + \mu_2\sin\omega m_2 t\}$$

So, $\mu_1 = 1/2$; $\mu_2 = 0.5$

$$\mu_t = \sqrt{0.5^2 + 0.5^2} = 1/\sqrt{2}$$

So, $\eta = \frac{\mu_t^2}{2 + \mu_t^2} = \frac{1/2}{1 + 1/2}$

$\eta = 20\%$ Ans

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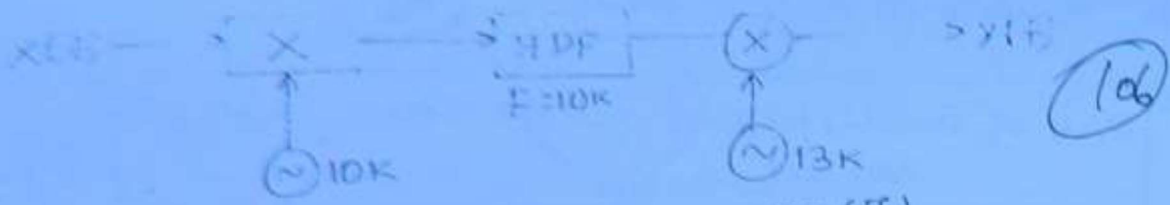
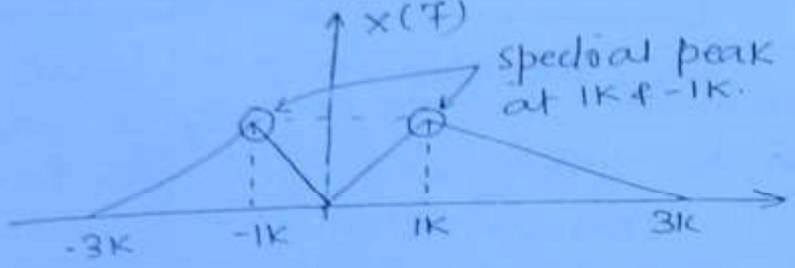
8) For the following ckt, find the +ve frequencies for which special peaks will be observed in y(t)

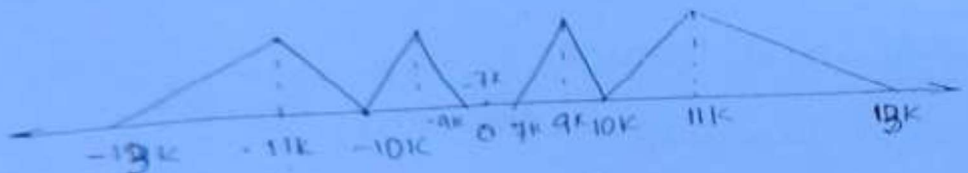


(10)

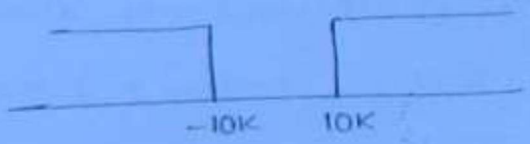
a) 2K, 24K
b) 1K, 22K
c) 1K, 2K, 24K
d) None



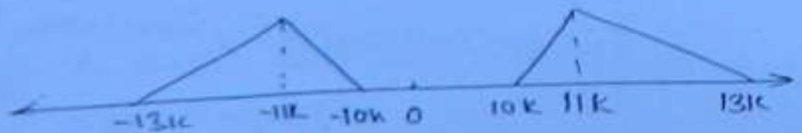
Soln: At o/p of multiplier ckt



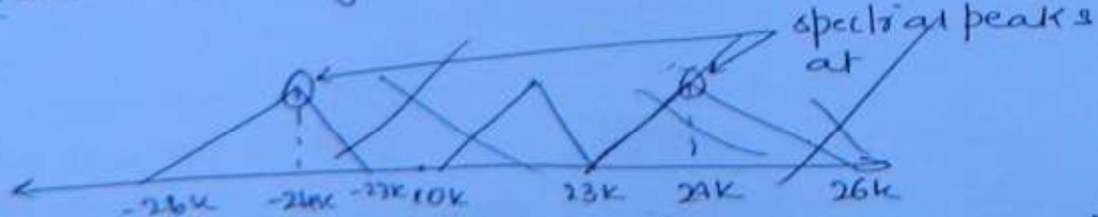
Spectrum of HPF:



So, O/P of HPF



At o/p of multiplier 2nd:
spectrum shifted by 13K



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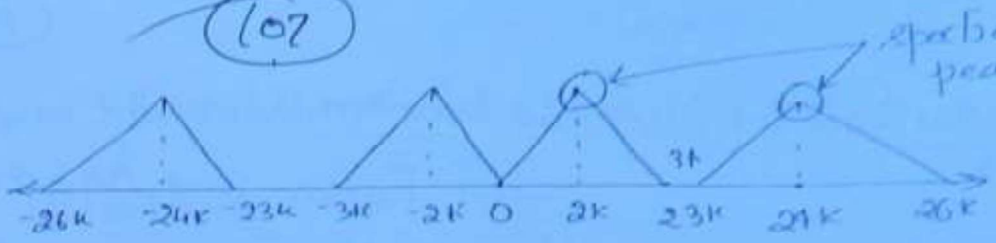
Multiplier 2nd diff → spectrum shifted by fc

(07)



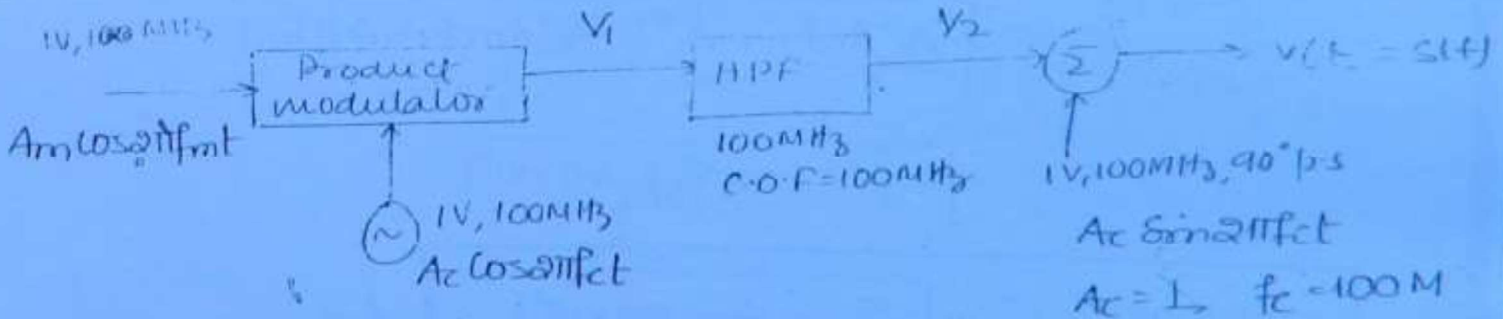
Q3. A sinusoidal carrier of 1V, 100MHz, is product modulated by a sinusoidal msg signal of 1V, 1MHz. The resulting signal is passed through HPF of fcutoff = 100MHz. Filtered o/p is added with sinusoidal signal of 1V, 100MHz, 90° phase shift. Find the envelope of o/p?

a) $\sqrt{\frac{5}{4} - \sin 2\pi \times 10^6 t}$ b) $\sqrt{1 + 2\cos 2\pi \times 10^6 t}$

c) $\sqrt{2 - \sin 2\pi \times 10^6 t}$ d) Const

Sol^{n}:

Block diagram



So, $A_m = 1$; $f_m = 1MHz$

$A_c = 1$; $f_c = 100MHz$

So, $V_1 = A_m \cos 2\pi f_m t \cdot A_c \cos 2\pi f_c t$ (Blocked by HPF)

$= \frac{A_c A_m}{2}[\cos 2\pi (f_c + f_m)t + \cos 2\pi (f_c - f_m)t]$

$(f_c + f_m)$ → 101M ; $(f_c - f_m)$ → 99M

So, (HPF) o/p $V_2 = \frac{A_c A_m}{2} \cos 2\pi (f_c + f_m)t$

So, summer o/p $= s(t) = \frac{A_c A_m}{2} \cos 2\pi (f_c + f_m)t + \sin 2\pi f_c t$

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Now, as

$A\cos 2\pi f_c t + B\sin 2\pi f_c t \xrightarrow{\text{envelope}} \sqrt{A^2+B^2}$

(108)

Now,

$$s(t) = \frac{A_c A_m}{2}\cos 2\pi f_c t \cos 2\pi f_m t - \frac{A_c A_m}{2}\sin 2\pi f_c t \sin 2\pi f_m t + A_c \sin 2\pi f_c t$$

$$= \underbrace{\frac{A_c A_m}{2}\cos 2\pi f_m t}_{A}\cos 2\pi f_c t + \underbrace{\left(A_c - \frac{A_c A_m}{2}\sin 2\pi f_m t\right)}_{B}\sin 2\pi$$

So, envelope $= \sqrt{A^2+B^2}$

$$= \sqrt{\frac{A_c^2 A_m^2}{4}\cos^2 2\pi f_m t + A_c^2 + \frac{A_c^2 A_m^2}{4}\sin^2 2\pi f_m t - 2\frac{A_c^2 A_m}{2}\sin 2\pi f_m t}$$

$$= \sqrt{\frac{A_c^2 A_m^2}{4} + A_c^2 - A_c^2 A_m \sin 2\pi f_m t}$$

$$= \sqrt{\frac{1}{4} + 1 - \sin 2\pi \times 10^6 t}$$

$$\boxed{\text{envelope} = \sqrt{\frac{5}{4} - \sin 2\pi \times 10^6 t}}$$ Ans

Q4 A non-linear device is characterised by

$V_o = aV_i + bV_i^3$

where $V_i = m(t) + \cos 2\pi f_i t$

by considering only DSB terms. find f_i such that resulting DSB signal carrier frequency is 1MHz

Soln: As, $V_o = a\{m(t) + \cos 2\pi f_i t\} + b\{m(t) + \cos 2\pi f_i t\}^3$

$V_o = am(t) + a\cos 2\pi f_i t + b\{m^3(t) + \cos^3 2\pi f_i t + 3m^2(t)\cos 2\pi f_i t + 3m(t)\cos^2 2\pi f_i t\}$

$(1+\cos 4\pi f_i t/2)$

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So, by considering only USB term we get

$$V_0 = b \cdot \frac{3m(t)\cos 4\pi f_1 t}{2}$$

$$V_0 = \frac{3b}{2} m(t) \cos 4\pi f_1 t$$

So, by standard DSB term

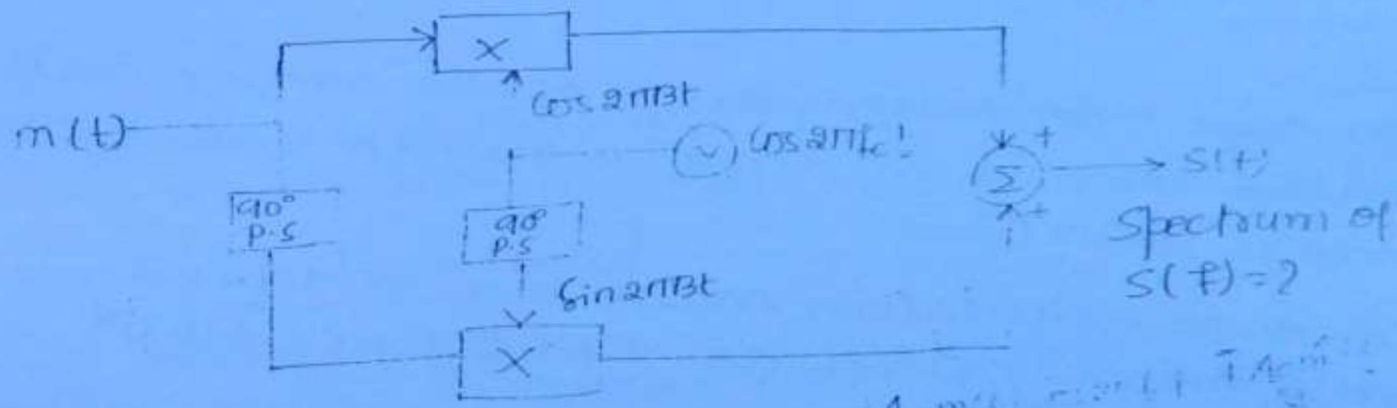
$$V_0 = A_c' m(t) \cos 2\pi f_c t$$

So, $f_c = 2f_1$

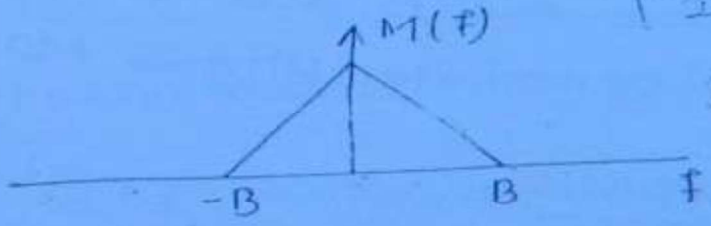
$f_1 = f_c/2$

$\boxed{f_1 = 0.5 \text{ MHz}}$ Ans.

Q5

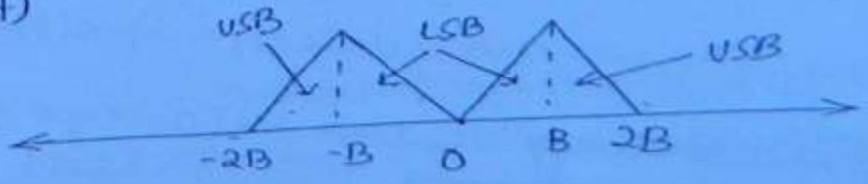


Spectrum of $S(f) = ?$

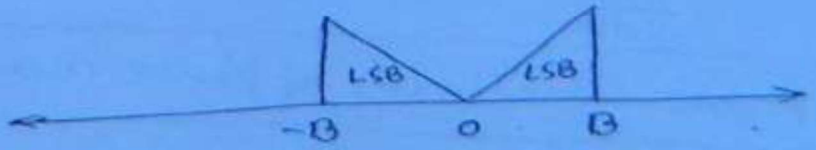


Sol^{n}:

$S_{DSB}(t)$



So, spectrum of $S(t) \longleftrightarrow S(f)$



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* ANGLE - MODULATION.

classified Broadly as:

1) Frequency Modulation.
2) Phase modulation

Now,

let,

carrier signal; $c(t) = A_c \cos\{2\pi f_c t + \phi\}$ (radians, radians)

Now, let

$2\pi f_c t + \phi = \theta(t)$

$A_c \cos\{\theta(t)\}$ ← total Angle

Note:-

1. In Angle modulation, angle of the carrier will be varied lineally in accordance with message signal voltage variations.

2. If Angle modulation, occurs due to dependence of f_c on $m(t)$, then is called as FREQUENCY MOD^N

3. If Angle modulation occurs, due to dependence of ϕ on $m(t)$; then is called as PHASE MODULATION.

* PHASE MODULATION:

Carrier before phase modulation $= c(t) = A_c \cos\{2\pi f_c t\}$

carrier after phase modulation $= S_{PM}(t) = A_c \cos\{2\pi f_c t + \phi\}$

* ϕ is varied according to the message signal amp.

So, $\phi = K_p m(t)$ (rad, volts)

K_p = phase sensitivity of phase modulator (rad/volt)

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Note:

1. Kp, specifies the amount of phase change in the carrier for 1 volt change in the message signal

if $m(t) = 0 \Rightarrow$ no modulation $\Rightarrow \phi = 0$

(III)

*FREQUENCY MODULATION:

Assume,

frequency of carrier before modulation = f_c

frequency of carrier after frequency modulation = f_i

Frequency of frequency modulated signal = Instantaneous frequency

* f_i is varied in accordance to variations in $m(t)$

so, mathematically

$$f_i = f_c + K_f m(t)$$

where,

K_f = frequency sensitivity of frequency modulation (Hz/volt)

If $m(t) = 0 \Rightarrow$ no modulation $\Rightarrow f_i = f_c$

also, let, $K_f = 25\text{KHz/volt}$ ← hence for 1 volt change frequency changes by 25 Hz

Note:

K_f; specifies the amount of frequency change in the carrier for 1 volt change in the message signal.

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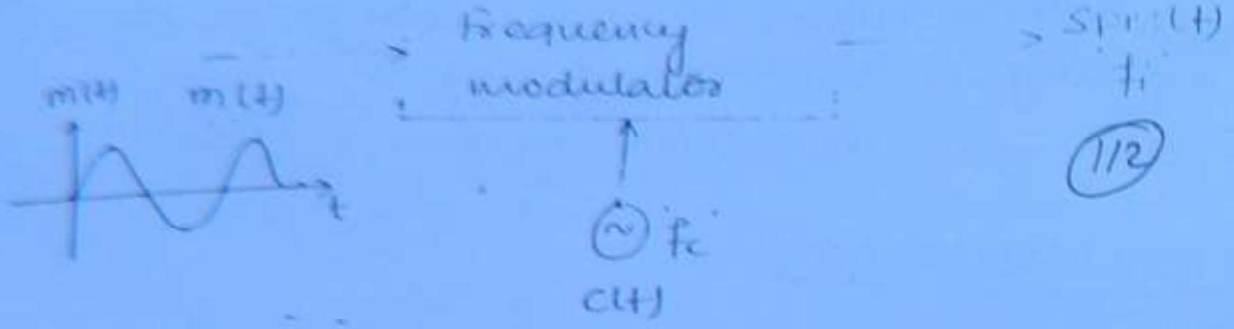
Kf = 25KHz/volt



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Now,

as $f_i = f_c + K_f m(t)$

So,

i) $m(t) = 0 \Rightarrow f_i = f_c + 0$

2) $m(t) = +5V \Rightarrow f_i = f_c + 125K$

3) $m(t) = -5V \Rightarrow f_i = f_c - 125K$.

Conclusion:

when,

1) $m(t) = 0 \Rightarrow f_i = f_c$.

2) $m(t) = +ve \Rightarrow f_i > f_c$.

3) $m(t) = -ve \Rightarrow f_i < f_c$.

Note:

1. By changing the amp. of message signal (voltage), variations in frequency is obtained. Hence F-M is also called as Voltage to Frequency Conversion.

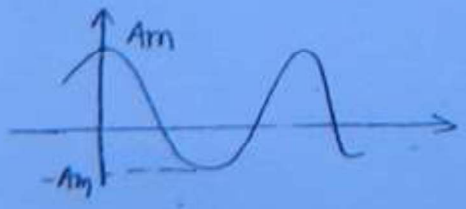
2. In FM, message signal voltage variation are converted as carrier signal frequency variation, so is also called as "VOLTAGE TO FREQUENCY CONVERSION."

*Analysis:

Assume, $m(t) = A_m \cos 2\pi f_m t$

$f_i = f_c + K_f m(t)$

frequency



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So,

max^m frequency of FM signal $\Rightarrow f_{max} = f_c + K_f A_m$

min^m frequency of FM signal $\Rightarrow f_{min} = f_c - K_f A_m$

So, max^m frequency deviation $= Max\{K_f m(t)\}$ (1/3)

So, $\Delta f = K_f A_m$

So, putting value of Δf in above eqn we get:

$$f_{max} = f_c + \Delta f$$
$$f_{min} = f_c - \Delta f$$

So, total frequency swing of FM signal $= f_{max} - f_{min}$
$= 2\Delta f$

Q1. An unmodulated carrier frequency is given by 1MHz. After frequency modulation, max^m frequency is given by 1.4 MHz. Find Δf and f_{min}?

Soln: Given:

$f_{max} = 1.4\,MHz$

$= f_c + \Delta f$

$= 1 + \Delta f = 1.4\,MHz$

$\Delta f = 0.4\,MHz$ Ans

And, $f_{min} = f_c - \Delta f$

$= 1 - 0.4$

$f_{min} = 0.6\,MHz$ Ans

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Q2. For an FM signal, fmax is given by 1.5 MHz. Total frequency swing is given by 900KHz. Find fc, Δf & fmin?

(114)

Soln: Given:

$2\Delta f$ = Total frequency swing = 900KHz

$$\boxed{\Delta f = 450\ KHz}$$ Ans

Now, $f_{max} = 1.5\ MHz$

$= f_c + \Delta f$

$f_c = f_{max} - \Delta f$

$= 1.5 \times 1000 - 450\ K$

$= 1050K$

$$\boxed{f_c = 1.05\ MHz}$$ Ans

And, $f_{min} = f_c - \Delta f$

$= 1.05 \times 1000 - 450$

$$\boxed{f_{min} = 600\ KHz}$$ Ans

Q3. A Sinusoidal carrier of 20V, 2MHz is frequency modulated by a mesg signal of $10\sin 4\pi \times 10^3 t$. Kf is given by 50KHz/volt. Find Δf; fmax & fmin?

Soln: Given, $A_c \cos 2\pi f_c t = c(t)$

So, $A_c = 20V$; $f_c = 2MHz$

$A_m \sin 4\pi \times 10^3 t = m(t)$

$A_m = 10$; $f_m = 2KHz$

$K_f = 50KHz/volts$

So, $f_{max} = f_c + K_f m(t)$

$= 2000 + 50 \times 10$

$$\boxed{f_{max} = 2500\ KHz}$$ fm

$f_{min} = f_c - K_f m(t)$

$= 2000 - 50 \times 10$

$$\boxed{f_{min} = 1500\ KHz}$$ Ans

$\Delta f = f_{max} - f_{min}$

$$\boxed{\Delta f = \frac{1000\ KHz}{2}}$$ fc

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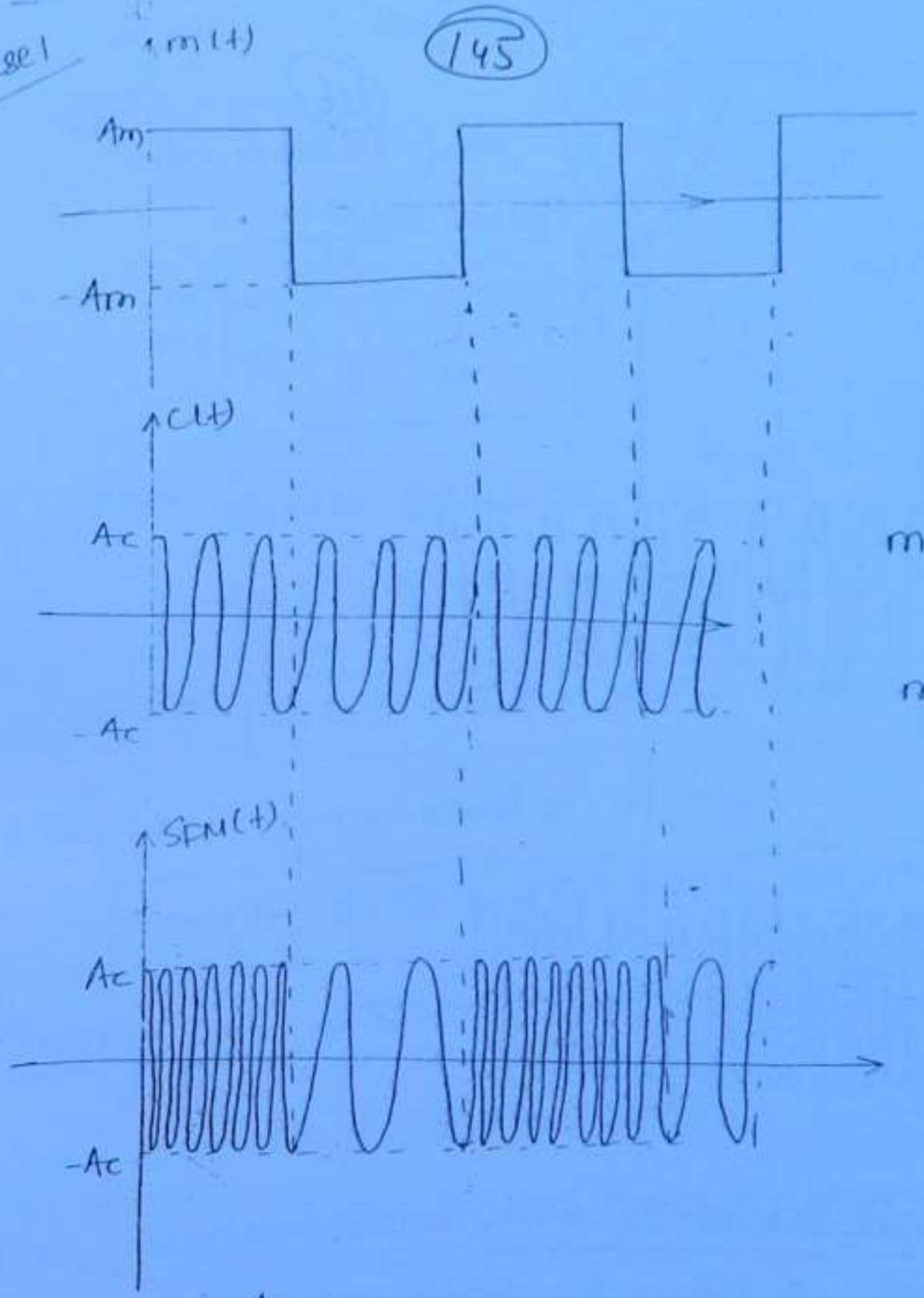
* Analysis:

Case 1



$f_i = f_c + K_f m(t)$

$m(t) = Am \Rightarrow f_i = f_c + K_f Am = f_1$

$f_1 > f_c$

$m(t) = -Am \Rightarrow f_i = f_c - K_f Am = f_2$

$f_2 < f_c$

As, $f_i = f_c + K_f m(t)$

NOW,

$m(t) = Am \Rightarrow f_i = f_c + K_f Am = f_1$

$\boxed{f_1 > f_c}$

$m(t) = -Am \Rightarrow f_i = f_c - K_f Am = f_2.$

$\boxed{f_2 < f_c}$

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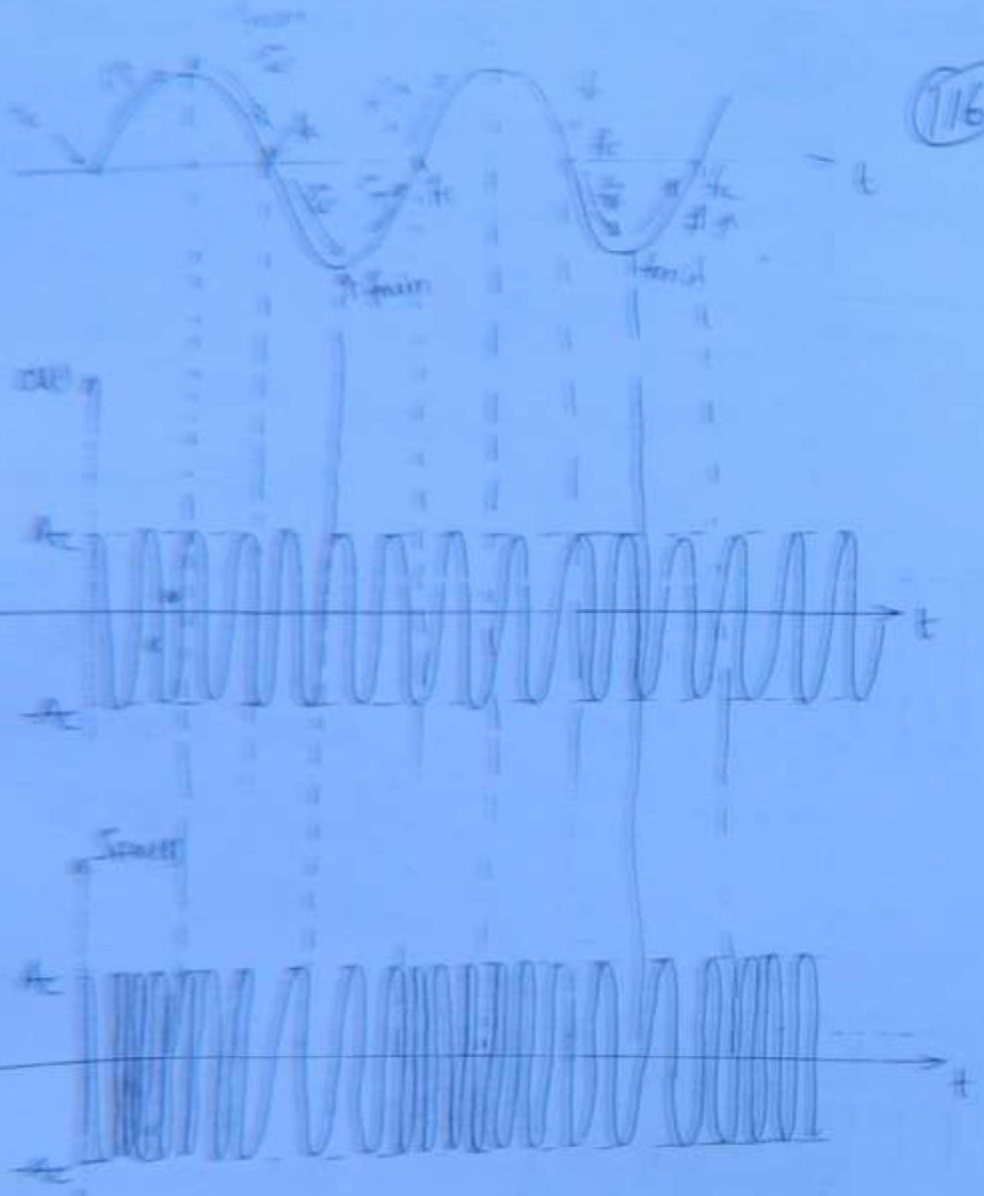

* GENERAL EXPRESSION OF FM:

Let, Carrier Signal $= c(t) = A_c \cos\{2\pi f_c t + \phi\}$

$= A_c \cos\{\theta(t)\}$

where, $\theta(t) = 2\pi f_c t + \phi$ ← no significance of ϕ.

To make ϕ insignificant;

$$\frac{d\theta(t)}{dt} = 2\pi f_c$$

f_c → before modulation

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Generalized expression of FM wave:

$$S_{FM}(t) = A_c \cos\{\theta_i(t)\} \quad \text{---(i)}$$

$\theta_i(t)$ ← Instantaneous Angle

So, after modulation

$$\frac{d\theta_i(t)}{dt} = 2\pi f_i$$

***

So, $$\boxed{f_i = \frac{1}{2\pi} \times \frac{d}{dt}\theta_i(t)}$$

NOW, $$\theta_i(t) = 2\pi \int f_i \, dt$$

$$\theta_i(t) = 2\pi \int \{f_c + K_f m(t)\} dt$$

So; $$\boxed{\theta_i(t) = 2\pi f_c t + 2\pi K_f \int m(t)\, dt}$$

Substituting in equation (i) we get:

$$\boxed{S_{FM}(t) = A_c \cos\{2\pi f_c t + 2\pi K_f \int m(t)\, dt\}}$$

* SINGLE TONE F-M:-

Assume, $m(t) = A_m \cos 2\pi f_m t$

So, $$S_{FM}(t) = A_c \cos\{2\pi f_c t + 2\pi K_f \int A_m \cos 2\pi f_m t \, dt\}$$

So, $$S_{FM}(t) = A_c \cos\{2\pi f_c t + 2\pi K_f \cdot A_m \frac{\sin 2\pi f_m t}{2\pi f_m}\}$$

***

NOW, let $$\boxed{\frac{K_f A_m}{f_m} = \frac{\Delta f}{f_m} = \beta}$$ ← modulation index of F-M

Also,

$$\boxed{\text{modulation index} = \beta = \frac{\text{maximum frequency deviation}}{\text{message signal freq}^n}}$$

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$S_{FM}(t) = A\cos\{2\pi f_c t + \beta \sin 2\pi f_m t\}$ — Single tone FM.

* Depending upon the value of β, the FM is classified as:

FM (118)

- NBFM ($\beta \leq 1$)
- WBFM ($\beta > 1$)

* Narrow Band FM (NBFM):-

The single tone FM is given as:

$$S_{FM}(t) = A_c \cos\{2\pi f_c t + \beta \sin 2\pi f_m t\}$$

$$S_{FM}(t) = A_c\{\cos(2\pi f_c t)\cos(\underbrace{\beta \sin 2\pi f_m t}_{\theta \approx 0\ (\because \beta \text{ small})}) - \sin(2\pi f_c t)\sin(\underbrace{\beta \sin 2\pi f_m t}_{\theta \approx 0})\}$$

Now, for NBFM; $\boxed{\beta \leq 1}$ (small value)

So, for small value of θ

$$\cos\theta \cong 1$$
$$\sin\theta \cong \theta$$

So,

$$S_{NBFM}(t) = A_c \cos 2\pi f_c t - A_c \sin 2\pi f_c t \times \beta \sin 2\pi f_m t$$

So,

$$S_{NBFM}(t) \cong A_c \cos 2\pi f_c t - A_c \beta \sin 2\pi f_c t \sin 2\pi f_m t$$

$$S_{NBFM}(t) = A_c \cos 2\pi f_c t - \frac{A_c \beta}{2}\{\cos 2\pi (f_c - f_m)t\} + \frac{A_c \beta}{2}\{\cos 2\pi (f_c + f_m)t\}$$

Now, Similarity b/w NBFM & AM: comparable; both ≤ 1

$$S_{AM}(t) = A_c \cos 2\pi f_c t + \frac{A_c \mu}{2}\cos 2\pi (f_c - f_m)t + \frac{A_c \mu}{2}\cos 2\pi (f_c + f_m)t$$

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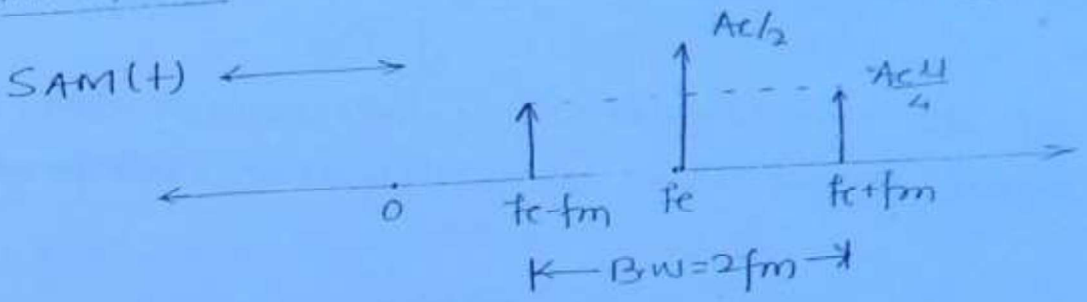
Note:

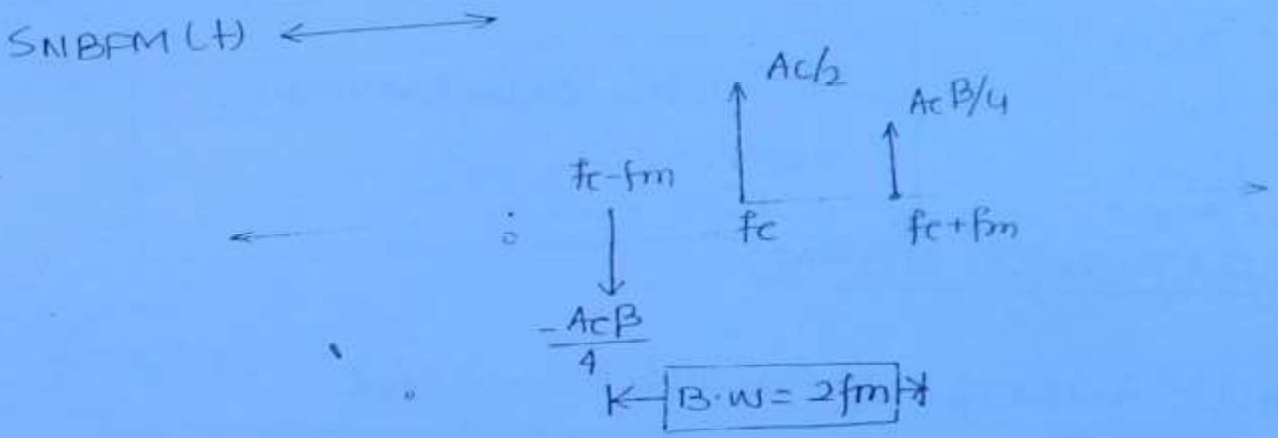
* General expression of AM & NBFM are same except 180° phase shift at LSB frequency component.

(1/9)

Spectrum of AM:



Spectrum of NBFM:



Note:

* Magnitude spectrum of AM & NBFM are same.

* Power of NBFM:

As,

$$P_t = P_c + P_{USB} + P_{LSB}$$

where,

$$P_c = \frac{A_c^2}{2R} \quad ; \quad P_{USB} = \left(\frac{A_c B}{2}\right)^2 / 2R = \frac{A_c^2 B^2}{8R} = P_{LSB}$$

So,

$$P_t = \frac{A_c^2}{2R} + 2 \times \frac{A_c^2 B^2}{8R} = \frac{A_c^2}{2R} + \frac{A_c^2 B}{4R}$$

$$\therefore P_t = \frac{A_c^2}{2R}\left\{1 + \frac{B^2}{2}\right\} \Rightarrow \boxed{P_t = P_c\left\{1 + \frac{B^2}{2}\right\}}$$

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CONCLUSION:

* NBFM has much similarity with AM, hence practical significance of NBFM is negligible. (120)
* Bandwidth and power requirements of NBFM will be same as AM.
* Because of its much similarities with AM, NBFM is given least practical significance compared to WBFM.

Q! AM and NBFM are having same modulation index were added. The Resulting signal will be.

a) DSB
b) SSB
c) SSB (USB with carrier).
d) NOT

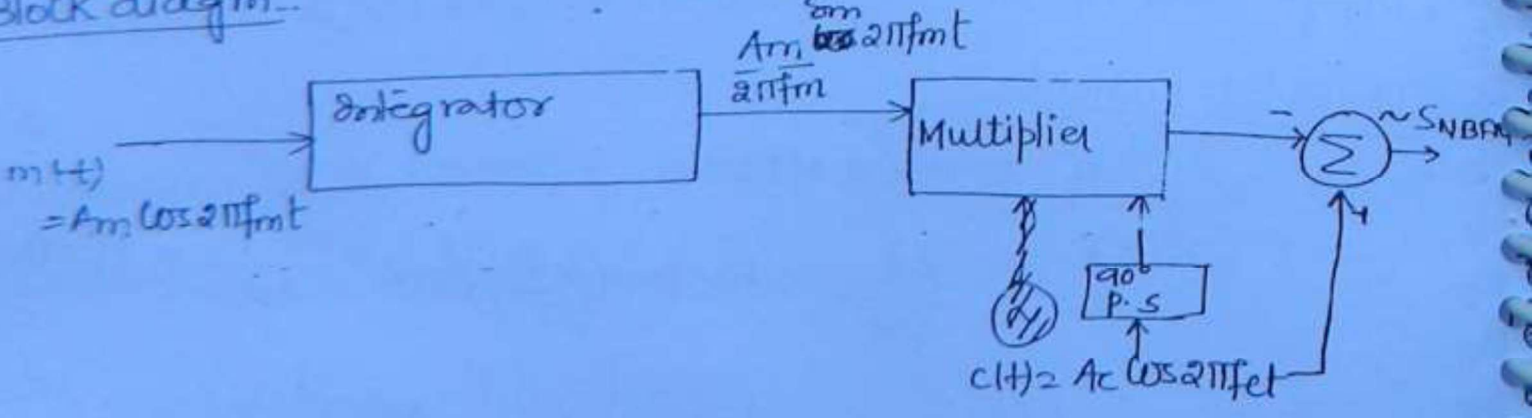
* GENERATION OF NBFM:

As

$$S_{NBFM}(t) = A_c \cos 2\pi f_c t - A_c \sin 2\pi f_c t \cdot \sin 2\pi f_m t$$

$$\cong A_c \cos 2\pi f_c t - \left(\frac{A_c K_f A_m}{f_m}\right) \sin 2\pi f_c t \sin 2\pi f_m t$$

Block diagram:



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WIDE BAND FM (WBFM)

* BESSEL FUNCTION:

Standard defn is given as:

$$J_n(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{j(x\sin\theta - n\theta)}\, d\theta \quad ; \theta = \text{dummy variable}$$

* Property of $J_n(x)$:-

i) $J_n(x)\downarrow$ decreases as $n\uparrow$ increases

So, $\boxed{J_0(x) > J_1(x) > J_2(x) > J_3(x) \ldots}$

ii) $J_n(+x) = (-1)^n J_n(x)$

So, $\boxed{\begin{aligned} J_{-n}(x) &= -J_n(x) \; ; n = \text{odd} \\ &= J_n(x) \; ; n = \text{even} \end{aligned}}$

iii) $\boxed{\sum_{n=-\infty}^{\infty} J_n^2(x) = 1}$

iv) $J_n(x)$ is a Real quantity.

* General expression of WBFM:

* General exp. is given as (of single tone):

$$SFM(t) = A_c \cos\{2\pi f_c t + \beta \sin 2\pi f_m t\}$$

Now, $\cos\theta = \text{Real}\{e^{j\theta}\}$

So, $SFM(t) = A_c \text{Real}\{e^{j(2\pi f_c t + \beta \sin 2\pi f_m t}\}$ → Fourier Series

$$= A_c \, Re\{e^{j2\pi f_c t} \cdot e^{j\beta \sin 2\pi f_m t}\} \quad \text{---(1)}$$

Now, $e^{j\beta \sin 2\pi f_m t}$ is a continuous periodic signal with

$\boxed{T = 1/f_m}$

Above signal is periodic. So, $x(t) = x(t+1)$

$$e^{j\beta \sin 2\pi f_m t} = e^{j\beta \sin 2\pi f_m (t+1/f_m)} \quad \{\because T = 1/f_m\}$$

* fourier series is used to find the frequency Analysis of continuous periodic signal. So, the exp Fourier series is given as:-

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \quad ; \omega_0 = \frac{2\pi}{T}$$

where,

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt$$

So,

$$e^{j\beta \sin 2\pi f_m t} = \sum_{n=-\infty}^{\infty} c_n e^{jn 2\pi f_m t} \quad \text{---- (2)}$$

Now,

$$c_n = \frac{1}{(1/f_m)} \int_{-1/2f_m}^{1/2f_m} e^{j\beta \sin 2\pi f_m t} \cdot e^{-jn 2\pi f_m t} dt$$

$$= f_m \int_{-1/2f_m}^{+1/2f_m} e^{j(\beta \sin 2\pi f_m t - n 2\pi f_m t)} dt$$

Now, as

$$J_n(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(x \sin\theta - n\theta)} dt$$

So, $\theta = 2\pi f_m t$

$2\pi f_m dt = d\theta$.

Now, when $t = -1/2f_m \Rightarrow \theta = -2\pi$.

$t = 1/2f_m \Rightarrow \theta = +\pi$.

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So,

$$C_n = f_m \int_{-\pi}^{\pi} e^{j(\beta \sin\theta - n\theta)} \cdot \frac{d\theta}{2\pi f_m}$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin\theta - n\theta)} \, d\theta = J_n(\beta) \quad \{\because x = \beta\}.$$

So,

$$\boxed{C_n = J_n(\beta).}$$

putting this value in eq^n (2) we get:

$$\boxed{e^{j\beta \sin 2\pi f_m t} = \sum_{n=-\infty}^{\infty} J_n(\beta) \cdot e^{jn2\pi f_m t}}$$

substituting this value in eq (1) we get:

$$SFM(t) = SWBFM(t) = A_c \, Re\left\{ e^{j2\pi f_c t} \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn2\pi f_m t} \right\}.$$

$$SWBFM(t) = A_c \, Re\left\{ \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi (f_c + nf_m) t} \right\}.$$

So, the General exp. of WBFM is given as:-

**

$$\boxed{SWBFM(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos 2\pi (f_c + n f_m) t}$$

Analysis:-

$$SWBFM(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos 2\pi (f_c + n f_m) t$$

$$= A_c J_0(\beta) \cos 2\pi f_c t + A_c J_1(\beta) \cos 2\pi (f_c + f_m) t +$$
$$A_c J_{-1}(\beta) \cos 2\pi (f_c - f_m) t + A_c J_2(\beta) \cos 2\pi (f_c + 2f_m) t$$
$$+ A_c J_{-2}(\beta) \cos 2\pi (f_c - 2f_m) t + \cdots$$

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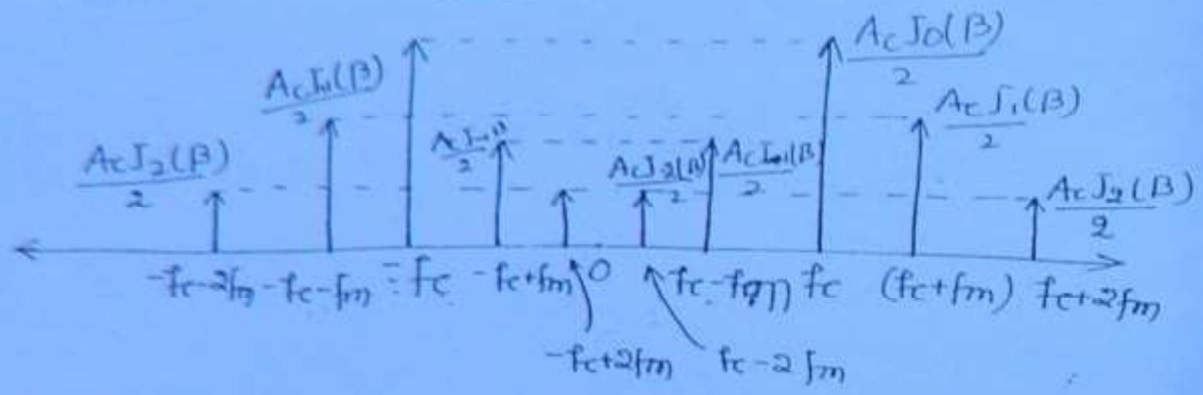

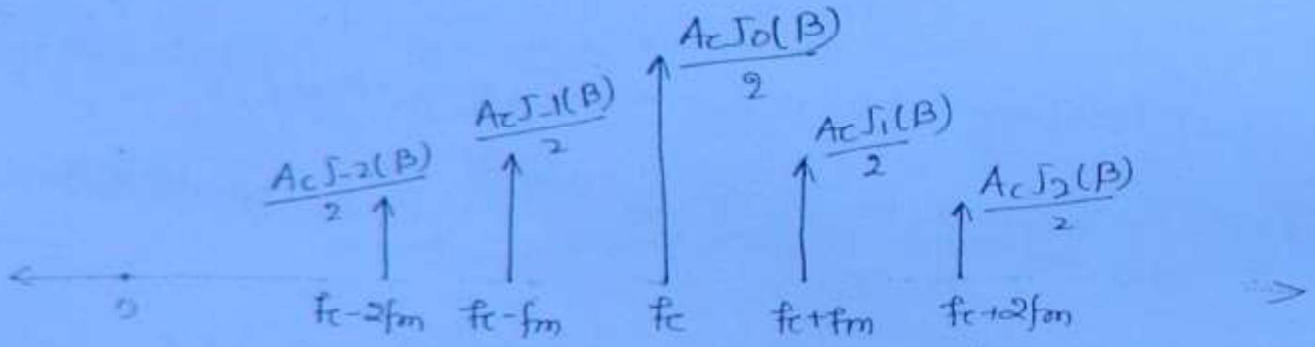

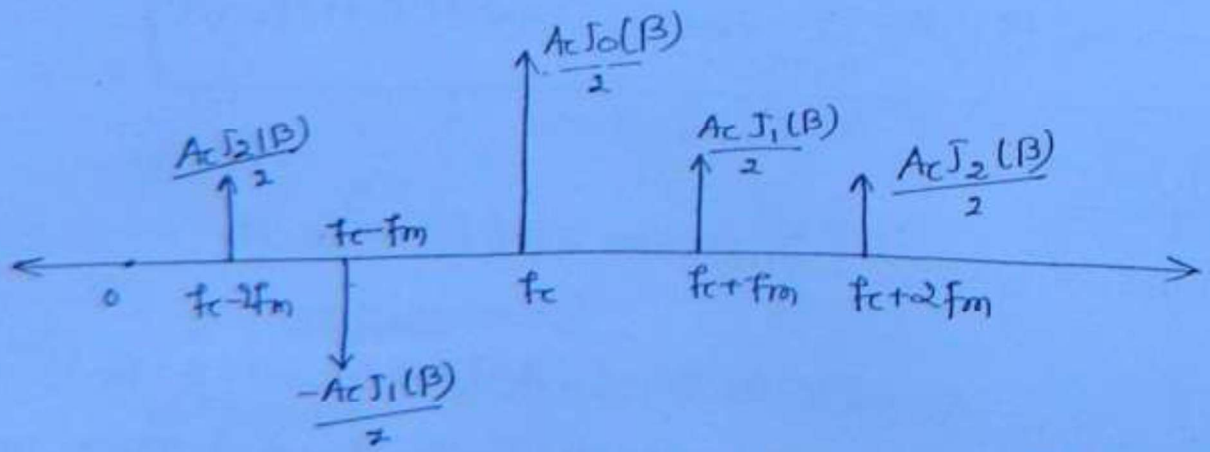

$J_0(\alpha) > J_1(\alpha) > J_2(\alpha) > J_3(\alpha) \ldots$

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Conclusion:-

1 WBFM consists of carrier frequency component, ∞ no of USB's and ∞ no of LSB's.

(125)

2 The Actual Bandwidth of WBFM is ∞

3 For WBFM, strength of higher order side Bands go on decreasing and finally becomes zero.

4 For WBFM, lower order sideBands are said to be significant sidebands and higher order sideBands are insignificant.

* Power of WBFM:-

As,

$$P_t = P_c + (P_{USB1} + P_{USB2} + \cdots) + (P_{LSB1} + P_{LSB2} + \cdots)$$

*** where,

$$P_c = \frac{A_c^2 J_0^2(\beta)}{2R} \quad ; \quad P_{USB1} = \frac{A_c^2 J_1^2(\beta)}{2R} \quad ; \quad P_{LSB1} = \frac{A_c^2 J_{-1}^2}{2R}$$

$$P_{USB2} = \frac{A_c^2 J_2^2(\beta)}{2R} \quad ; \quad P_{LSB2} = \frac{A_c^2 J_{-2}^2}{2R}$$

So,

$$P_t = \cdots + \frac{A_c^2 J_{-2}^2(\beta)}{2R} + \frac{A_c^2 J_{-1}^2(\beta)}{2R} + \frac{A_c^2 J_0^2(\beta)}{2R} + \frac{A_c^2 J_1^2(\beta)}{2R} + \frac{A_c^2 J_2^2(\beta)}{2R} + \cdots$$

So,

$$P_t = \frac{A_c^2}{2R} \left\{ \cdots + J_{-2}^2(\beta) + J_{-1}^2(\beta) + J_0^2(\beta) + J_1^2(\beta) + J_2^2(\beta) + \cdots \right\}$$

$$P_t = \frac{A_c^2}{2R} \left\{ \sum_{n=-\infty}^{\infty} J_n^2(\beta) \right\}.$$

**

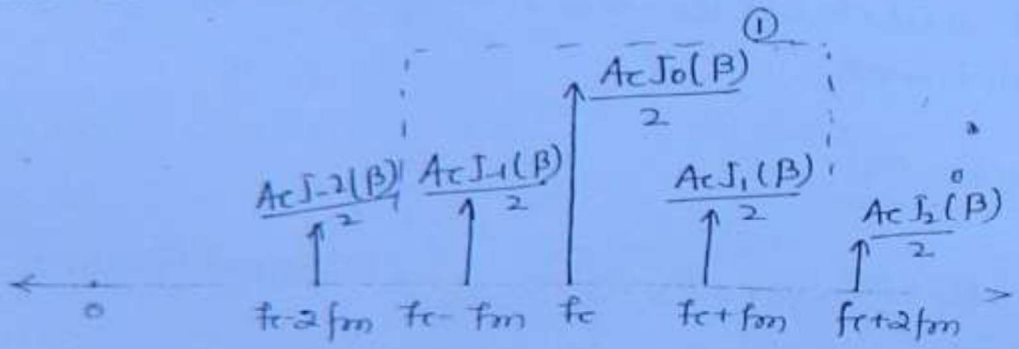
$$\boxed{P_t = \frac{A_c^2}{2R}} \quad \left[\because \sum_{n=-\infty}^{\infty} J_n^2(\beta) = 1 \right]$$

Note:

1. For WBFM, the power of carrier before modulation same as after modulation.

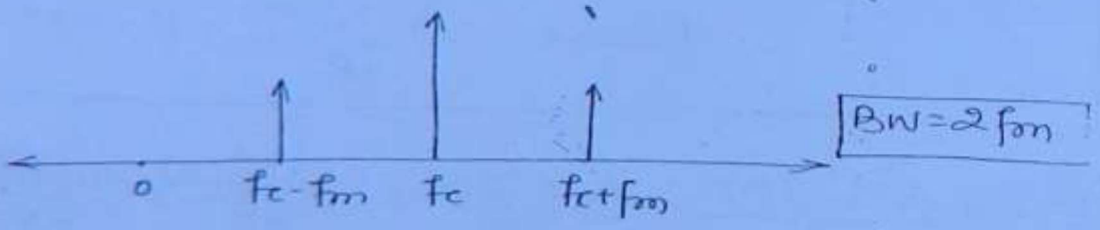
* Practical Bandwidth of WBFM: {CARSON'S RULE}

* The Actual B.W of WBFM is ∞. For transmission of signal, it should be band limited by retaining only significant sidebands and eliminating insignificant sidebands



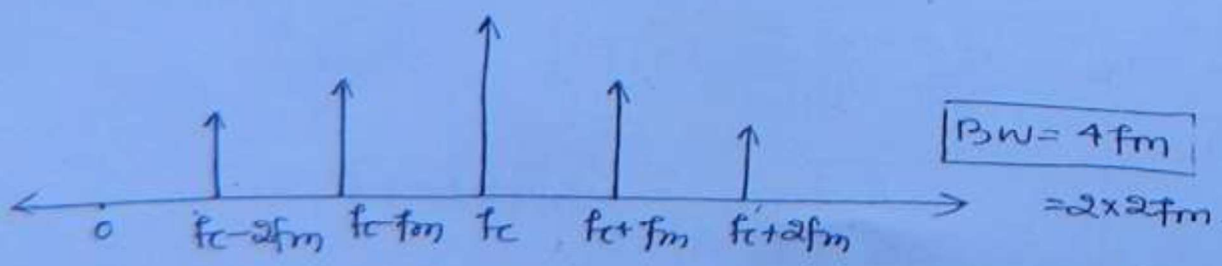
Case 1 (WBFM consist of significant SB's upto 1st order):-

After passing through Band limiting we get



Case 2 (WBFM consist of significant SB's upto 2nd order):-

After Band limiting the signal we get:



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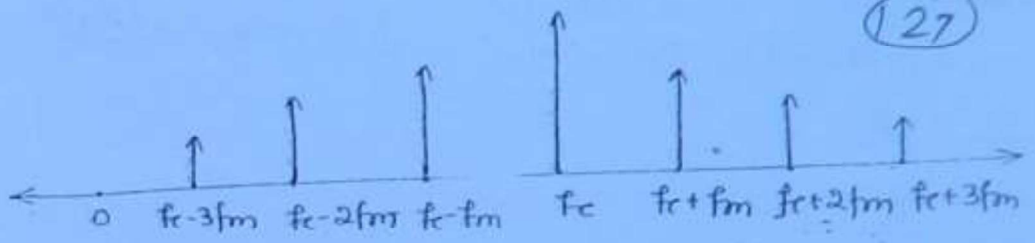
Case 3 (upto 3rd order)

After passing through I and limiting filter we get



$B.W = 6f_m$
$= 3 \times 2f_m$

CARSON'S RULE:

According to carson, WBFM consist of the significant sideband upto order "B+1"; when the modulation index is β.

So,

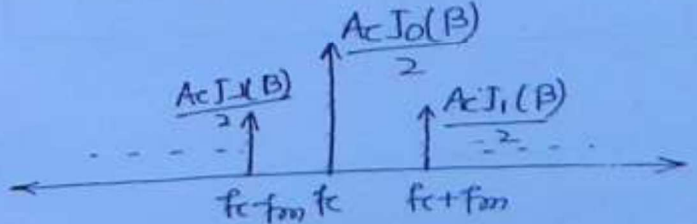
$$B.W = (\beta+1) \times 2f_m$$

Also

$$B.W = (\beta+1) \times 2f_m$$
$$= \left(\frac{\Delta f}{f_m} + 1\right) \times 2f_m$$

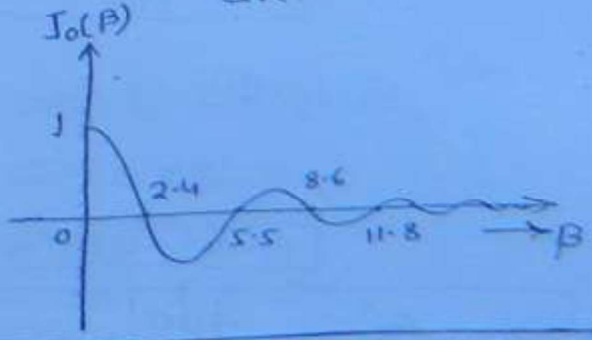
$$B.W = 2(\Delta f + f_m)$$

* MODULATION EFFICIENCY (η):



NOW,

$$P_c = \frac{A_c^2 J_0^2(\beta)}{2R}$$



Standard values → $J_0(\beta) = 0; \beta = 2.4, 5.5, 8.6, 11.8, \ldots$

for $\beta = 2.4, 5.5, 8.6, 11.8$, hence $P_c = 0$, are called as eigen values of β.

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NOTE:

* For above values of β, power taken by carrier freq component will be zero so, that modulation index will become 100%.

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$$J_n(\beta) = 0 \; ; \; \beta = 2.4, 5.5, 8.6, 11.8, \ldots$$

$$P_c = 0$$

$$\eta = 100\%$$

Q1. A sinusoidal carrier of 20V, 2MHz is frequency modulated by a sinusoidal msg signal of 10V, 50KHz.

$K_f = \frac{25\text{KHz}}{\text{volts}}$.

a) find Δf; β; Bandwidth & power.

b) repeat above if msg signal amplitude is doubled.

Soln: Given:

$A_c = 20\text{V}$; $f_c = 2000\text{ KHz}$

$A_m = 10\text{V}$; $f_m = 50\text{ KHz}$

$K_f = \frac{25\text{KHz}}{\text{Volts}}$

So, $f_1 = \{f_c + K_f m(t)\}$; $f_2 = f_c + K_f m(t)$

$= (2000 + 25 \times 10)$ $= 2000 - 25 \times 10$

$= 2250\text{K}$ $= 1750\text{ KHz}$

Now, $\boxed{\beta = \frac{\Delta f}{f_m} = \frac{K_f . A_m}{f_m} = \frac{25 \times 10}{50\text{K}} = 5 \text{ (WBFM)}}$

So, $B.W = 2(\beta + 1) f_m$

$= 2 \times 6 \times 50$

$\boxed{B.W = 600\text{K}}$

$\boxed{P_t = \frac{A_c^2}{2R} = \frac{400}{2} = 200\text{W}}$

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2. Am: 2x10 = 20V

So, $\uparrow \Delta f = K_f A_m \uparrow$

Am is doubled Δf is doubled.

$$\Delta f = 25K \times 20 = 500K$$

$$\uparrow \beta = \frac{K_f A_m \uparrow}{f_m}$$

$$\beta = \frac{25K \times 20}{50} = 10$$

* B.W = (β+1) x 2fm

= 11 x 2 x 50

B.W = 1100K

* $P_t = \frac{A_c^2}{2R} = 200W$

Conclusion:

* when mesg signal amp. is doubled, both Δf and β will also be doubled.

Q2. An FM signal is given by:

$$s(t) = 10\cos\{2\pi \times 10^6 t + 8\sin 4\pi \times 10^3 t\}$$

i) Find β; Δf; B.W & Power.

ii) Repeat above if mesg. signal frequency is doubled.

Soln: Given, $s(t) = 10\cos\{2\pi \times 10^6 t + 8\sin 4\pi \times 10^3 t\}$

Comparing with:

$$s(t) = A_c\cos\{2\pi f_c t + \beta\sin 2\pi f_m t\}.$$

So, $A_c = 10V$; $f_c = 1000K$; $\beta = 8$ $f_m = 2K$.

Now, $\Delta f = \beta \times f_m = 16K$

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* B.W = $2(\beta+1)f_m$

$= 2 \times 9 \times 2K$

$\boxed{B.W = 36K}$

* $\boxed{P_t = \frac{A_c^2}{2R} = \frac{100}{2} = 50W}$

2. Frequency of msg signal doubled. So, new f_m

$f_m = 4KHz$

Now, $\Delta f = K_f \cdot A_m$

$= 16KHz$ (no change)

Also, $\Delta f = \beta \times f_m$ (2 ↓, 2 ↑)

Now, $\beta = \frac{K_f A_m}{f_m}$ (2 ↓, 2 ↑) $= 4$ (halfed)

So,

* B.W = $2(\beta+1)f_m$

$= 2 \times 5 \times 4K$

$\boxed{B.W = 40K}$

* $\boxed{P_t = 50W}$

Conclusion:

when message signal frequency doubles; "Δf" will not be changed and "β" will halfed.

Q3. A carrier is frequency modulated with max^m frequency deviation of 16KHz. Msg signal freq^n is given by 4KHz

a) Find β & B.W?

b) Repeat above if message signal Amp. is doubled and frequency is reduced to 1KHz?

Soln:- Given:

$\Delta f = 16KHz$

$f_m = 4KHz$

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Now, $\Delta f = \beta f_m$; $\Delta f = \frac{K_f A_m}{f_m}$

$\boxed{\beta = 1}$

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Now, $B.W = 2(\beta+1) f_m$

$= 2 \times 5 \times 4$

$\boxed{BW = 40K}$

b) Given:

$\Delta f = 16KHz$ (from previous case)

$f_m = 1KHz$.

So, $\Delta f = \frac{K_f A_m}{f_m} \uparrow^2$

So, $\boxed{\Delta f = 32K.}$

Now, $\boxed{\beta = \frac{\Delta f}{f_m} = \frac{32}{1} = 32}$

So, $B.W = 2(\beta+1) f_m$

$= 2 \times 33 \times f_m$

$\boxed{B.W = 66K}$

Q4. A carrier is frequency modulated with max^m frequency deviation of 100K. Find β & B.W if msg signal frequency is a) 10K b) 1 MHz

Soln: Given:

a) $\Delta f = 100K$

$f_m = 10K$

Now, $\Delta f = \beta f_m$

$\beta = \frac{\Delta f}{f_m} = 10$ (WBFM)

So, $B.W = 2 \times (\beta+1) f_m$

$= 2 \times 11 \times 10K$

$\boxed{BW = 220K}$

b) $\Delta f = 100K$ {NBFM}

$f_m = 1000K$

$\Delta f = \beta f_m$

$\beta = \frac{\Delta f}{f_m} = \frac{100}{1000}$

$\boxed{\beta = 0.1}$

$B.W = 2 f_m$

$= 2 \times 1 M$

$\boxed{B.W = 2M}$

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Q5 Given:

$c(t) = 5\cos 2\pi \times 10^6 t$

$m(t) = \cos 4\pi \times 10^3 t$

a) c(t) & m(t) are used to generate AM with $\mu = 0.707$. Find B.W & power?

b) c(t) & m(t) are used to generate FM; with max^m freq^n deviation is 3 times B.W of AM. Find the coeff of $\cos 2\pi \times (016 \times 10^3)t$ in F.M expression.

a) $5J_6(8)$ c) $5J_3(6)$

b) $5J_8(6)$ d) $5J_3(8)$

Soln: Given,

$m(t) = \cos 4\pi \times 10^3 t$

$Am = 1$; $fm = 2K$ Hz

$c(t) = 5\cos 2\pi \times 10^6 t$

$Ac = 5$; $fc = 1000K$.

Now,

$B.W = 2fm = 2 \times 2K$

$\boxed{B.W = 4K}$

$\text{Power} = P_t = P_c\left\{1 + \frac{\mu^2}{2}\right\}$

$P_c = \frac{A_c^2}{2R} = \frac{25}{2}$

$P_t = \frac{25}{2}\left\{1 + \frac{1}{4}\right\} = \frac{25}{2} \times \frac{5}{4}$

$\boxed{P_t = \frac{125}{8} W}$

b) $\Delta f = 3 \times B.W$ of AM

$= 3 \times 4K$

$\boxed{\Delta f = 12K}$

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Now, $\beta = \frac{\Delta f}{f_m}$

$\boxed{\beta = \frac{12}{2} = 6}$ (WBFM) $\quad \beta = 3 \times 2 f_m$

Now,

$$S_{WBFM}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos 2\pi (f_c + n f_m).$$

Now, $\cos 2\pi (2016 \times 10^3 t)$.

So, $f_c + n f_m = 1016$

$n f_m = 1016 - 1000$

$8n = \frac{1016 - 1000}{2}$

$\boxed{n = 8}$ Ans

So $\boxed{\text{Coeff.} \Rightarrow A_c J_n(\beta) = 5 J_8(6)}$

Q6 A Sinusoidal carrier of frequency f_c is used for both AM & FM xmitter; msg signal freqn is given by 5KHz. Max freqn deviation = 2 x B.W of AM. Find Modn index of Both AM & FM such that strength of freqn component $f_c + 5K$ is same in both AM & FM spectrum.

$J_1(0) = 1$; $J_1(2) = 0.57$; $J_1(4) = 0.37$; $J_1(8) = 0.08$

Solⁿ: B.W of AM = $2 f_m$; $\mu = ?$

$= 10$ KHz ; $\beta = ?$

$\Delta f = 2 \times 10K = 20K$. ; $\frac{\Delta f}{f_m} = \beta \Rightarrow \beta = \frac{20}{5} = 4$

Strength of $f_c \pm 5K$ is same in both AM & FM

Now, $\frac{A_c \mu}{2} = \frac{A_c J_1(\beta)}{2} \Rightarrow \mu = J_1(4)$

$\mu = 0.37 \times 2 \Rightarrow \boxed{\mu = 0.74}$

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Q1 A Sinusoidal carrier is frequency modulated by a sinusoidal msg signal of frequency m and amp A_m. Conducting and experiment with $f_m = 1$ KHz and increasing A_m from 0V, it is observed that the strength of the carrier frequency component in the spectrum becomes 0 for the 1st time with $A_m = 2V$.

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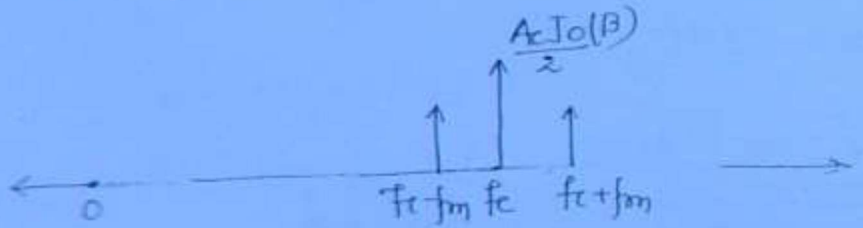
Find i) K_f

ii) Find msg signal amplitude for which strength of carrier frequency component becomes zero for the 2nd time?

Soln: Strength of frequency component = $\frac{A_c J_0(\beta)}{2}$

$f_m = 1$ KHz ; $A_m = 0V \uparrow 2V$



NOW, $J_0(\beta) = 0$; $\beta = 2.4, 5.5, 8.6, 11.8$

NOW, $\beta_1 = \frac{K_f A_m}{f_m}$

NOW; $2.4 = \frac{K_f \times 2}{1K}$

$$\boxed{K_f = 1.2 \text{ KHz/Volt}}$$ Ans

NOW, $\beta_2 = \frac{K_f A_m}{f_m}$

$5.5 = \frac{1.2 \times 10^3 \times A_m}{1 \times 10^3}$

$$\boxed{A_m = 4.58 V}$$ Ans

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Q6 An unmodulated FM xtr power is given by 100W; with modulation; it is observed that strength of 1st order side Band is 0. Find

i) the power of carrier frequency component
ii) Total side Band power (13)
iii) total second order side Band power.

$J_0(2.4) = 0$; $J_1(2.4) = 0.3$; $J_2(2.4) = -0.28$
$J_0(3.8) = 0.4$; $J_1(3.8) = 0$; $J_2(3.8) = 0.2$
$J_0(5.1) = -0.36$; $J_1(5.1) = -0.23$; $J_2(5.1) = 0$

Soln: Given:- $P_t = 100W$ — $A_c^2/2R = A_c = \sqrt{200}$

1st order SideBand $= \frac{A_c J_1(\beta)}{2} = 0$

$J_1(\beta) = 0 = J_1(3.8)$

$\boxed{\beta = 3.8}$

Now, $\frac{A_c J_0(\beta)}{2}$ = Carrier frequency component

Power $= \frac{A_c^2 J_0^2(\beta)}{2}$

$= \frac{200 \times J_0^2(3.8)}{2} = \frac{200 \times 0.16}{2}$ (100)

$\boxed{\text{Power} = 16\ W}$ Ans.

Now,
ii) Total power $= \frac{A_c^2}{2R} = 100W = P_t$

So, $\boxed{\text{Total side Band power} = 100 - 16 = 84\ W}$ Ans

iii) Second order power $= \frac{A_c^2 J_2^2(\beta)}{2R} + \frac{A_c^2 J_{-2}^2(\beta)}{2R}$

$= 2 \frac{A_c^2}{2} \{J_2^2(\beta) + J_{-2}^2(\beta)\}$

$= A_c^2 \{J_2^2(\beta)\} = 200 \times J_2^2(3.8)$

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$P_{2nd\ order} = 2 \times \frac{0.04}{100}$

$P_{2nd\ order} = 8W$ Ans

* GENERATION OF FM:

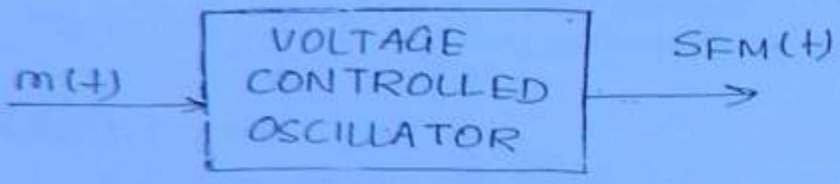
Generation of FM done by:-

1) Direct Method.

2) Indirect Method or Armstrong Method.

* DIRECT METHOD:

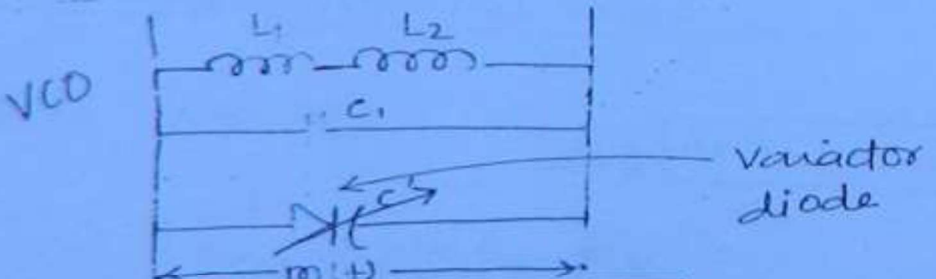
FM ↔ VCO



Hartley osc

$f_i = \frac{1}{2\pi\sqrt{(L_1+L_2)\cdot C}}$

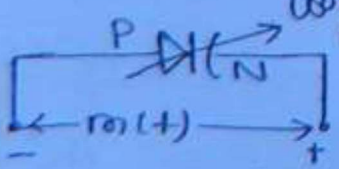
tank ckt of VCO :-



$f_i = \frac{1}{2\pi\sqrt{(L_1+L_2)(C_1+C')}}$

As, $f_i = f_c + K_f m(t)$

Case 1 :- $m(t) = +ve$



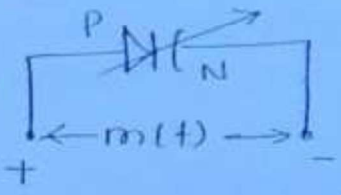
connected in Reverse mode

$C \propto \frac{1}{V}$

When connected in the Reverse mode

Hence $[C(t) \uparrow \rightarrow C \downarrow \rightarrow f_i \uparrow]$

Case2 $m(t) = -ve$



when connected in the Forward mode

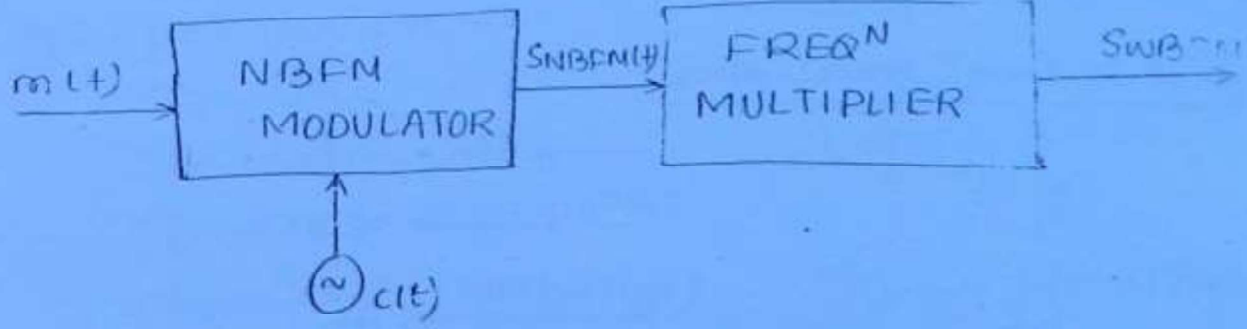
$[W \downarrow \rightarrow C \uparrow \rightarrow f_i \downarrow]$

Hence, the frequency is varied in accordance to the msg signal voltage variations.

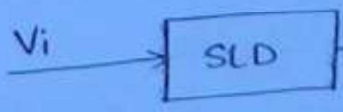
* INDIRECT METHOD (ARMSTRONG METHOD):-

* In this method WBFM is generated from NBFM.

Block diagram:



* Frequency multiplier is nothing but Square law device followed by proper Band pass filter



$V_2 = a_1 V_i + a_2 V_i^2 + a_3 V_i^3 + \cdots$

* Pass Band of BPF is such that it allows the V_i^2 of the Square law device

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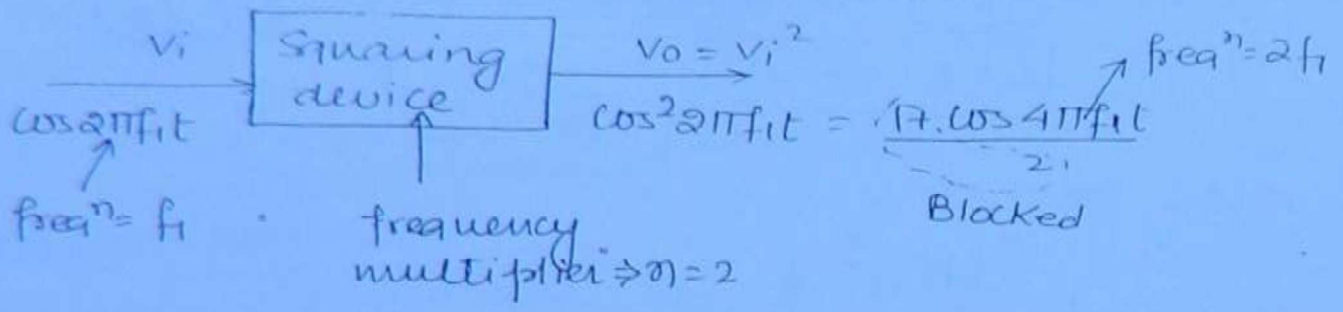
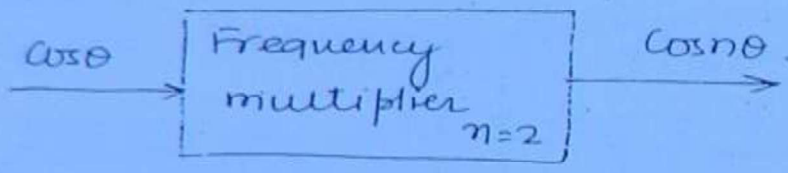


Note :-



* The General eqn of Single tone FM is

$$S_{NBFM}(t) = \bar{A}_c \cos\{2\pi f_c t + \underbrace{\beta \sin 2\pi f_m t}_{\theta}\} : \beta \le 1$$

$$(\text{Freq}^n \text{ mul})_{O/P} = S_{NBFM} = A_c \cos\{2n\pi f_c t + n\beta \sin 2\pi f_m t\}$$

'n' should such that

$$\boxed{n\beta > 1}$$

Input of Freq^n multiplier		O/P
f_c	⟶	nf_c
β	⟶	$n\beta$
f_m	⟶	f_m

$$\boxed{\uparrow^n \Delta f = \uparrow^n \beta f_m}$$

So, when β is increased by n

Hence, Δf is increased by n

So $\boxed{\text{new } \Delta f = n\Delta f}$

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Q) An FM is given by:-

$$s(t) = 10\cos\{2\pi\times10^6 t + 0.2\sin 2\pi\times2\times10^3 t\}$$

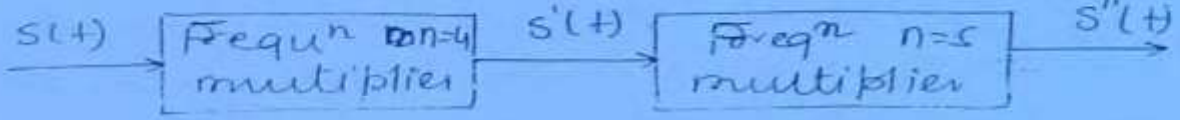
It is passed through cascaded frequency multiplier of having multiplying const of 4 and 5 Respt. Find all the parameters of FM signal at the O/P of each of the multiplier?

Soln: Given:

$$s(t) = 10\cos\{2\pi\times10^6 t + 0.2\sin 4\pi\times10^3 t\}.$$

Compaiing with standard eqn:

$$s(t) = A_c\cos\{\cos 2\pi f_c t + \beta\sin 2\pi f_m t\}.$$



$A_c = 10V$; $\beta = 0.2$

$f_c = 1MHz$; $f_m = 2KHz$

$\Delta f = \beta f_m$
$= 0.4KHz$

After passing: (n=4)

✓ $A_c = 10V$

✓ $f_c = 4\times1 = 4MHz$

✓ $\beta = 4\times0.2 = 0.8$ (NBFM)

✓ $f_m = 2KHz$ (no change)

✓ $\Delta f = 4\times0.4$ ($\beta\times f_m$)
$= 1.6K$ ✓

✓ $B.W = 2f_m$
$= 4K$

✓ $P_t = \frac{A_c^2}{2R}\left\{1+\frac{\beta^2}{2}\right\}$

$= \frac{100}{2}\left\{1+\frac{0.64}{2}\right\}$

$P_t = 66W$ ✓

After Passing (n=5):

✓ $A_c = 10V$

✓ $f_c = 5\times4M = 20MHz$

✓ $\beta = 0.8\times5 = 4$ (WBFM)

✓ $f_m = 2K$

✓ $\Delta f = 5\times1.6K$
$= 8K$

✓ $B.W = 2(\beta+1)f_m$
$= 2\times5\times2K$
$= 20K$

✓ $P_t = \frac{A_c^2}{2R} = 50W$

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* Demodulation of FM:

i) slope detector.
ii) Balanced slope detector.

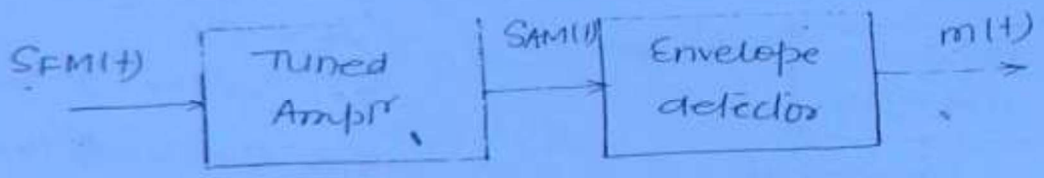
* Phase discrimination method:

i) Foster Seley detector.
ii) Ratio method.
iii) PLL-method (mostly used) (IES < subj. / obj.

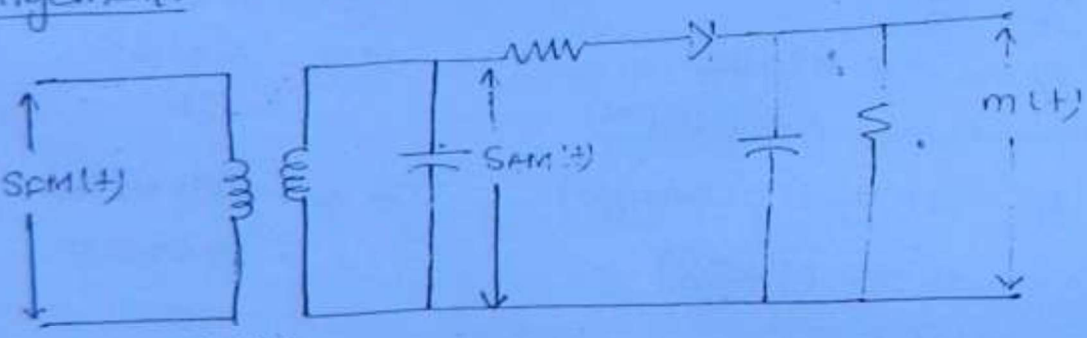
* Every other demodulator except PLL, needs xmer for their constn, so for FM demodulation; generally PLL method will be used.

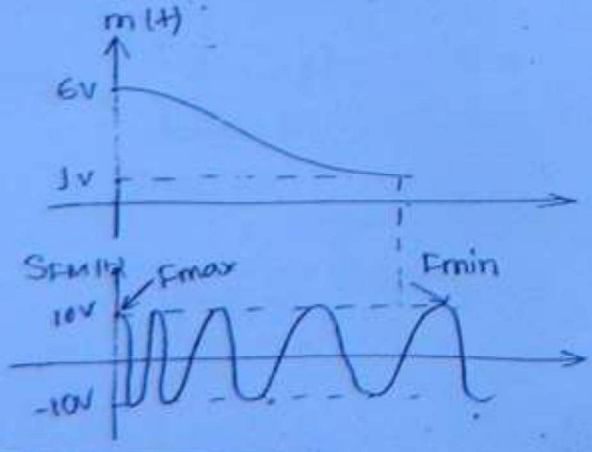
i) Slope detector:

Block diagm:



ckt arrangement:





Message signal Amp. is decreased by 6 times so frequency is also decreased by 6 times.
Fmax = 1200K
Fmin = 200K

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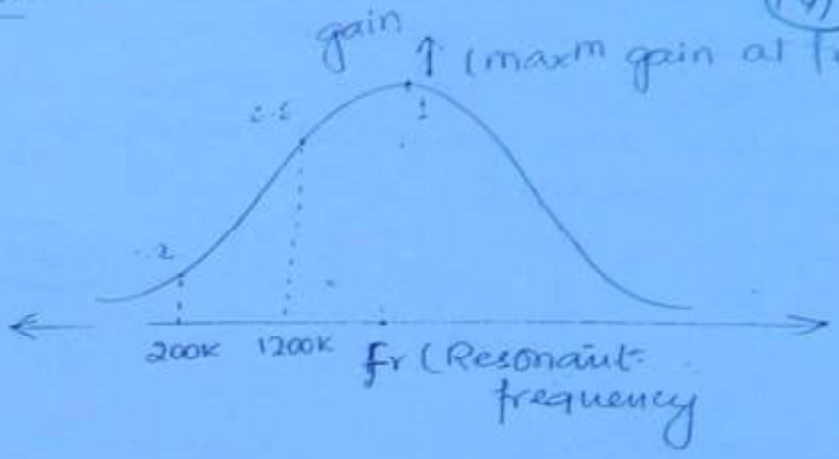
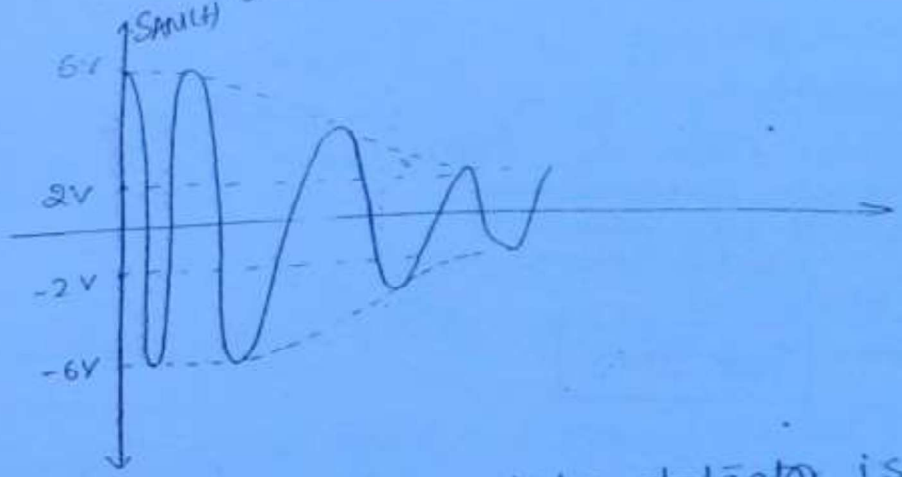
Tuned Amp Char:



* The fr should be such that $f_0 > f_{0max}$
(of Tuned Ampl^r)

* The gain offered by Tuned ampl^r at 1200K is 0.6; so the resulting signal peak amp. is decreased from 10V to 6V and for fmin = 200K; the peak amp. is 2V

So, O/P is given as:



So, the O/P of envelope detector is

The above doesn't resemble to m(t) but has some deviation and error is obtained.

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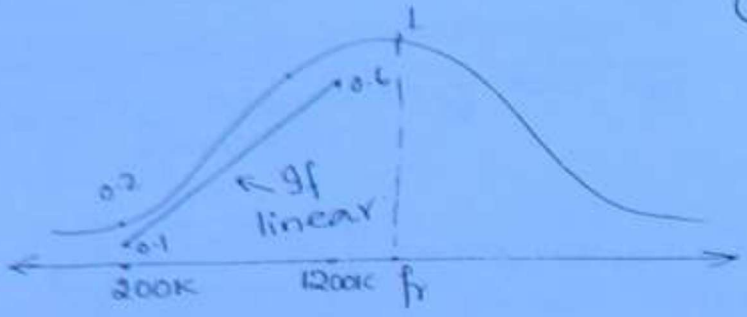
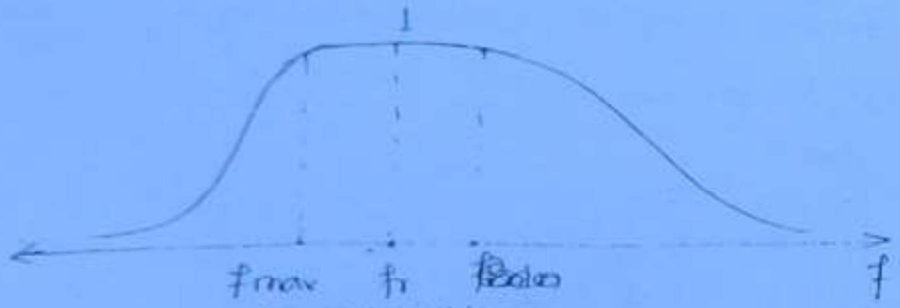
* The gain freq^{n} characteristics was linear if the same amount of gain is obtained as the frequency decreases, the msg. signal o/p of ED would be same as the original msg. signal and no error was present



* To avoid above a practical tuned ampl^{r} is constructed of gain frequency char as following:



(taken corresponding where the gain starts decreasing)

So, $\boxed{f_r > f_{max}}$

Note:-

1. Since gain frequency char. of Tuned Ampl^{r} is nonlinear in nature, some amount of non-linearity will be introduced in frequency to voltage conversion.
So the Reconstructed msg. signal is not perfectly corresponds to transmitted msg. signal and is called as SLOPE ERROR.

2. In Balanced slope detector, 2 slope detectors are connected in Balanced to decrease the slope Error, to the min^{m} possible extent.

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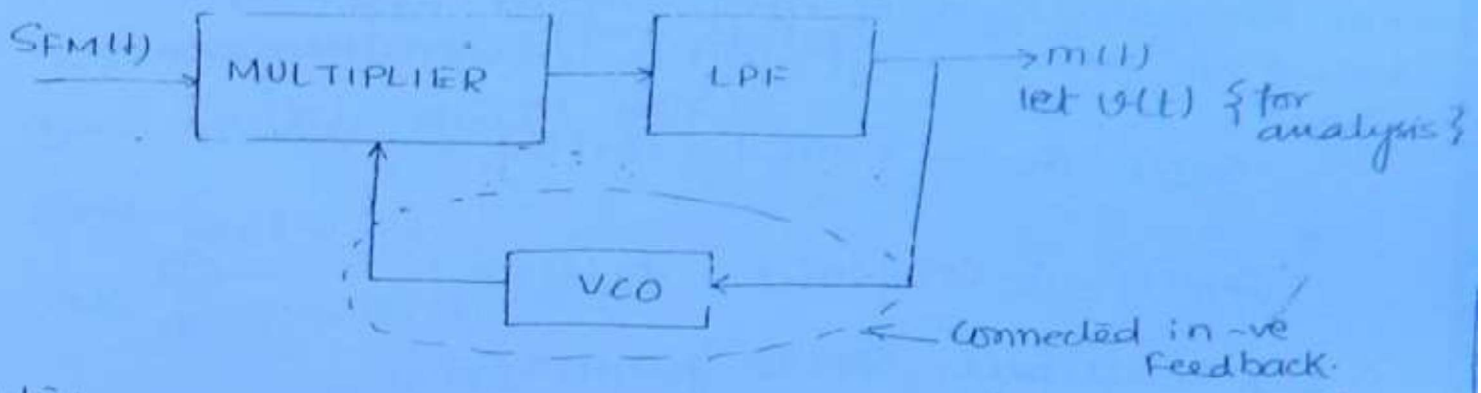
* PLL METHOD:

Block diagram:



Note:

1. The frequency of FM signal changes continuously w.r.t msg. signal voltage variations. So to maintain freqn synchronisation L.O of Synchronous detector is replaced by VCO
2. For the VCO, the msg. signal is taken as input, so that VCO o/p frequency changes continuously w.r.t m(t) voltage variations, frequency synchronisation can be achieved.

* As,

$$S_{FM}(t) = A_c \cos\left\{2\pi f_i t + \underbrace{2\pi K_f \int m(t)\,dt}_{\phi_1}\right\}$$

$$\underset{S_v(t)}{\phi_{(VCO)o/p}} = A_v \cos\left\{2\pi f_2 t + \underbrace{2\pi K_v \int v(t)\,dt}_{\phi_2}\right\}$$

For perfect Reconstruction of msg. signal,

i) f_i should be made equal to f_2 ie

$$\boxed{f_1 = f_2 = f_c}$$

Then PLL is said to be working in the LOCK MODE.

ii) $\phi_1(t)$ should be made equal to $\phi_2(t)$

$$\boxed{\phi_1(t) = \phi_2(t)}$$

Then PLL said to be working in the CAPTURE MODE.

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Note

Because of ve feedback, within no time f_1 will be made equals to f_2 and PLL will go to LOCK MODE.

(144)

* For Reconstruction of msg signal VCO output should have 90° phase shift w.r.t transmitter carrier; so that

$$S_V(t) = A_V \sin\{2\pi f_c t + \phi_2(t)\} \leftarrow \text{VCO o/p} \quad -(1)$$

&

$$S_{FM}(t) = A_c \cos\{2\pi f_c t + \phi_1(t)\} \quad -(2)$$

So the multiplier o/p is given as:

$$S_V(t) \times S_{FM}(t) = \frac{A_c A_V}{2}\left\{\sin(4\pi f_c t + \phi_1(t) + \phi_2(t) + \sin\{\phi_2(t) - \phi_1(t)\}\right\}$$

$$= \frac{A_c A_V}{2}\sin(4\pi f_c t + \phi_1(t) + \phi_2(t)) - \frac{A_c A_V}{2}\sin(\phi_1(t) - \phi_2(t))$$

let $\phi_1(t) - \phi_2(t) \simeq \phi_e(t) \leftarrow$ phase error

For PLL, the care will be taken to make $\phi_1(t)$ very close to $\phi_2(t)$ so that $\phi_e(t)$ will be very small.

So,

$$S_V(t) \cdot S_{FM}(t) = \frac{A_c A_V}{2}\sin\{4\pi f_c t + \phi_1(t) + \phi_2(t)\} - \frac{A_c A_V}{2}\sin\{\phi_e(t)\}$$

(Mul) o/p.

(no effect as blocked by LPF)

The 1st term in the above exp corresponds to very high frequencies, so it is not allowed by the LPF and since $\phi_e(t)$ is very small:

$$\sin\{\phi_e(t)\} \simeq \phi_e(t)$$

So

$$(Mul)_{o/p} \simeq -\frac{A_c A_V}{2}\phi_e(t)$$

$$(Mul)_{o/p} \simeq -\phi_e(t).$$

This is given to the LPF.

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So, $v(t) = \phi_e(t) \times h(t)$

So, taking Fourier xform on both side we get:

$$\therefore V(F) = \phi_e(F) \cdot H(F) \quad ---- (A)$$

Now, $\phi_E(F) = ?$

as, $\phi_e(t) = \phi_1(t) - \phi_2(t)$

$$= \phi_1(t) - 2\pi K_v \int v(t)\,dt.$$

where, $v(t) = \phi_e(t) \times h(t)$.

$$\frac{d\phi_e(t)}{dt} = \frac{d\phi_1(t)}{dt} - 2\pi K_v \cdot v(t)$$

$$\frac{d}{dt}\phi_e(t) = \frac{d}{dt}\phi_1(t) - 2\pi K_v \{\phi_e(t) \times h(t)\}.$$

taking F.T on both sides

$$j \cdot 2\pi f \phi_e(f) = j2\pi f \phi_1(f) - 2\pi K_v \{\phi_e(f) \cdot h(f)\}$$

$$\left\{\because \frac{d}{dt}\phi(t) \longleftrightarrow j\omega\phi(f)\right\}$$

$$\phi_e(f)\{jf + K_v H(f)\} = jf\phi_1(f)$$

$$\boxed{\phi_e(f) = \frac{jf\phi_1(f)}{\{jf + K_v H(f)\}}}$$

Also,

$$\phi_e(f) = \frac{\phi_1(f)}{1 + \frac{K_v H(f)}{jf}}$$

Now when [pass Band gain of LDF = ∞]

so, [$\phi_e(f) = 0 \Rightarrow \phi_e(t) \Rightarrow \phi_1(t) = \phi_2(t)$]

But practically it is not possible, so the pass Band gain of LPF is made high.

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Note:

1. If the open loop gain of LPF is ∞ ie

$$|H(f) = \infty| \quad (146)$$

then, $\phi_e(t) = 0$

but for a practical LPF, $H(f)$ will be finite

2. By making $H(f)$ = very large; $\phi_e(t)$ will be made very small quantity.

Now, substituting value of $\phi_e(f)$ in eqn (A) we get

So,

$$V(f) = \frac{\phi_1(f)}{1 + \frac{K_v}{jf} H(f)} \cdot H(f)$$

$H(f)$ is very large

So,

$$1 + \frac{K_v}{jf} H(f) \simeq \frac{K_v}{jf} H(f)$$

So,

$$V(f) = \frac{\phi_1(f)}{\frac{K_v}{jf} H(f)} \cdot H(f)$$

$$\boxed{V(f) = \frac{jf}{K_v} \phi_1(f)}$$

$$V(f) = \frac{jf\,\phi_1(f)}{K_v} \times \frac{2\pi}{2\pi}$$

By inverse F.T we get

$$v(t) = \frac{1}{2\pi K_v} \cdot \frac{d}{dt} \phi_1(t)$$

$$= \frac{1}{2\pi K_v} \cdot \frac{d}{dt} \times \left\{ 2\pi K_f \int m(t)\,dt \right\}$$

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So,

$$v(t) = \frac{K_f}{K_v} m(t)$$

Now.

Note:

1. If $S_v(t) = A_c \cos\{2\pi f_c t + \phi_2(t)\}$

$S_{FM}(t) = A_c \cos\{2\pi f_c t + \phi_1(t)\}$

So,

$$(Mul)\ o/p = \frac{A_c A_v}{2} \cos\{4\pi f_c t + \phi_1(t) + \phi_2(t)\} + \frac{A_c A_v}{2} \cos\{\phi_1 - \phi_2(t)\}.$$

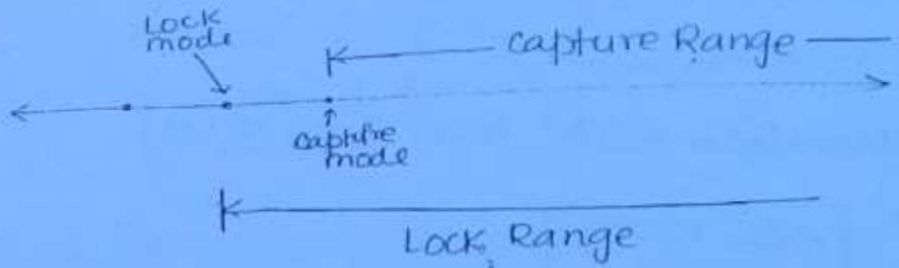
$$\phi_e(t) = \phi_1(t) - \phi_2(t) \approx 0$$

So, $\cos^0\{\phi_e(t)\} \approx 1$

So, Input of LPF = 1

Hence msg signal could n't be obtained.

2.



So,

$$L.R \geq C.R$$

Note:

1. For PLL; -ve feedback is Responsible for LOCK MODE.
2. LPF is responsible for CAPTURE MODE of PLL.
3. The frequencies produced by VCO corresponds to LOCK MODE of PLL.
4. For PLL:

$$L.R \geq C.R$$

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* PLL demodulation is based on the differentiation process.

* PHASE MODULATION:

* Carrier before phase modulation = $A_c \cos\{2\pi f_c t\}$.

* Carrier after phase modulation = $S_{PM}(t) = A_c \cos\{2\pi f_c t + \phi\}$
(Phase modulated signal)

↑ Phase deviation

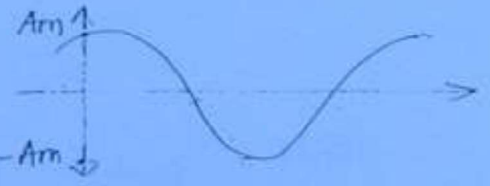
where, $\phi = K_p m(t)$

K_p = Phase sensitivity of phase modulator (rad/volt).

So,

$$S_{PM}(t) = A_c \cos\{2\pi f_c t + \underbrace{K_p m(t)}_{\text{Phase deviation}}\}$$ ← General exp. of phase modulation

let, $m(t) = A_m \cos\{2\pi f_m t\}$



So,

max^m phase deviation $= \Delta\phi = \max\{K_p m(t)\}$

$\therefore$ So, $\boxed{\Delta\phi = K_p A_m}$

So,

$S_{PM}(t) = A_c \cos\{2\pi f_c t + K_p \cdot A_m \cos 2\pi f_m t\}$.

let, $K_p \cdot A_m = \beta$ = modulation index of P.M.

Now,

Corresponding to F.M we have,

$$\beta = \frac{K_f A_m}{f_m} = \frac{\Delta f}{f_m}$$

$$\boxed{M.I = \frac{\text{Max}^m \text{ freq}^n \text{ deviation}}{\text{msg. signal freq}^n}}$$

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And corresponding to PM we have:

$$\beta = K_p A_m = \Delta\phi$$

$$\boxed{M.I = \max^m \text{ phase deviation}}$$

Note:

For phase modulation, β & $\Delta\phi$ are independent of msg. signal frequency variations

Now, let $\beta = K_p A_m$

So,

$$\boxed{S_{PM}(t) = A_c \cos\{2\pi f_c t + \beta \cos 2\pi f_m t\}}$$

&

$$\boxed{S_{FM}(t) = A_c \cos\{2\pi f_c t + \beta \sin 2\pi f_m t\}.}$$

Note:

1. General expressions of FM & PM are same except 90° phase shift at msg frequency component.
2. The magnitude spectrum of PM will be same as FM so that B.W and power requirements of PM & FM will be the same.

* B.W of PM (WBFM):

$$\boxed{B.W = 2(\beta+1) f_m} \Rightarrow \boxed{2(\Delta\phi+1) f_m = B.W}$$

* Power of PM:

$$\boxed{P_t = \frac{A_c^2}{2R}}$$

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Q1 A phase modulated signal is given by:

$$S_{PM}(t) = 10\cos\{2\pi \times 10^6 t + 6\sin 6\pi \times 10^3 t\}$$

(15b)

i) Find all parameters of PM.

ii) Repeat above if msg signal fo eqn is doubled

Soln:

$$S_{PM}(t) = 10\cos\{2\pi \times 10^6 t + \beta \sin 6\pi \times 10^3 t\}.$$

$$S_{PM}(t) = A_c \cos\{2\pi f_c t + \beta \sin 2\pi f_m t\}$$

i)

$A_c = 10$

$f_c = 1M = 1000K$

$f_m = 3K$

$\beta = 6\,rad = \Delta\phi.$

So, $B.W = 2(\beta+1) f_m$; $\boxed{\beta = \Delta\phi = 6\,rad.}$ Ans.

$= 2 \times 7 \times 3$

$\boxed{B.W = 42K}$ Ans

$\boxed{P_t = \frac{A_c^2}{2R} = \frac{100}{2} = 50W}$ Ans.

Now,

ii) let $f_m = 6K$; $\boxed{\beta = \Delta\phi = K_p A_m = 6\,rad}$ Ans.

So, $B.W = 2(\beta+1) f_m$

$= 2 \times 7 \times 6$

$\boxed{B.W = 84K}$ Ans

$\boxed{P_t = \frac{A_c^2}{2R} = \frac{100}{2} = 50W.}$ Ans.

Note :-

For PM; as the msg signal frequency doubles the corresponding B.W will also be doubled.

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* Generation of FM from PM & PM from FM:

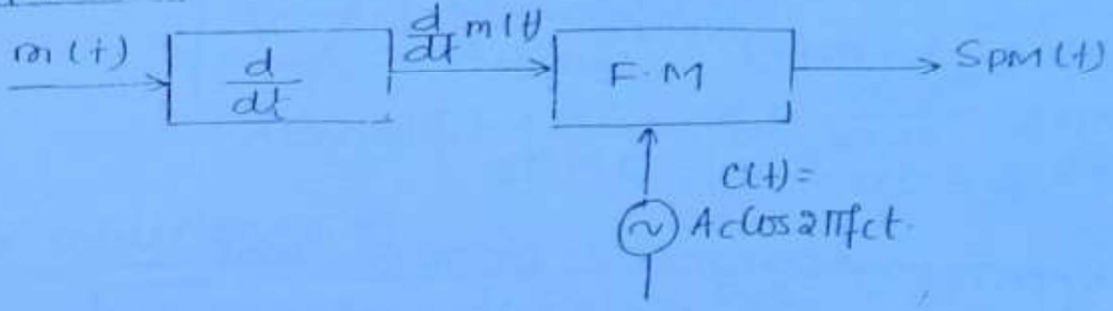
As,

$$S_{PM}(t) = A_c \cos\{2\pi f_c t + K_p m(t)\}$$ (15)

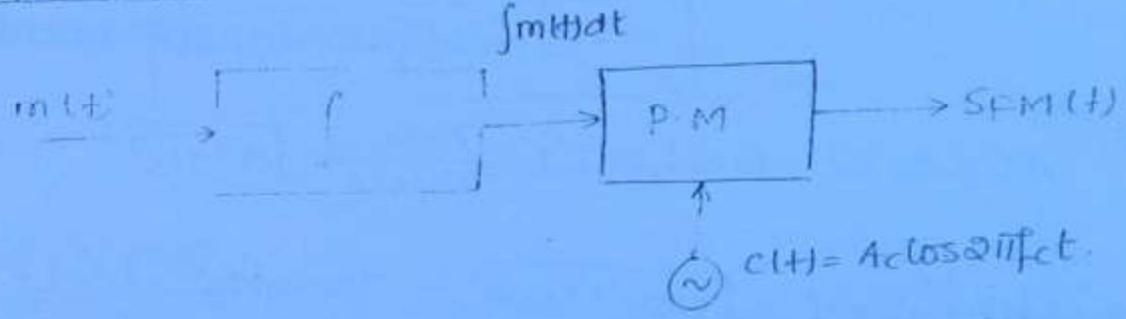
&

$$S_{FM}(t) = A_c \cos\{2\pi f_c t + 2\pi K_f \int m(t)dt\}$$

* PM from FM:



* FM from PM:



Note:

1. Phase modulation of $m(t)$ is nothing but frequency modulation of $d/dt\, m(t)$.

Q1. An Angle modulated signal is given by

$$S(t) = \cos\{2\pi(10^6 \times 2t + 30\sin 150t + 40\cos 150t\}.$$

Find max^m freq^n deviation & max^m phase deviation?

Sol^n: Given:

$$S(t) = \cos \underbrace{2\pi\{2\times10^6 t + 30\sin 150t + 40\cos 150t\}}_{\Phi i(t)}.$$

$\therefore$ Angle modulated signal, so $\Phi i(t)$

$$\Phi i(t) = 2\pi\{2\times10^6 t + 30\sin 150t + 40\cos 150t\}$$

FOR FM:

So, $f_i = \frac{1}{2\pi}\cdot\frac{d}{dt}\Phi i(t)$

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$$f_i = \frac{1}{2\pi}\cdot\frac{d}{dt}\{2\pi(2\times10^6 t + 30\sin 150t + 40\cos 150t)\}$$

$$= \frac{1}{2\pi}\{2\pi(2\times10^6 + 4500\cos 150t - 6000\sin 150t)\}$$

Now,

$f_i = f_c + K_f m(t)$ ← frequency deviation.

So, $f_c = 2\times10^6$ Hz $= 2$ MHz

& $K_f m(t) = 4500\cos 150t - 6000\sin 150t$

So, max^m freq^n deviation, $\Delta f = \max\{k_f m(t)\}$.

$= \max\{4500\cos 150t - 6000\sin 150t\}$

as. $A\cos 2\pi f_0 t + B\sin 2\pi f_0 t \xrightarrow{\text{max value}} \sqrt{A^2+B^2}$

So, $\Delta f = \sqrt{4500^2 + 6000^2} = 7500$ Hz

$\boxed{\Delta f = 7.5\text{ KHz}}$ Ans.

For PM:

As, $s_{pm}(t) = A_c\cos\{2\pi f_c t + K_p m(t)\}$. ← phase deviation

Now,

$s(t) = A_c\cos 2\pi\{2\times10^6 t + 30\sin 150t + 40\cos 150t\}$

$= A\cos\{4\pi\times10^6 t + 60\pi\sin 150t + 80\pi\cos 150t\}$

So, $2\pi f_c = 4\pi\times10^6$

$k_p m(t) = 60\pi\sin 150t + 80\pi\cos 150t$

$\Delta\phi = \max\{k_p m t\} = \sqrt{(60\pi)^2 + (80\pi)^2}$

$\boxed{\Delta\phi = 100\pi\text{ rad.}}$ Ans.

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Work
Pg: 16/6 21

$\Delta f_{max} = 1000$ → $\phi(t)$ $10\cos\{\omega_c t + 5\sin 3000t + 10\sin 2000\pi t\}$ $A\cos 2\pi f_c t + B\sin 2\pi f_c t$

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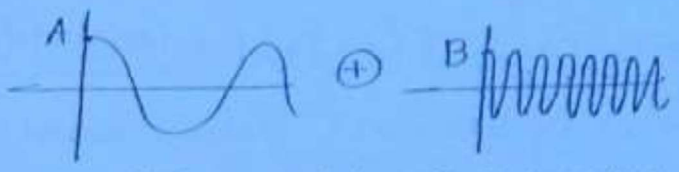
Now,

$A\cos 2\pi f_1 t + B\cos 2\pi f_2 t \xrightarrow[\text{value}]{\text{max}^m} (A+B)$



→ when added max^m value (A+B).

$A\sin 2\pi f_1 t + B\sin 2\pi f_2 t \xrightarrow[\text{value}]{\text{max}^m} A+B)$

* For multitone FM:

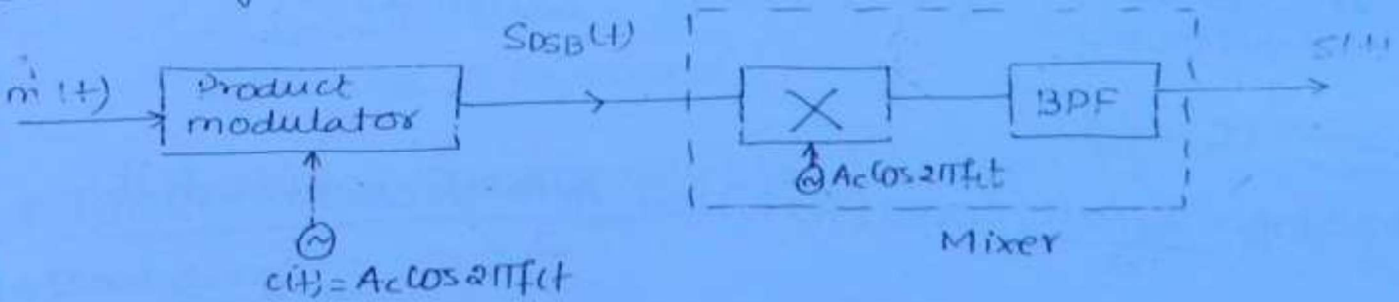
$$\text{Mod}^n \text{ Index} = \text{deviation Ratio} = \frac{\Delta f}{f_{max}}$$

$$\text{B.W} = 2(\beta+1) f_{max}$$

* MIXER:

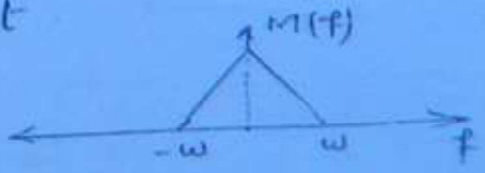
Mixer is used for Frequency station of Modulated signal.

Block diag^m:

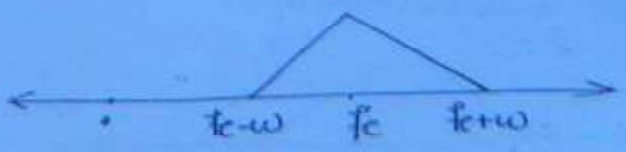


$S_{DSB}(t) = A_c m(t) \cos 2\pi f_c t$

let $m(t) \longleftrightarrow M(f)$



$S_{DSB}(t) \longleftrightarrow$



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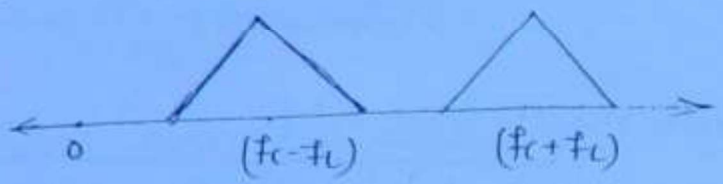
$(Mul)_{O/P} = S_{DSB}(t) \cdot (LO)_{O/P}$

$= A_c m(t) \cos 2\pi f_c t \times \cos 2\pi f_L t$

Case i $(f_c > f_L)$

$$(Mul)_{O/P} = \frac{A_c m(t)}{2}\left\{\cos 2\pi(f_c + f_L)t + \cos 2\pi(f_c - f_L)t\right\}.$$

(mul) O/P ⟷



So,
(Mixer) O/P (depending upon Pass Band of filter)

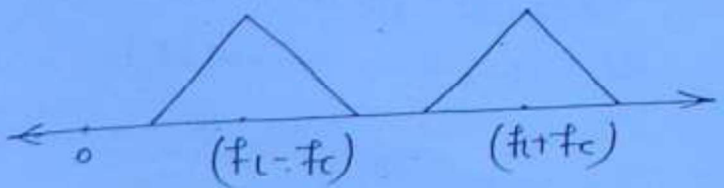
when B.P.F O/P is this, it is called as DOWN-CONVERSION

when B.P.F allows the $(f_c + f_L)$ component; it is called as UP-CONVERSION

Case ii : $(f_c < f_L)$.

$$(Mul)_{O/P} = \frac{A_c m(t)}{2}\cos 2\pi(f_L + f_c)t + \frac{A_c m(t)}{2}\cos 2\pi(f_L - f_c)t$$

(mul) O/P ⟷



(Mixer) O/P (down conversion) or (up conversion)

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Note:

1. Down conversion will give subtraction of the frequencies and upconversion gives sum of the frequencies

(55)

Date 11-11-11 *PSU's)

* RECIEVERS:

- Tuned Radio Frequency (TRF).
- Super heterodyne (SHD).

* Depending on modulation scheme:

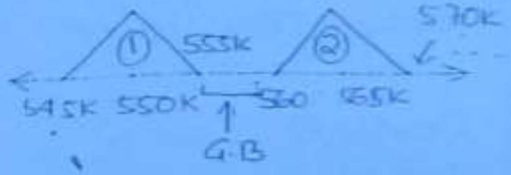
1) AM Reciever.
2) FM Reciever.

AM standards:

According to federation commitee of comm(FCC):

Carrier freq^n = 550 K to 1650K.

A.M B.W = 10K.



FM standards:

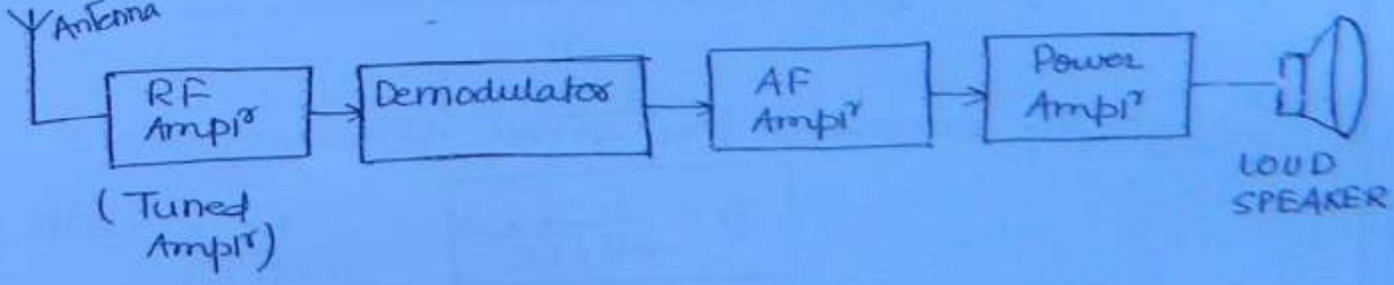
According to FCC:

Carrier Freq^n = 88 to 108 MHz.

FM ~~FB~~ BW = 200K.

* TUNED RADIO FREQUENCY (TRF) Reciever:

* Block diagm:



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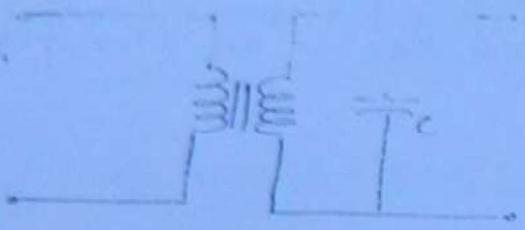
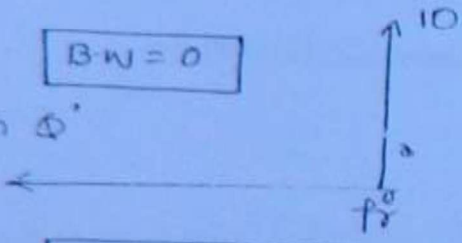
Tuned CKt (of Ampl^r):



* Gain Frequency Char. of Tuned Ampl^r:

Ideal Tuned Ampl^r → "high Q"

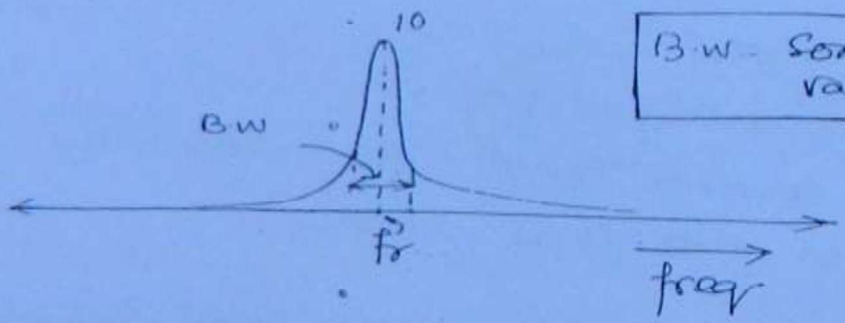


← allows only single frequency ie fr and rejects other frequency

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

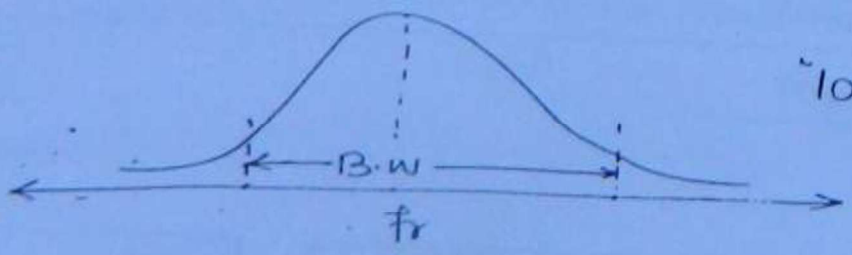
* The Range of frequencies which the Tuned ampl^r, passes w/o any Attenuation is called as BW of Tuned Ampl^r.

Practical Tuned Ampl^r → (highly Qualitative)



B.W. Some finite value

Practical Tuned Ampl^r (General case)



"low Q"

The Resonant freq^n given as:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Quality factor given as:

$$Q = \frac{1}{2\pi}\sqrt{\frac{L}{C}}$$

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Note:

1. Quality factor is also said as the sharpness of the GAIN-FREQ^N CHAR.

   Sharpness high = Q high.
   Sharpness low = Q low.

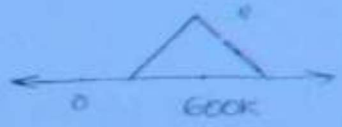
2. Also, B.W is inversely ~~proportional~~ proportional to Q.

$$B.W \propto 1/Q$$
$$B.W = f_r/Q$$

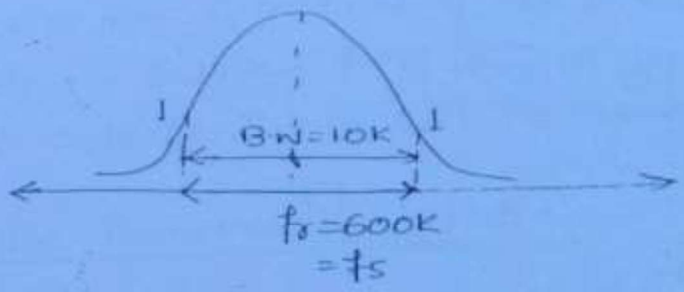
*Opn of TRF Rx:

Case 1:

1) Assume Rx is tuned to 600K station
   ie f_c = 600K



2) Assume Tuned Ampl^n should be



3) f_r should be equal to 600K ; f_r=600K

4) To get B.W of 10K ; Q should be

$$B.W = \frac{f_r}{Q}$$

$$Q = f_r/B.W = \frac{600K}{10K}$$

$$\boxed{Q = 60}$$

5) Now, $f_r = \frac{1}{2\pi\sqrt{LC}} = 600K$.

$$Q = \frac{1}{2\pi} \times \sqrt{\frac{L}{C}} = 60.$$

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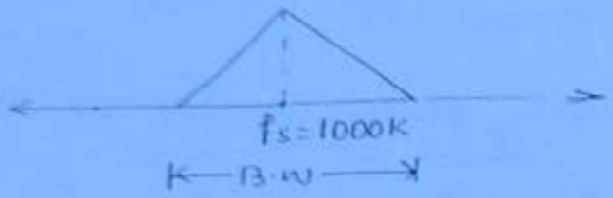
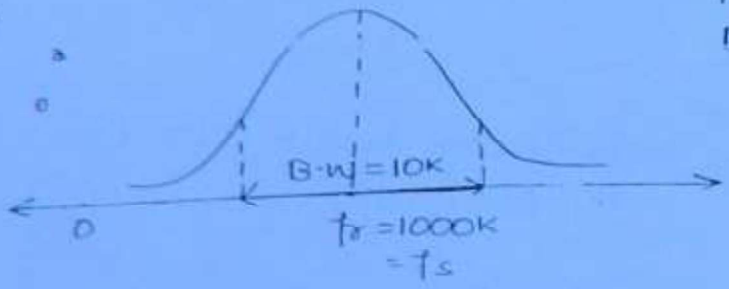
Case 2:

1. Assume the Rx is tuned to 1000K station
   i.e $f_s = 1000K$.



2. Tuned Ampl^r should be as:-

$f_0 = 1000K$
B·W = 10K



3. To get B·W of 10K; Q should be

$$\boxed{Q = 100}$$

4. As. $f_0 = \frac{1}{2\pi\sqrt{LC}}$; $Q = \frac{1}{2\pi}\sqrt{\frac{L}{C}}$

for changing the freqency. Capacitance is Varied.
Such that $f_0 = 1000K$
but correspondly due to diff formulaes of f_r & Q, the value of Q is not exactly 100.

So, let for $f_0 = 1000K \Rightarrow Q = 120$

So,
$$B.W = \frac{f_0}{Q} = \frac{1000}{120}$$

$$\boxed{B.W = 8.3K}$$

Hence, Some B.W of Tuned Ampl^r is lost and the msg signal correspondly can't be Reconstructed.

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Conclusion:

1. If B.W of T.Ampl^r : 100K ⇒ m(t) can be perfectly Reconstructed

2. If B.W of Tuned Ampl^r < 10K ⇒ m(t) can't be Reconstructed perfectly & freq^n attenuated

3. If B.W of T.Ampl^r > 10K ⇒ m(t) can't be Reconstructed perfectly & some unwanted freq^n are allowed.

* Due to the above discussion the selectivity of the TRF Rx is poor.

Note:

1. For TRF Rx, as the Tuning changes B.W of the Tuned Ampl^r will be changed accordingly.

2. For excellence in selectivity, to what may be the station, Rx is Tuned for B.W of Tuned Ampl^r should be 10K only. So selectivity of TRF Rx is worst.

* Characteristic Parameters of Rx:

1. Selectivity: It is the ability of the Rx to allow only the desired freq^n components and Rejecting undesired frequency components.
It mainly depends on characteristics of Tuned Ampl^r.

2. Sensitivity: It specifies the min^m strength of the signal to be maintained at the Rx input; to produce faithful output.

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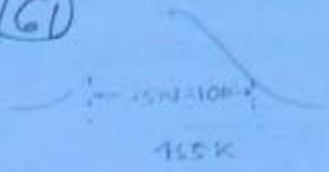
× IF Ampl^r = Tuned Ampl^r always tuned to 455 K and B.W = 10K



So, $\boxed{Q = \frac{f_r}{B.W} = 45.5}$

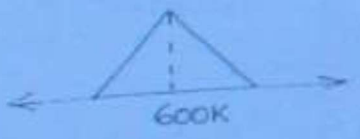
× Mixer is designed to work as a down convertor and f_L is preferred to take as greater than f_S ($f_L > f_S$); so that Mixer O/P is equal to:

$$\boxed{f_L - f_S = 455\ K = IF = \text{Intermediate freqn}}$$

Opn:

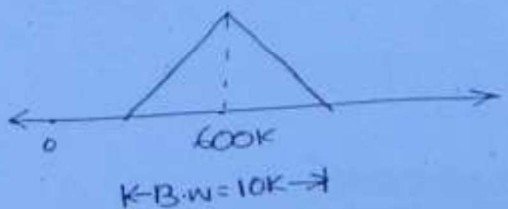
case 1:

1. Assume Rx is tuned to 600KHz station.



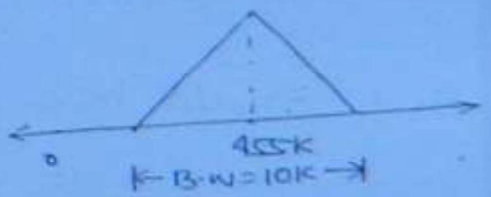
2. I.F ampl^r tuning →

3.



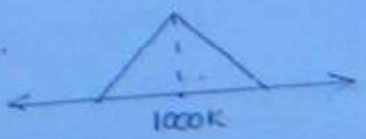
mixer

f_L = 1055 K

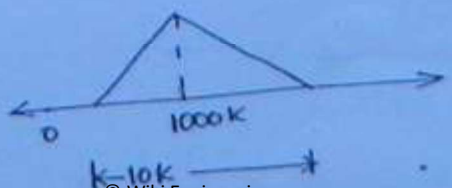


case 2:

1. Assume Rx is tuned to 1000 KHz station.

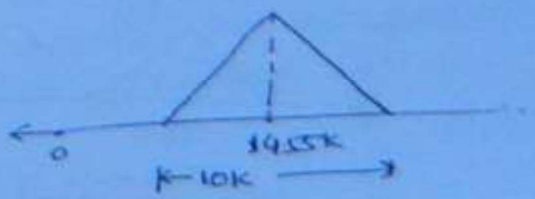


2.



mixer

f_L = 1455 K



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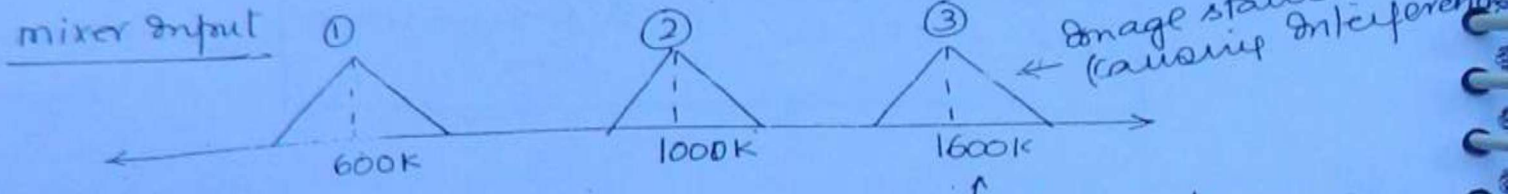
Note:

1. In TRF Rx, tuning is achieved by changing f_r of the tuned Ampl^r.
2. In SHD Rx; tuning is achieved by changing "f_L" of the local oscillator.

*** Image Frequency:

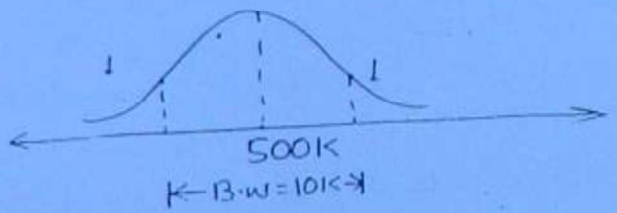
let the messages spectrum be:



2. Assume Rx is tuned to 600K

$f_s = 600K$

3. let intermediate frequency. IF = 500K

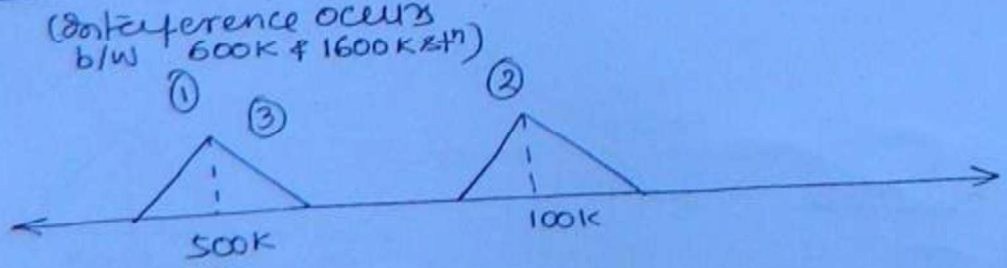


4. f_L should be, $f_L - f_s = I.F$

$f_L = I.F + f_s = 500K + 600K$

$\boxed{f_L = 1100K}$

Mixer O/P:



1100 - 600 = 500K

1600 - 1100 = 500K

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*1600K station is causing interference to 600K station so 1600K is said to be image frequency of 600K

Now,

Image freq^n. $f_{si} = f_s + 2\,I.F$

let,

$I.F = 500K$

$f_s = 600K$

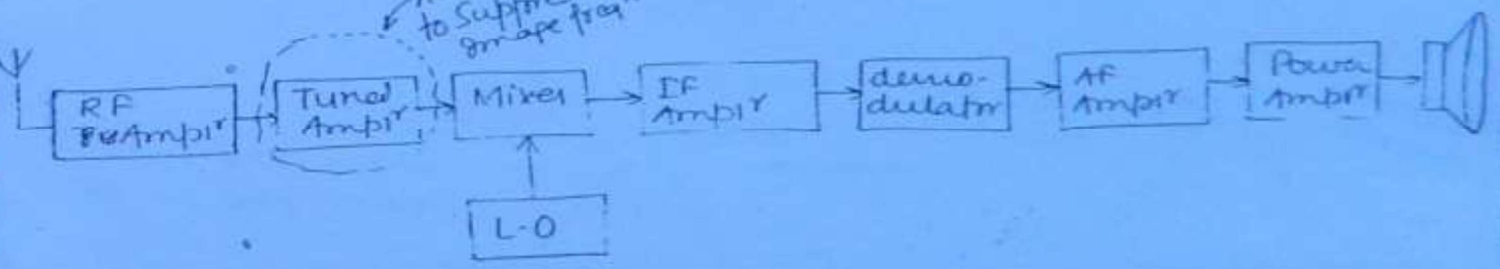
$f_{si} = 600 + 1000$

$\boxed{f_{si} = 1600K}$

*Suppression of Image frequency:

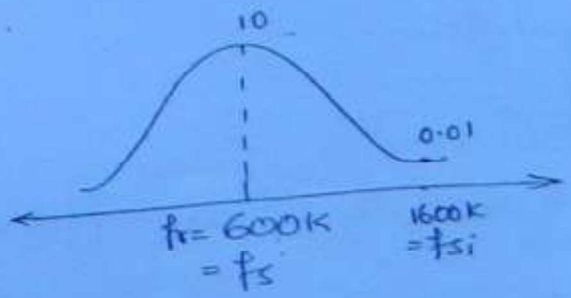
1. Since, desired station and image station are occupying same frequency Band at Mixer output, so for comfortable Reconstruction of desired signal image frequency should be suppressed.

2. To suppress image frequency, a tuned Ampl^r will be used before mixer.



"Practical SHD"

*Gain-Freq^n char. of Tuned Ampl^r:



10V ↑ 10V ↑

$f_s = 600K$; $f_{si} = 1600K$

the Resonant freq^n of this Ampl^r is not fixed as that of IF Ampl^r. But it can be varied in accordance to the desired st^n.

So, o/p of Tuned Ampl^r ; $f_s = 600K \rightarrow 10V$

$f_{si} = 1600K \rightarrow 0.1V$

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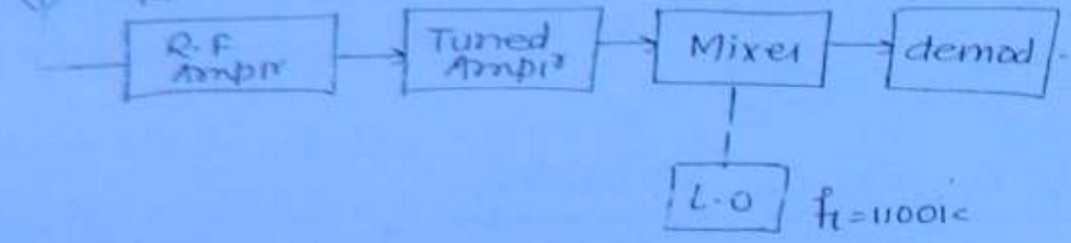
fs = 1600K - 10V 500K - 10V } same strength hence interference is high & m(t) is distorted

RF Ampl' → Mixer → demod

fl = 1100K

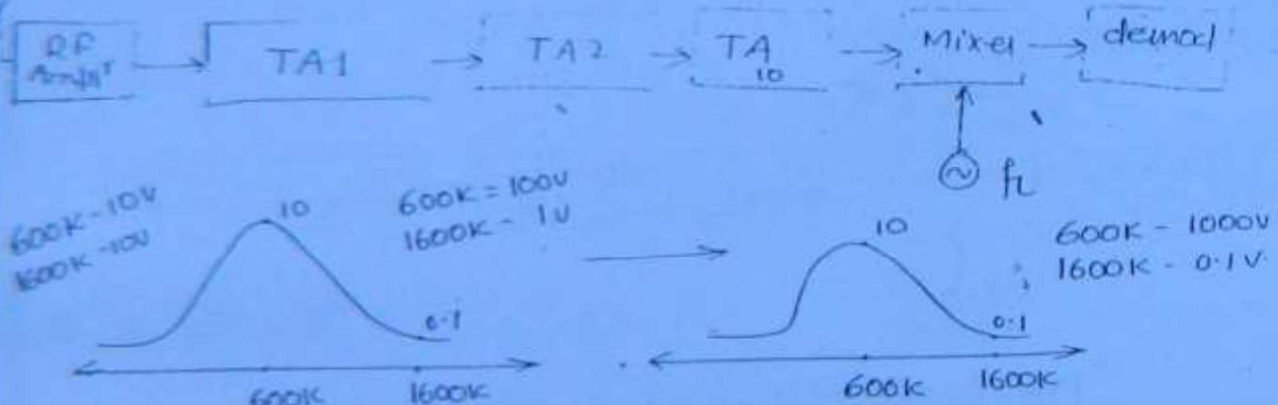
(64)

fs = 600K - 10V 500K - 100V } diff strength; hence less
fs = 1600K - 10V 500K - 0.1V } interference; so m(t) easily Reconstructed



*For much separation of image freq"; Cascaded Tuned Ampl' will be used at the output of Mixer.

So,



Note:

*For getting optimum suppression, Cascaded Tuned Ampl' should have same characteristics.

* Image Rejection Ratio (IRR):

It specifies, effectiveness of tuned Ampl' in suppressing image frequency.

OR

It specifies how many times, image frequency is attenuated w.r.t desired frequency.

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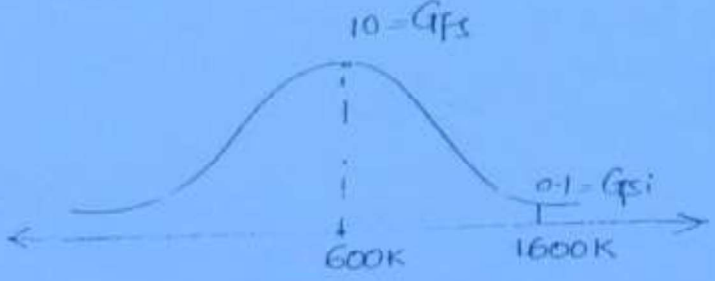
Mathematically.

$$IRR = \frac{\text{Gain offered by TA to fs}}{\text{Gain offered by TA to fsi}}$$

or $$\alpha = IRR = \frac{G_{fs}}{G_{fsi}}$$

So,



$$\alpha = \frac{G_{fs}}{G_{fsi}} = 10/0.1$$

$$\alpha = 100$$

If Tuned Ampl^r are connected in cascade then,

$$\text{effective } IRR = \alpha_1 \cdot \alpha_2 \cdot \alpha_3$$

So, from above discussion

$\alpha_1 = 100$; $\alpha_2 = 100$

$\therefore$ effective $\alpha = 100 \times 100$

$= 10000$

* If G_{fs} & G_{fsi} are not known then;

$$\alpha = \sqrt{1 + P^2 Q^2}$$

where, Q = Quality factor of TA

$$P = \frac{f_{si}}{f_s} - \frac{f_i}{f_{si}}$$

* If 2 tuned Ampl^r of having diff. char are connected in CASCADE, then

$$\alpha = \sqrt{P^2 Q_1^2 + 1} \times \sqrt{1 + P^2 Q_2^2}$$

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× If tuned Ampl^n having same char. then;

$$\alpha = \sqrt{1+P^2\Phi^2} \times \sqrt{1+P^2\Phi^2}$$

$$= 1+P^2\Phi^2$$

Q1. A Rx is tuned to 600KHz, I.F = 450KHz
Find f_L & f_{si} ?

Soln: Given: $f_s = 600K$
$f_i = 450K$

So, $f_i = f_L + f_s$

$f_L = f_i + f_s = 600 + 450$

$\boxed{f_L = 1050K}$ Ans

And, $f_{si} = f_s + 2I_f$
$= 600K + 900K$

$\boxed{f_{si} = 1500K}$ Ans

Q2. A Rx is tuned to 500K; local osc. freqⁿ is given b
1050K. Find i) I.F and f_{si}
ii) Find IRR, if $\Phi = 50$.

Soln: Given:
$f_s = 500K$
$f_L = 1050K$

; $f_i = f_L - f_s$
$= 1050 - 500$

$\boxed{f_i = 550K}$

So, $f_{si} = f_s + 2I_f$
$= 500 + 1100K$

$\boxed{f_{si} = 1600K}$

Now, $P = \frac{f_{si}}{f_s} - \frac{f_s}{f_{si}} = \frac{1600}{500} - \frac{500}{1600}$

$P = 2.8$

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So, $\alpha = \sqrt{1+P^2Q^2}$

$= \sqrt{1+(2.8)^2 \times 100}$

(127)

$\boxed{\alpha = 144.3}$ Ans

Q3. A Rx is tuned to 750K; Corresponding Jonage freqn is given by 1750K. Find i) f_L & I.F
ii) Find IRR if 2 tuned ampl^r of having Q = 50 & 70 are connected in cascade

Soln: Given: $f_s = 750K$

$f_{si} = 1750K$

So, $f_{si} = f_s + 2I_f$

$I_f = \frac{1750 - 750}{2} = 500K$

Now, $f_i = f_L - f_s$

$500 = f_s - 750K$

$f_s = 1250K$

Now, $P = \frac{f_{si}}{f_s} - \frac{f_s}{f_{si}} = \frac{1750}{750} - \frac{750}{1750} = 1.9$

Now, $\alpha = \sqrt{1+P^2Q_1^2} \times \sqrt{1+P^2Q_2^2}$

$= \sqrt{1+1.9^2 \times 2500} \times \sqrt{1+1.9^2 \times 4900}$

$\boxed{\alpha = 12636.05}$

Q4. A Rx is tuned to 1MHz, I.F = 455 KHz and Q = 100.
i) Find IRR
2) Find IRR if the Rx is tuned to 25MHz

Soln: Given, $f_s = 1000K$

I.F = 455K

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$f_{si} = f_s + 2f$

$= 1000 + 910$ K

$f_{si} = 1910$ K

So, $P = \frac{f_{si}}{f_s} - \frac{f_s}{f_{si}} = \frac{1910K}{1000} - \frac{1000}{1910}$

$P = 1 - 386$

So, $\alpha = \sqrt{1 + P^2 Q^2}$

$= \sqrt{1 + 1386^2 \times 10000}$

$\boxed{\alpha = 138.6}$

Now,

$f_s = 25M = 25000$ K

So, $f_{si} = f_s + 2f$

$= 25000 + 910$

$= 25910$ K

$P = \frac{f_{si}}{f_s} - \frac{f_s}{f_{si}} = \frac{25910}{25000} - \frac{25000}{25910}$

$P = 0.072$

So, $\alpha = \sqrt{1 + P^2 Q^2}$

$= \sqrt{1 + 0.072^2 \times 10000}$

$\boxed{\alpha = 7.26}$

Note:

1. In the 2nd case, when $f_s = 25$ MHz; α is very small ie 7.
2. For comfortable reconstrn of signal α has to be increased.

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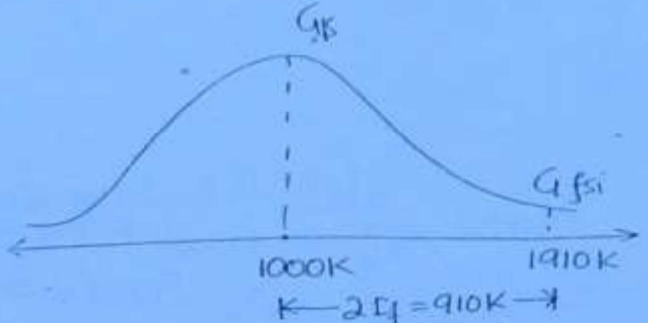
So, f_{si}

Case 1 (when $f_s = 1M$):



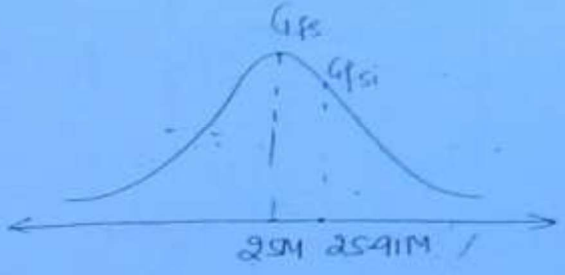
$f_s = 1M$

$f_s = 1000K$

$Q = 100$

$\boxed{B.W = 10K}$

Case 2 (when $f_s = 25M$)



$f_s = 25M$

$f_s = 25000K$

$Q = 100$

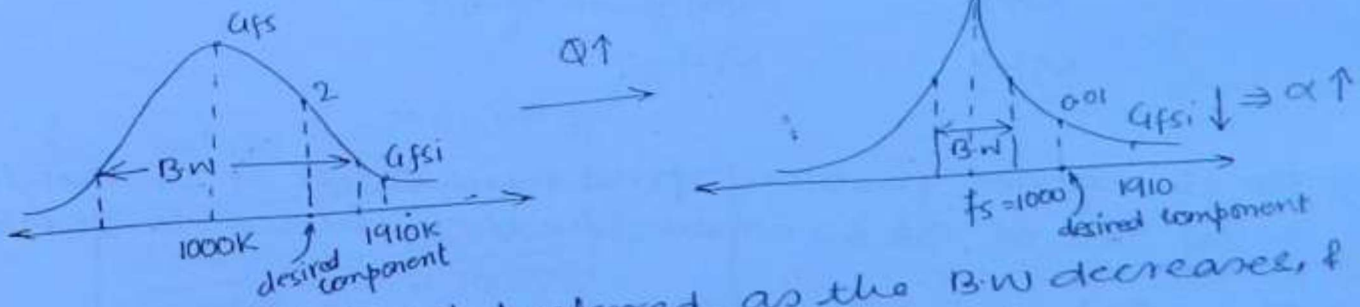
$\boxed{B.W = 250K}$

Drawbacks

*Measures to increase α:

As $\alpha = \sqrt{1 + P^2 Q^2}$

i) by increasing Q:



Practically it is not preferred, as the B.W decreases, & hence the selectivity of the Rx is affected.

Hence, it may be concluded that some of the desired frequency components will be attenuated. Hence the desired msg. signal (audio signal) cannot be perfectly Reconstructed.

Note:- The above is not suggested because B.W of the Tuned Ampl^r will be very much decreased and it affects the selectivity of the Rx.

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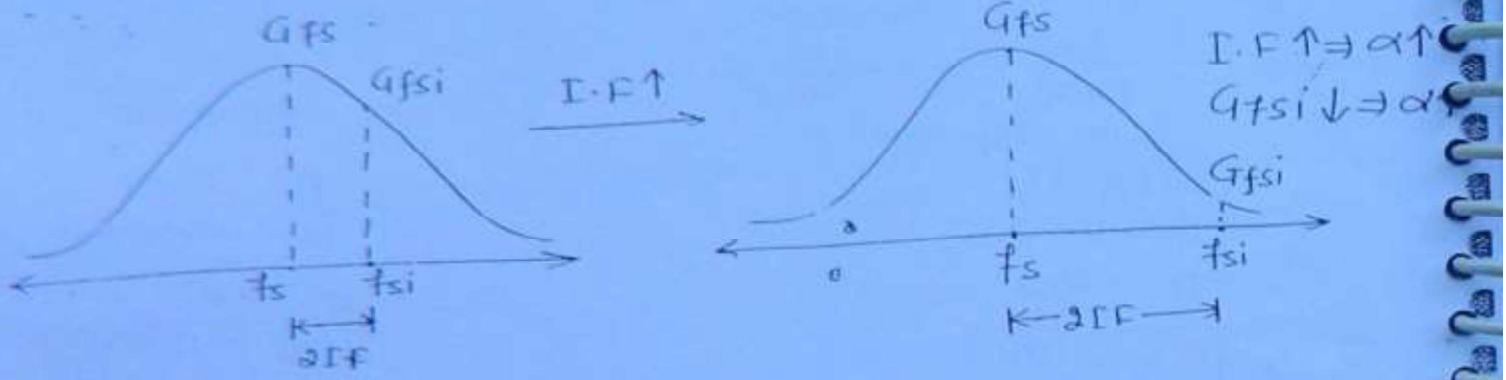
As $\uparrow P = \frac{f_{si}\uparrow}{f_s} - \frac{f_s}{f_{si}}$

Hence, $\uparrow f_{si} = f_s + 2I.F \uparrow$



Conclusion:

To get high value of IRR; both fs and IF should be in the same order.

fs	IF
KHz	KHz
MHz	MHz

Q. For the above problem, find new value of IF required to get IRR of 138.6? when the Rx is tuned to 25MHz?

Soln: Given, $\alpha = 138.6$

$f_s = 25000K$

So, $\alpha = \sqrt{1+P^2Q^2} \Rightarrow 138.6$

$1+P^2Q^2 = (138.6)^2$

$PQ = 138.6$

$P = 1.385$

NOW, $\frac{f_{si}}{f_s} - \frac{f_s}{f_{si}} = 1.385$

$\frac{f_{si}^2 - f_s^2}{f_s \times f_{si}} = 1.385$

$\Rightarrow \frac{f_{si}^2 - 625}{f_{si}} = 34.65$

$f_{si}^2 - 34.65 f_{si} - 625 = 0$

$f_{si} = 47.74\,M$

$\Rightarrow f_s + 2I_f = 47.74$

$\boxed{I_f = 11.37\,M}$ Ans

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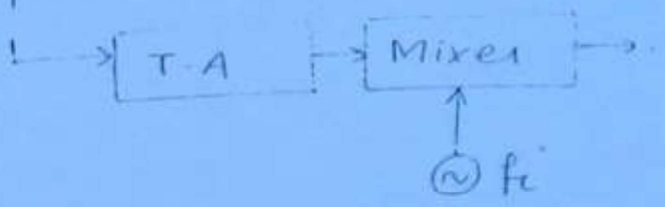

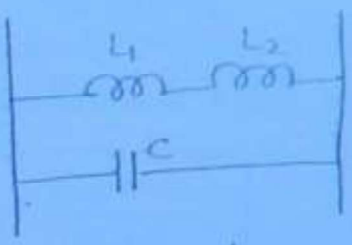

$$f = \frac{1}{2\pi\sqrt{(L_1+L_2)\cdot C}}$$

Case 1: $(f_L > f_S)$

Assume $I_f = 500K$

$IF = f_L - f_S = 500K$

Now, for AM, Range is $f_S \rightarrow 550K$ to $1650K$

i. when $f_S = 550K \Rightarrow f_L = 1050K$

So, $f_r = \frac{1}{2\pi\sqrt{(L_1+L_2)C_1}} \Rightarrow 1050 = \frac{1}{2\pi\sqrt{(L_1+L_2)\cdot C(max)}}$ — (1)

2. when, $f_S = 1650K \Rightarrow f_L = 2150K$

So, $2150 = \frac{1}{2\pi\sqrt{(L_1+L_2)C(min)}}$ — (2)

So, eqn (2) ÷ eqn (1) we get:

$$\boxed{\frac{C_{max}}{C_{min}} \simeq \left(\frac{2150}{1050}\right)^2 \simeq 4}$$

Case 2: $(f_L < f_S)$

let $I_F = 500K$

$IF = f_S - f_L = 500K$

i. $f_S = 550K \Rightarrow f_L = 50K$

So, ~~$50K = \frac{1}{2\pi\sqrt{(L_1+L_2)C_{min}}}$~~ — (1)

$50K = \frac{1}{2\pi\sqrt{(L_1+L_2)C_{max}}}$ — (1)

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2 when $f_s = 1650K$ $\Rightarrow f_1 = 1150K$

So, $1150K = \frac{1}{2\pi\sqrt{(L_1+L_2)C_{min}}}$ ... (2)

eq(2) ÷ eq(1) we get

$$\frac{C_{max}}{C_{min}} \approx \left(\frac{1150}{50}\right)^2 \approx 500$$

Conclusion:

Tuning of capacitor will be easy for $f_1 > f_s$. So, it is preferred.

* AM – For long distance
  FM – For short distance

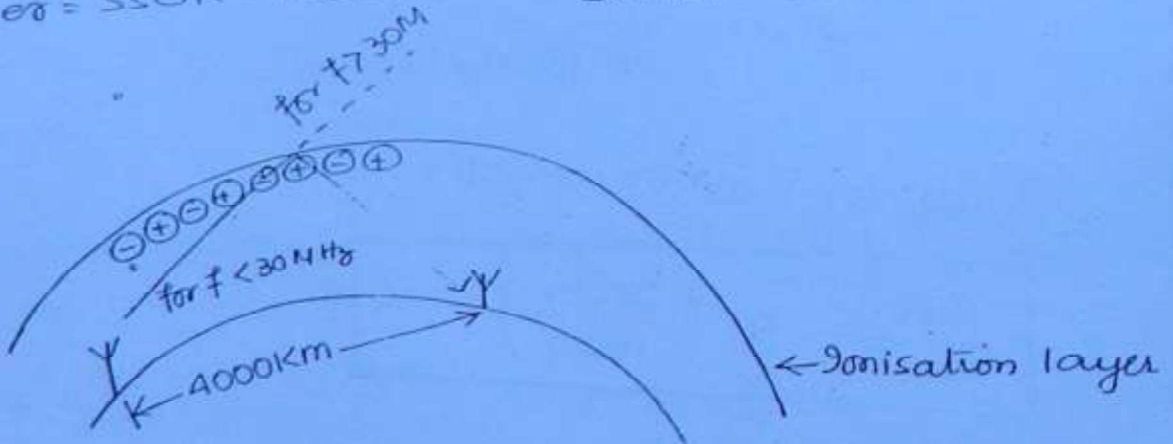
Discussion :-

As, for AM:

Carrier = 550K – 1650K

For FM

Carrier – 88M to 108M.



* Due to the property of Ionisation layer, if the freqn < 30M, it will reflect back the signal.
* But if f > 30M, escapes the layer and never comes back.
* If the distance b/w xmitter and Rx is 4000 Km (upto) it is called as Single HOP propagation.

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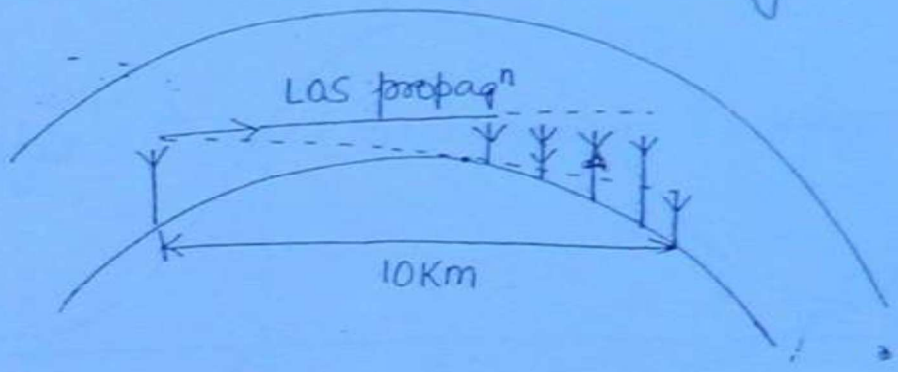
* If for distance >1000Km, transceivers are used to regenerate the signal, and is called as Ionospheric layer propagation.

For FM: For FM, line of sight propagation used. The max^m distance that may be covered by FM is 10Km.



* If LOS propagation is not maintained, then the FM signal is blocked by curvature of Earth and signal is lost.

* Freq^n Reuse Technique can be implemented.

Note:

1. For AM; Sky wave propagation is used where long distance comm^n is possible.
2. In FM; LOS propagation is used, where short dist comm^n is possible.
3. Freq^n Reuse concept can be effectively implemented in FM xmission.

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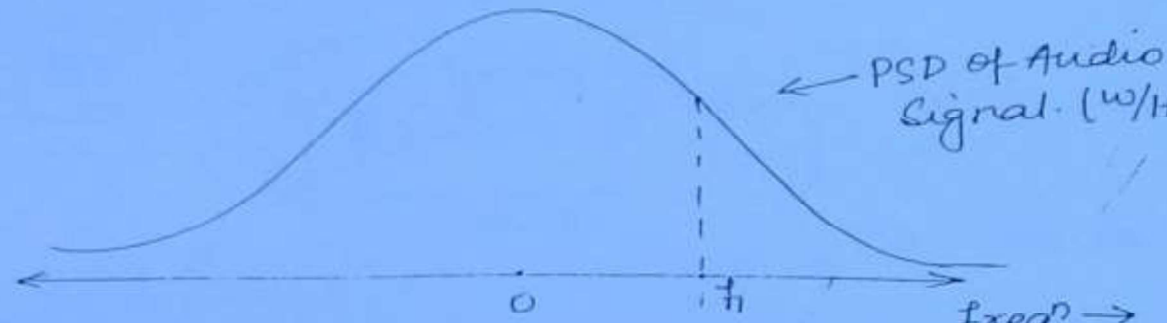
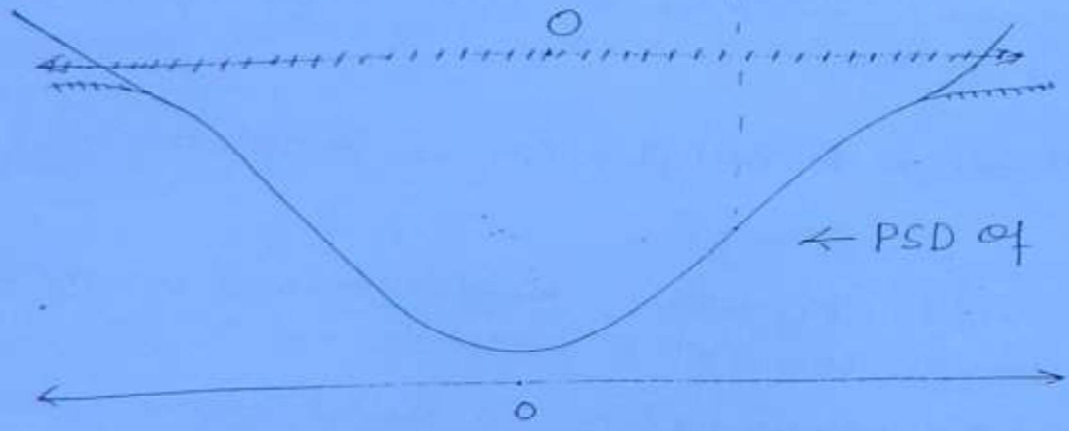
* PRE-EMPHASIS & DE-EMPHASIS:

As we know that, the PSD of Audio Signal is given by:



Also, PSD of Noise (undesired signal)



Analysis:

* upto f_1; $\frac{S}{N} > 1$; these freqⁿ components can be Reproduced at the Rx output.
* Above f_1; $S/N < 1$; these high freqⁿ components cannot be Reproduced at the Rx output
* Hence, too high freqⁿ, ∴
  i) either S power has to be increased
  ii) decreasing N power (but it is Indeterministic).
* So, artificially increasing the signal strength at high freqⁿ is called as PRE-EMPHASIS.

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*PRE-EMPHASIS:-

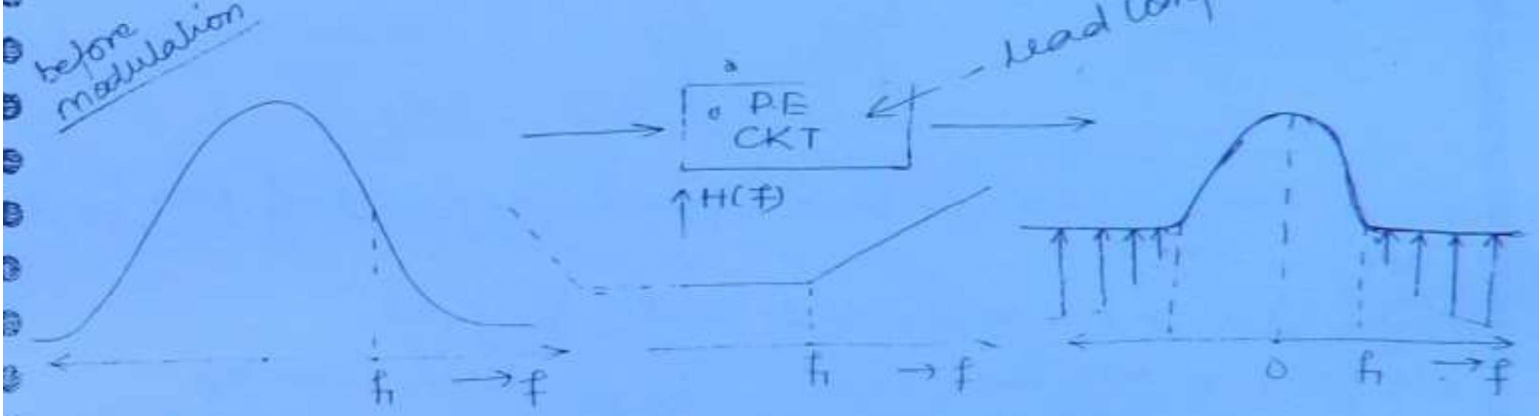
* To improve fidelity S/N of high freqn components of the Audio signal has to be increased.

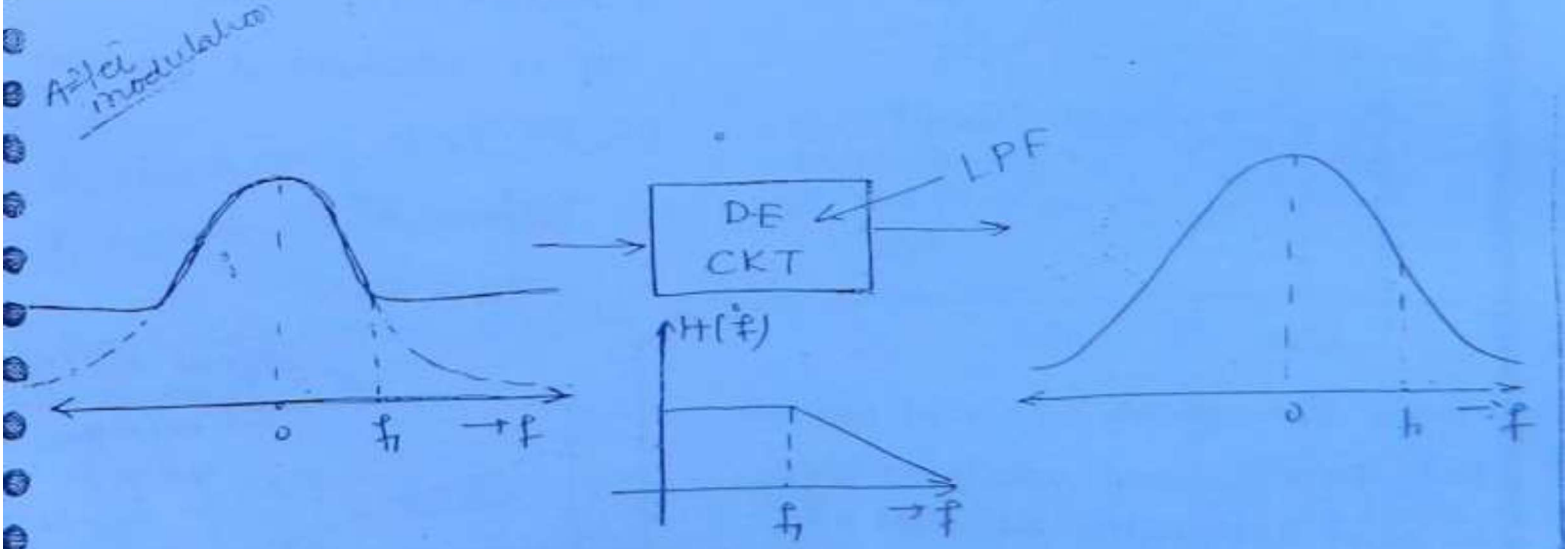
(175)

* The process of increasing the strength of high freqn components of Audio signal is called as PRE-EMPHASIS

* Pre-Emphasis will be done in Tx before modulation

So,





DE-EMPHASIS:-

* It is the process of decreasing the strength of High freqn components of Audio signal.

* De-emphasis will be done in the Rx after demodulation.

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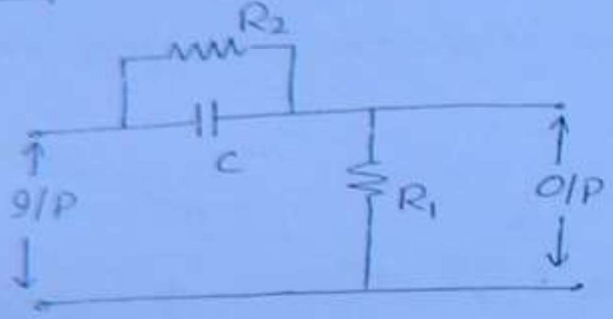
×LPF:

×Lead compensator:

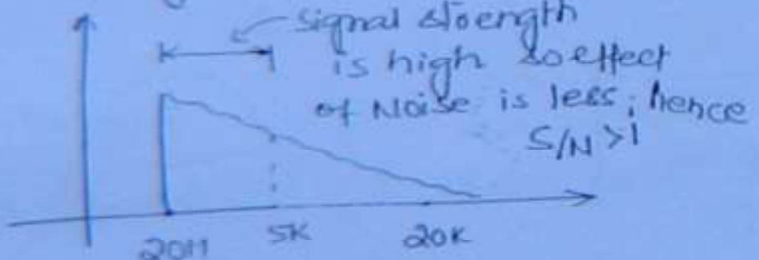


Q. Why Pre-emphasis & De-emphasis needed in FM but not in AM?

As for AM the standard AM B.W is 10KHz

AM B.W = 10KHz

So, Mesg B.W = 5KHz.



Hence, Pre-emphasis not needed.

But, freqn above 5K has less signal strength. So S/N < 1.

Now for voice signal Range is 300Hz to 3.5K, hence, the S/N > 1. So, it can be xmitted very easily w/o any effect of Noise

As, for F.M

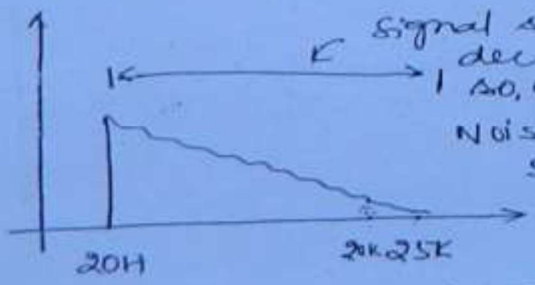
B.W = 200KHz

$2(\Delta f + f_m) = 200K$

Δf ie standard for FM is 75KHz.

$So\ 2f_{dev} = 50K$

$f_m = 25K$



Hence, Pre-emphasis needed to increase S/N.

Note:

In AM transmission, low freqn of Audio signal is only considered for transmission. So, PE & DE are not Req.

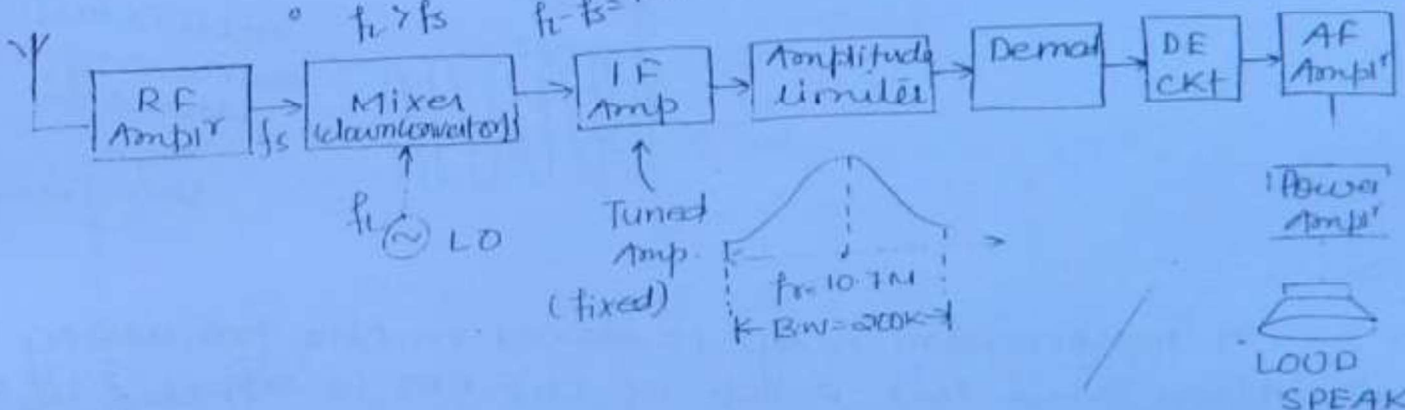
In FM transmission the high freq of audio signal also will be considered for xmission, and for high freqn S/N < 1 so PE & DE are Required.

* F.M RECIEVERS:

For FM, the standards are:

Carrier freqn = 88M to 108 MHz.
F.M BW = 200 KHz.
I.F = 10.7 MHz.

* Block diagram:

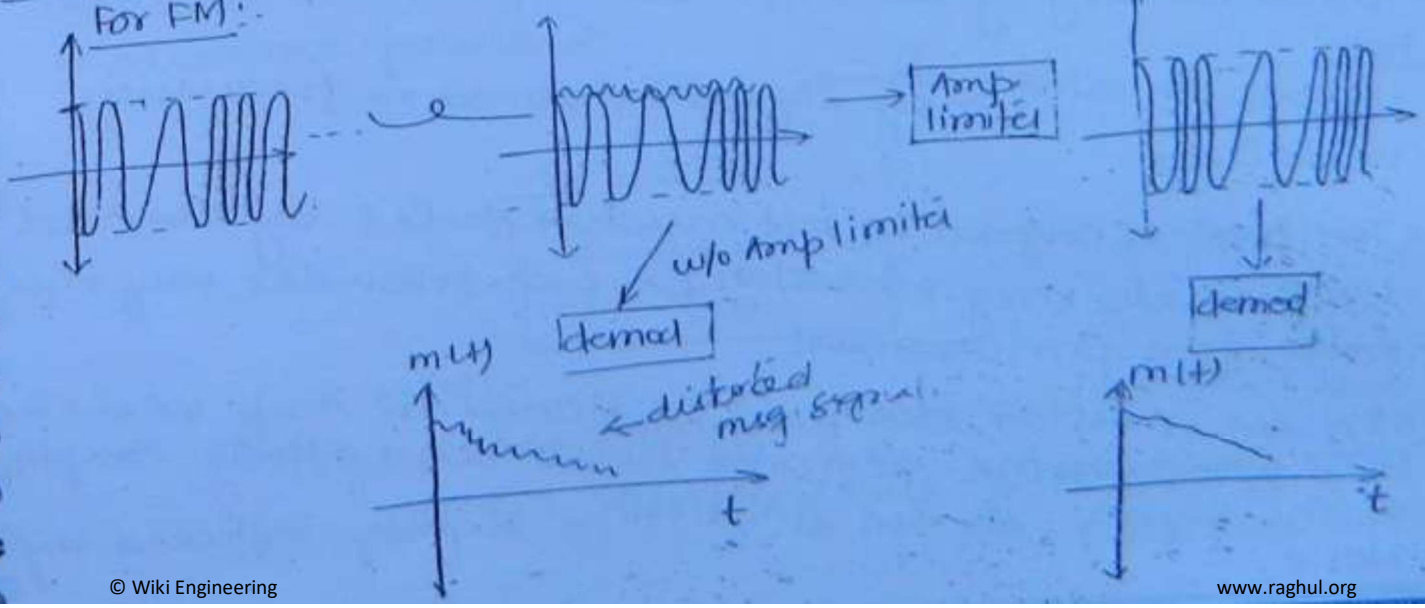


Note:

As the tuning of the Rx (ie f_s) changes, correspondingly f_i has to be changed such that; $f_i - f_s = I.F = 10.7\ MHz$

* Amplitude limiter: ← gives a const amplitude

For FM:



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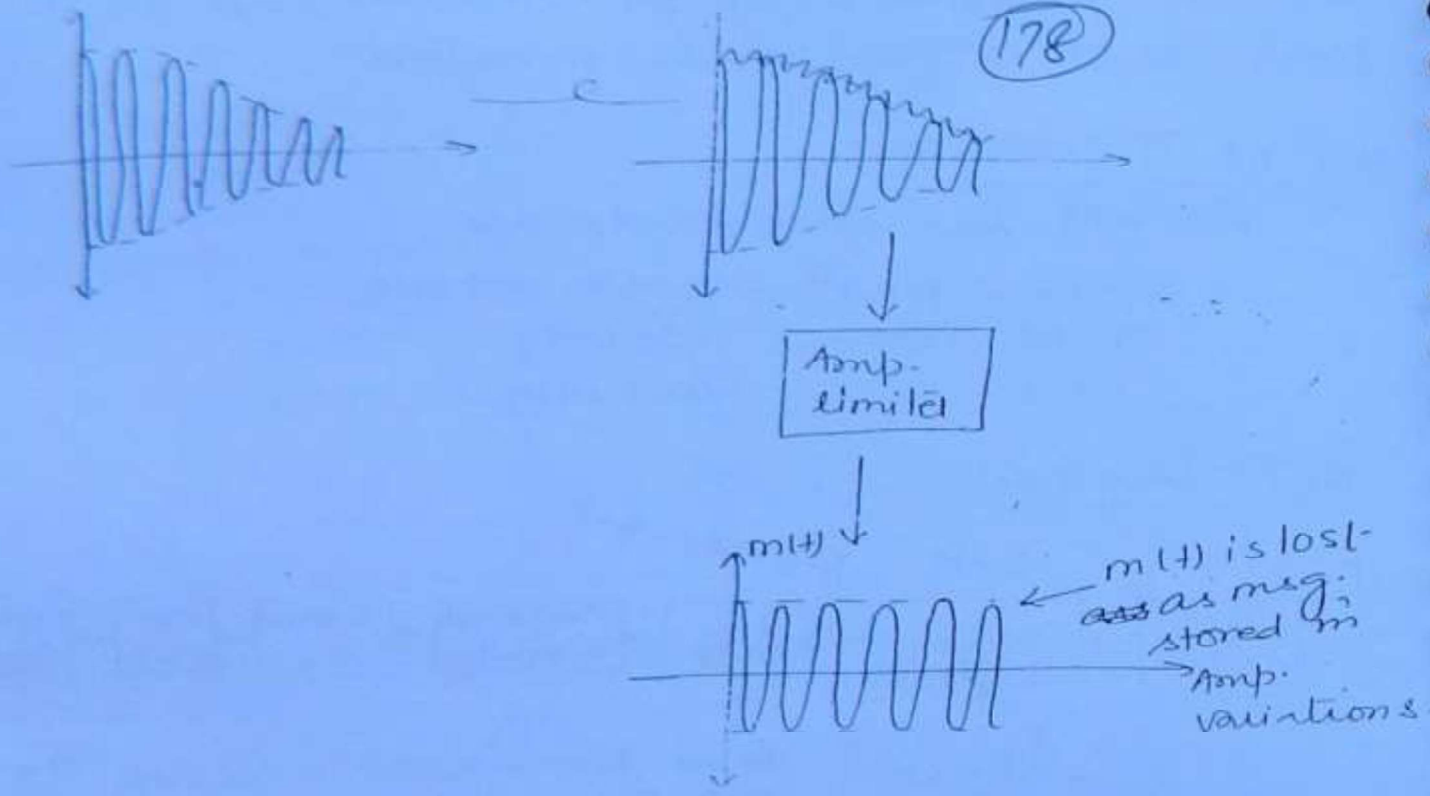
* If Amp. limiter is used in FM, then



Note:

* In FM transmission, msg. is stored in the frequency variation and the amp. of the FM is affected by Noise during channel xmission.
* The msg can be Reconstructed, as msg stored in freqn variation, but the amp. limiter is used to Remove the variation of Amp. in channel xmission. This is done to overcome the drawback of demodulation which is highly sensitive to Amp variation.

Note:

* In FM xmission, msg signal is stored in frequency variations.
* The freqn of signal is not much affected by channel Noise. So, the msg xmission in FM provides very much Noise free environment
* In AM xmission, the msg is stored in Amp. variation and most of the channel Noise also affects Amplitude of the signal, so AM xmission is highly affected by Noise

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